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DEPARTMENT OF INDUSTRIAL ENGINEERING

ADP PROJECT

SAGITTARIUS

MBDA MISSILE SYSTEMS

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Abstract

Project Sagittarius is an educational exercise developed by a team of seven students from the University Federico II of Naples in collaboration with MBDA Italy as part of the Aerospace Design Project exam held by the university. The purpose of the project is to learn design phases of a complex missile system in which students saw themselves involved in the design of the munition. In the following report, is going to be analyzed the preliminary sizing of a Surface-to-air missile, which is designed to defend a zone from UAV flight and loitering munitions attacks; several scenarios have been investigated.

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Chapter 1

Introduction

The rapid proliferation of unmanned aerial vehicles (UAVs), commonly known as drones, has revolutionized modern warfare, offering unprecedented capabilities in surveillance, reconnaissance, and targeted strikes. These agile and versatile platforms have become integral to military operations, providing real-time intelligence and enhancing situational awareness on the battlefield. However, the ease of use and high cost-efficiency of drones also pose significant security threats, challenging the integrity of military operations and national defense. Hostile drones can be used for espionage, delivering payloads, or even as loitering munitions to perform precise attacks, necessitating the development of robust countermeasures.

Addressing these emerging threats requires innovative solutions, and one such solution is the development of anti-drone missile systems. Anti-drone technology represents a critical component of modern defense strategies, aimed at neutralizing aerial threats and ensuring the operational security of military forces. The complexity of countering drone threats lies in their diverse range of sizes, speeds, and flight patterns, which necessitates advanced detection, tracking, and interception capabilities.

This report details the design and development process of a cutting-edge anti-drone missile, specifically engineered to effectively neutralize unauthorized or hostile drones in combat scenarios. The project encompassed a comprehensive study of existing drone technologies, threat scenarios, and countermeasures, culminating in the creation of a missile system tailored to intercept and destroy drones with high precision in dynamic and hostile environments.

The development process began with an extensive threat assessment, analyzing various types of drones used by potential adversaries, their operational characteristics, and potential attack vectors. This analysis informed the design requirements for the anti-drone missile, emphasizing the need for rapid response times, high maneuverability, and precision targeting. Key components of the missile system, including the seeker, guidance system, propulsion, and warhead, were meticulously designed and integrated to achieve optimal performance.

The following sections of the report will cover the project's objectives, design methodology, system specifications, testing procedures, and performance evaluation. Objec-

tives include ensuring the missile's ability to detect, track, and intercept drones under various combat conditions. The design methodology section will delve into the engineering principles and technological innovations employed in the missile's development. System specifications will provide a detailed overview of the missile's capabilities, including its range, speed, and targeting accuracy. Simulation procedures will outline the rigorous trials conducted to validate the munition capabilities in a multitude of possible scenarios.

Chapter 2

Concept of Operation

2.1 Mission Characteristic

The mission to be performed is of paramount importance, focusing on the protection of a designated zone with a radius of 3 kilometers. Designating the type of mission as Ground-based air defence. This protected area may encompass critical infrastructure and assets that that must be protected. Ensuring the safety and integrity of this zone against aerial threats, particularly hostile drones, and loitering munitions are the primary objective.

To achieve this, the anti-drone missile system must possess the capability to detect, track, and intercept potential threats at significant distances. Specifically, the missile must be able to engage and neutralize hostile drones at a range of up to 8000 meters from the missile launch point. This extended engagement range is crucial to provide a substantial buffer zone, allowing for the interception of threats well before they can penetrate the protected perimeter.

Moreover, the system must be adept at handling threats at various altitudes, with the capability to intercept targets at heights of up to 5500 meters. This high-altitude interception ability ensures that drones operating at different flight levels, whether they are conducting high-altitude surveillance or low-flying attack missions, can be effectively neutralized.

The requirement for such extensive range and altitude capabilities is driven by the need to preemptively address threats and prevent them from coming anywhere near the protected area. To provide the previously expressed defensive characteristics the anti-drone missile system has been adapted to multiple scenarios other than the design one, set to an initial target position of 8000 m and an initial high of 5000 m with an initial velocity of 60 m/s flying towards the area to protect horizontally.

In summary, the mission necessitates an advanced and highly capable missile system that can deliver precise and reliable performance over considerable distances and altitudes. This ensures comprehensive protection of the 3-kilometer radius zone, safeguarding critical assets from the ever-evolving threats posed by modern drone technol-

ogy.

2.2 Target Characteristics

As previously said an inquire on the possible target has been conducted, analyzing maximum load factor, speed range, weight range, armour capabilities and average dimensions. The drones to be targeted have a maximum load factor of n=2, so that with the "rule of three" this sets the missile to have a load factor at least of six. The speed range of the drones goes from a minimum of 30 m/s to a maximum of 60 m/s. The weight of the targets also has been studied and it was found to be in a range from a minimum of 100 kilograms to a maximum of 300. The dimensions are the most variable parameters going from bigger aircraft that have a wing span of 2.5 meters to small drones of a 0.5 meters wingspan.

As previously mentioned, a comprehensive inquiry into the potential targets has been conducted to inform the design and operational parameters of the anti-drone missile system. This analysis included an examination of various factors such as maximum load factor, speed range, weight range, armor capabilities, and average dimensions of the drones that are likely to be encountered.

One of the critical parameters studied was the maximum load factor of the drones. It was determined that the target drones possess a maximum load factor of

$$n = 2$$

To ensure the missile can effectively engage and outmaneuver these drones, the "rule of three" was applied, setting the missile's load factor to at least six. This ensures that the missile has three times the maneuvering capability of the target, allowing it to perform sharp turns and rapid adjustments necessary for successful interception.

The speed range of the target drones was also analyzed. These drones can operate at speeds ranging from a minimum of 30 meters per second to a maximum of 60 meters per second. This speed range necessitates that the missile system is versatile and capable of engaging targets moving at both low and medium speeds. The missile must therefore be designed to have a proper velocity to intercept faster targets effectively.

In addition to speed, the weight of the target drones was assessed. The analysis found that the drones vary significantly in weight, with a range from a minimum of 100 kilograms to a maximum of 300 kilograms. This variation in weight affects the drones' flight characteristics and their susceptibility to different interception strategies. The missile system must be able to detect and engage targets across this weight spectrum, requiring a robust targeting and propulsion system.

The dimensions of the target drones were identified as one of the most variable parameters. The drones range from larger aircraft with a wingspan of 2.5 meters to smaller drones with a wingspan of just 0.5 meters. This variation poses a challenge for the missile's guidance and targeting systems, as they must be capable of identifying and accurately targeting drones of vastly different sizes. Larger drones present a bigger target but may have more advanced armor capabilities, while smaller drones are harder to detect and require precise targeting.

Furthermore, the analysis included an assessment of the armor capabilities of the

target drones. Although many drones are lightly armored to maximize their payload capacity and maneuverability, some may have reinforced structures to protect critical components. This necessitates that the missile be equipped with a warhead capable of penetrating the armor to ensure the destruction of the target.

In summary, the detailed analysis of the potential targets provided critical insights that shaped the design and functionality of the anti-drone missile system. By understanding the maximum load factors, speed ranges, weight variations, and dimensions of the target drones, the missile system has been tailored to effectively counter a wide array of threats, ensuring robust defense capabilities in diverse combat scenarios.

Parameter	**Details**
Maximum Load Factor (n)	Target drones: $n = 2$ Missile load factor: ≥ 6
Speed Range	Minimum speed: 30 m/s Maximum speed: 60 m/s
Weight Range	Minimum weight: 100 kg Maximum weight: 300 kg
Dimensions	Largest wingspan: 2.5 meters Smallest wingspan: 0.5 meters
Armor Capabilities	Lightly armored

Table 2.1 Analysis of Target Drone Parameters

2.3 Interceptor Characteristics

To execute the ground-based air defense mission described earlier and to neutralize the identified threats, the missile has been meticulously designed with specific characteristics while keeping costs as low as possible. The primary goal was to develop an effective yet economical solution.

Before beginning the design process, it was decided to develop a subsonic missile. This decision was based on the relatively low speed of the target drones and the desire to enhance the missile's maneuverability without encountering complications induced by shock waves due to supersonic speeds. This approach allows for better control and precision during engagements.

In the initial phase of the project, a thorough investigation was conducted to study existing missile systems. This research informed the establishment of a set of requirements for our missile. It was determined that the missile would be composed of two stages: a booster stage and a sustain stage. The booster is designed to propel the missile to a velocity of nearly 200 meters per second, depending on the scenario, providing the necessary initial thrust to reach engagement speeds quickly. The sustain stage provides a constant thrust to maintain the missile's speed without further acceleration into supersonic velocities. The cruise speed is intended to remain between 200 and 275 meters per second, avoiding issues associated with shock waves and ensuring stable flight dynamics.

After these decisions were made, it was determined that the missile would be mounted on a lightweight vehicle to enhance mobility and deployment speed. The missile's max-

imum weight was set at 10 kilograms to ensure compatibility with various launch platforms and ease of transport. The maximum diameter of the missile is specified to be 0.08 meters, and the length is set at 1.20 meters. These dimensions ensure the missile is compact enough for efficient storage and deployment while maintaining the necessary aerodynamic properties.

These design choices and specifications were made to create a highly effective ground-based air defense missile system tailored to neutralize specific drone threats. By focusing on a subsonic design and carefully balancing speed, weight, and dimensions, the missile achieves a high level of maneuverability and precision while maintaining cost-efficiency. This ensures that the defense system can be widely deployed and reliably protect critical areas from aerial threats.

Parameter	**Details**
Missile Mission Type	Ground-based Air defence.
Launching platform	Lightweight vehicle.
Missile Type	Subsonic missile .
Number of stages	2 .
Stages	**Booster Stage:** Propels missile to 200 m/s. **Sustain Stage:** Maintains speed between 200 and 275 m/s.
Missile Load Factor	6
Expected Missile Weight	Maximum of 10 kg to ensure
Expected Missile Dimensions	Diameter: 0.08 metres Length: 1.20 metres

Table 2.2 Missile Design Requirements

Chapter 3

Reference Scenario

In this chapter will be presented the reference scenario which is the scenario that the mission aims to face. The initial positions and the results will be presented here, while the Simulink model used to obtain these results and the missile design will be presented in the in the following chapters. As already mentioned in the previous chapter, this scenario is chosen so that the missile that comes out of the design process is capable of adapting to any other scenario. The reference scenario is devised as follows: the position of the missile at the beginning of the simulation is in the origin, while the target one is set to be at a distance on the ground of 8 kilometers at an altitude of 5 kilometers, and it travels at a constant horizontal speed of 60 m/s towards the initial position of the missile, that corresponds with the center of the zone to be protected by this ground-based air defence mission.

Initial distance on the ground	$8000m$
Initial altitude	$5000m$
Velocity	60 m/s

Table 3.1 Target parameters

3.1 Trajectory overview

By means of the Simulink model that will be described in chapter 5, the following simulation has been obtained. The results will now be presented to provide an overview of the events, thereby making the design choices more comprehensible later on. It is necessary to note here that the graphics shown in this chapter refer only to the second stage flight, and the initial conditions for this phase are the final ones of the boosting phase, which are not reported due to its brevity.

In fact, it is possible to notice that the interceptor has a non-zero speed at the first timestep of the Simulink simulation and it is not placed at the origin, where it actually

starts at the beginning of the whole simulation, comprehensive of the boosting phase.

The first aspect to examine in this simulation is the flight trajectory shown in Figure 3.1, where the missile trajectory is depicted in blue, and the target's trajectory is shown in red. In Figure 3.1, it is evident that the missile starts with a correct angle, and the trajectory is nearly a straight line. This indicates a good design and appropriate selection of flight, aerodynamic, and guidance parameters for the missile, which will be elaborated upon with proper engineering choices.

It can be noted that the missile tends to predict the target's final position due to its navigation capabilities, as will be explained in more detail in chapter 6.

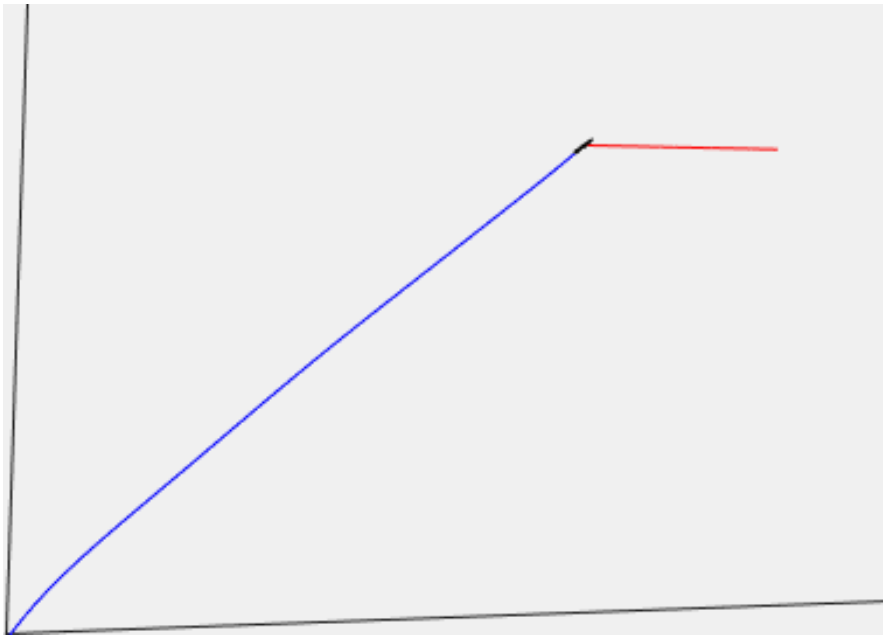


Figure 3.1 Missile trajectory

3.2 Trajectory simulation and result analysis

During the mission the *distance to go*, shown in fig 3.2, decreases as the missile approaches the target. The mass (3.3) decreases due to the burning of propellant until $t = 30$ seconds, at which point the mass stabilizes and does not change further because the maximum burn time of the propellant grain has been reached. For the same reason, the airspeed increases until $t = 30$ seconds, and then begins to decrease as the thrust is nullified, leaving only the contributions of drag and gravity.

The Probability to Kill (P_k) value is plotted in the final moments of the simulation (3.5). Initially, P_k is zero, but as the missile reaches an appropriate distance from the target, P_k increases to nearly 1, before decreasing again as the missile moves away from the target. It is observed that there is a moment when P_k is at its highest, which is not necessarily when the missile and target are closest, as will be discussed in chapter 7.

As previously mentioned, the missile follows an almost straight line, which is evident from the fact that it does not need to turn, and the angles of attack are practically always zero as shown in figure 3.6. Examining the values of the pitch angle θ (3.7), that is the angle between the horizontal axis and the body reference line, it is noted that initially, the starting angle, dictated by the auto-aiming function described in chapter 5, is slightly greater than the angle maintained during the cruise phase. Although the trajectory is very clean, and the P_k is high, there is always some error in the guidance system. This error causes the demanded acceleration by the guidance system to diverge in the final moments because the system wants the missile to "translate" along the axis normal to it in order to hit the target as shown in figure 3.8. Additionally, the look angle also diverges in the final moments (3.11), as once the target is passed, it is practically behind the missile, causing the look angle to strongly increase.

The demanded acceleration is transformed into fin deflection through an autopilot explained in chapter 8, resulting in an actuated acceleration (3.10). It is seen that the autopilot is well-calibrated as the actuated acceleration closely follows the demanded acceleration, as visible in the detailed graphics that cut out the last moments.

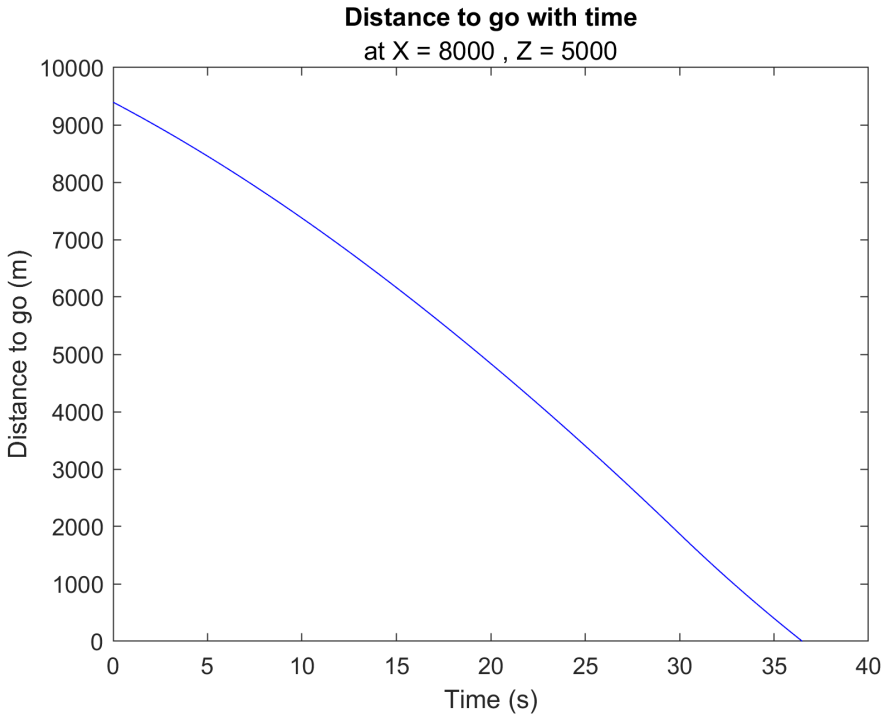


Figure 3.2 Distance to go

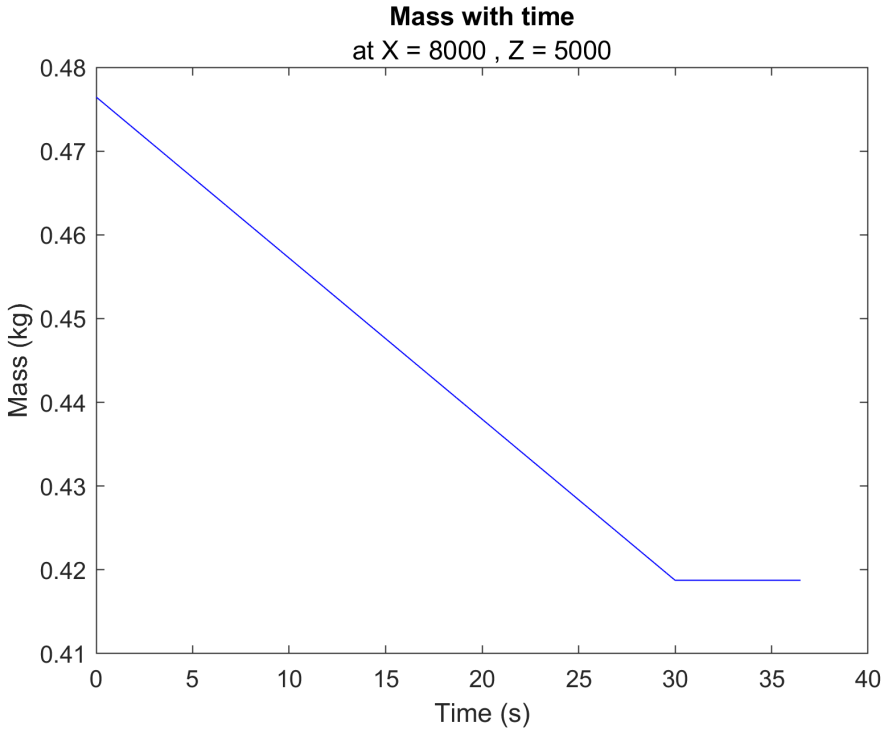


Figure 3.3 Mass

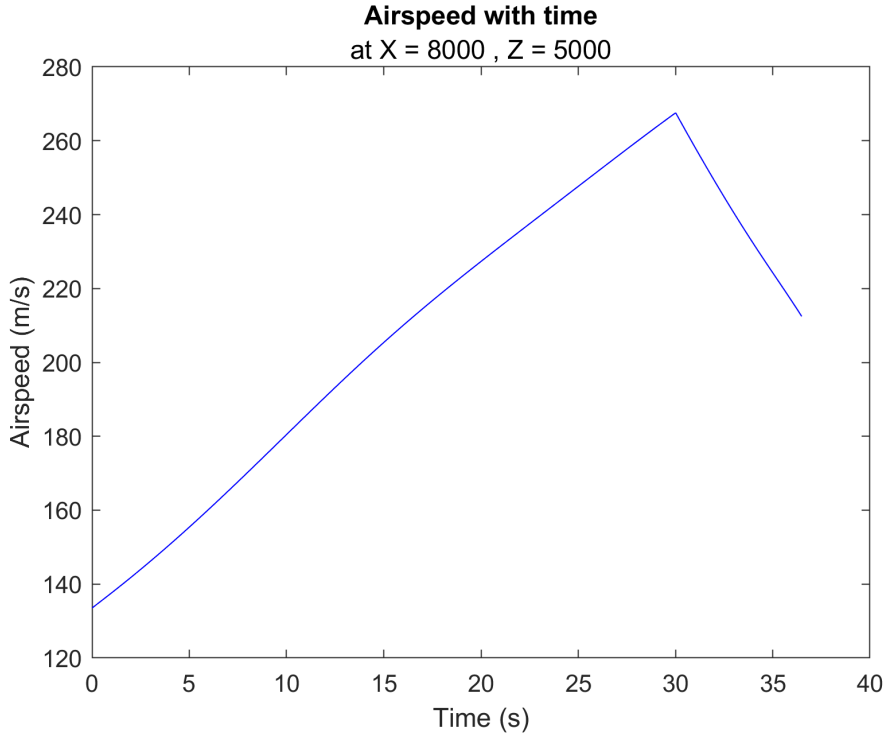


Figure 3.4 Airspeed

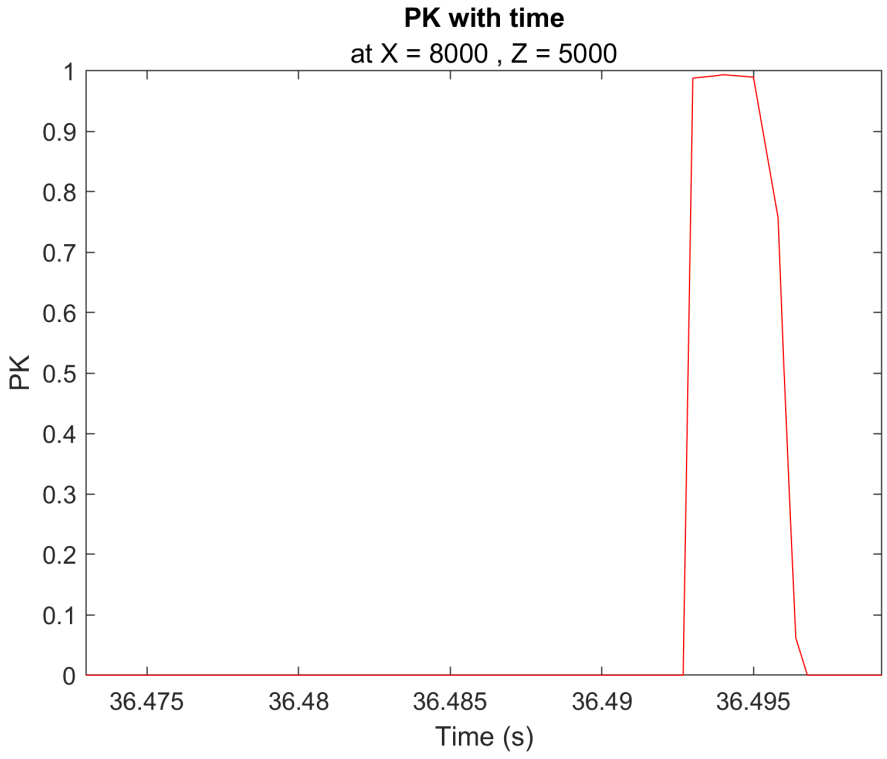


Figure 3.5 PK

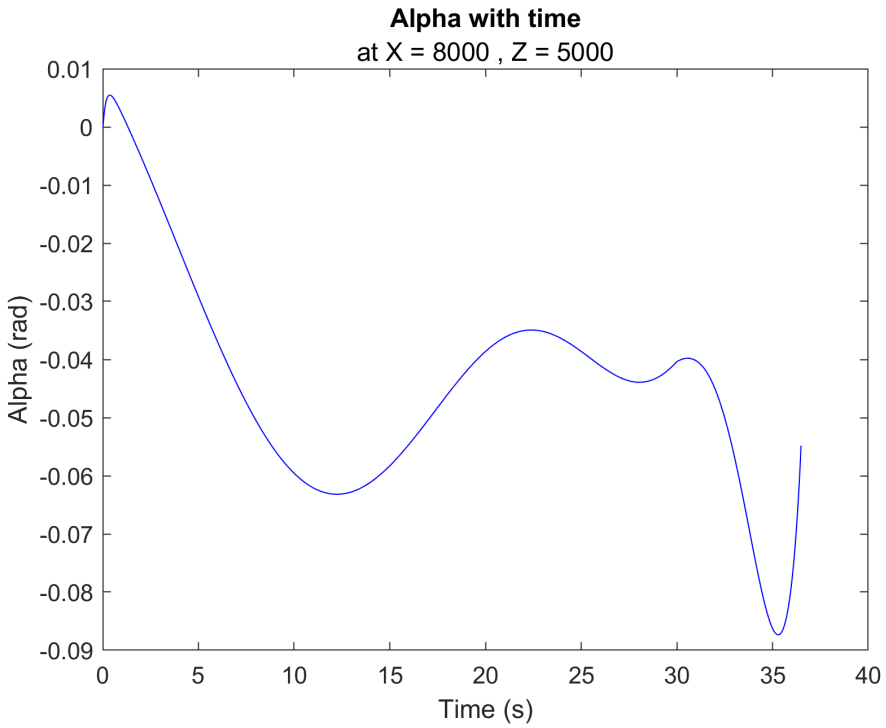


Figure 3.6 Alpha

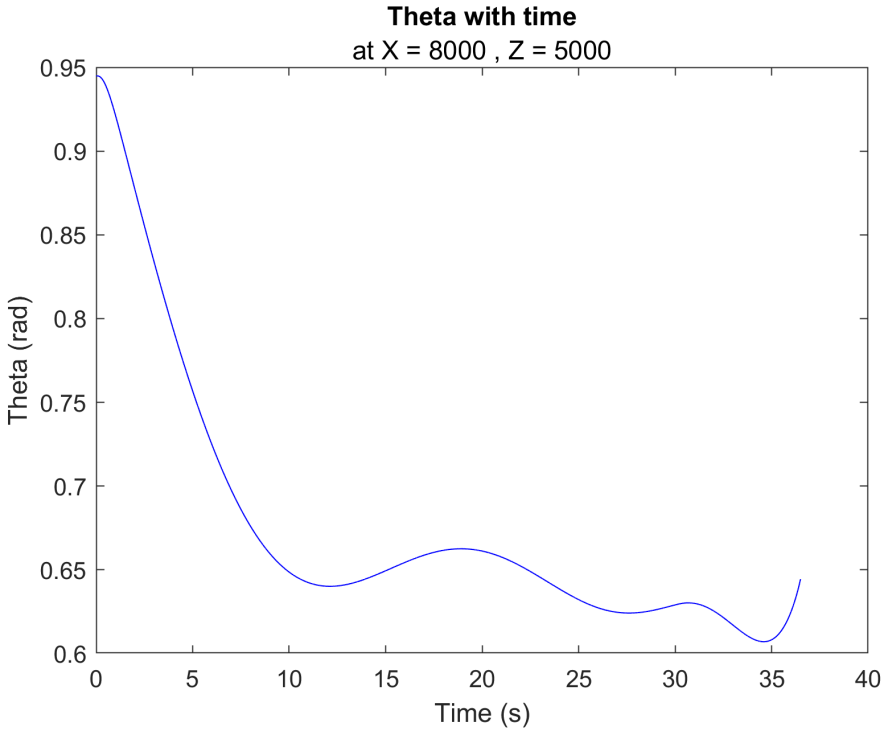


Figure 3.7 Pitch angle

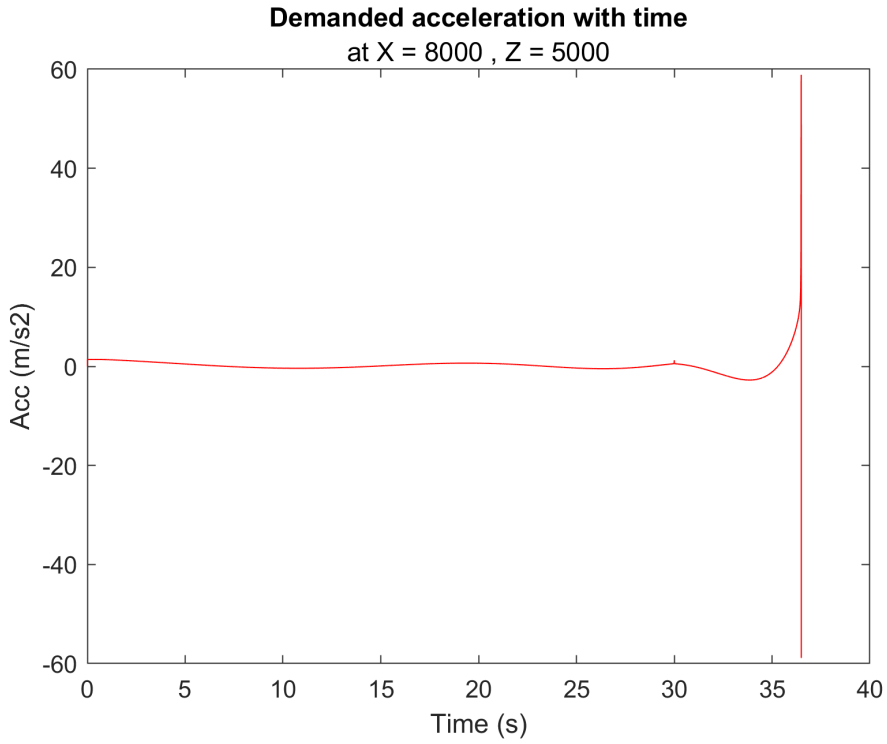


Figure 3.8 Demanded acceleration

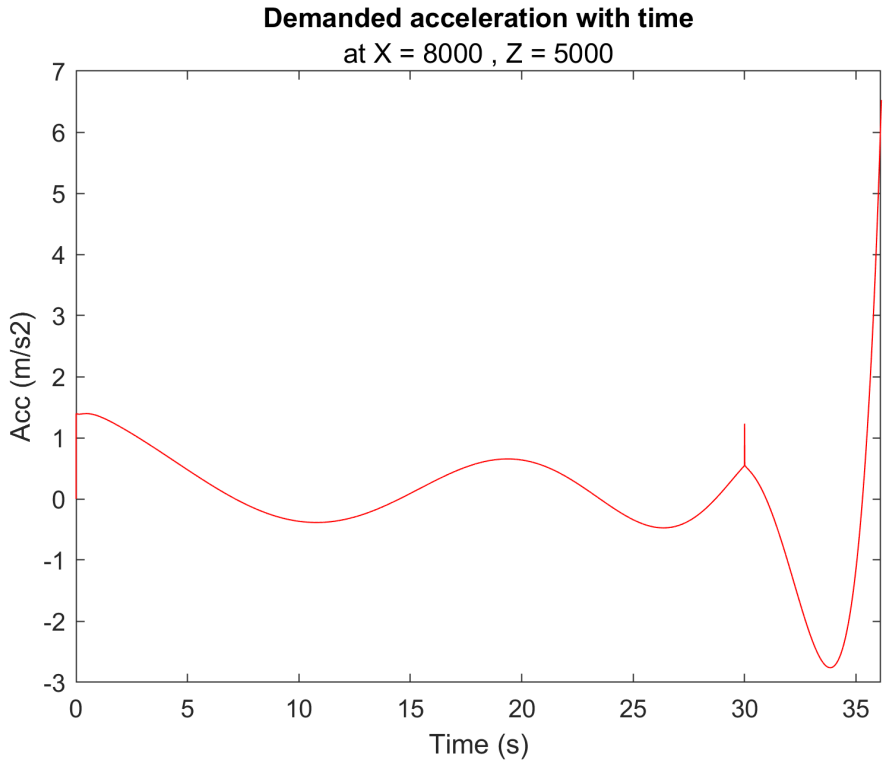


Figure 3.9 Detail of the Demanded acceleration before the divergence

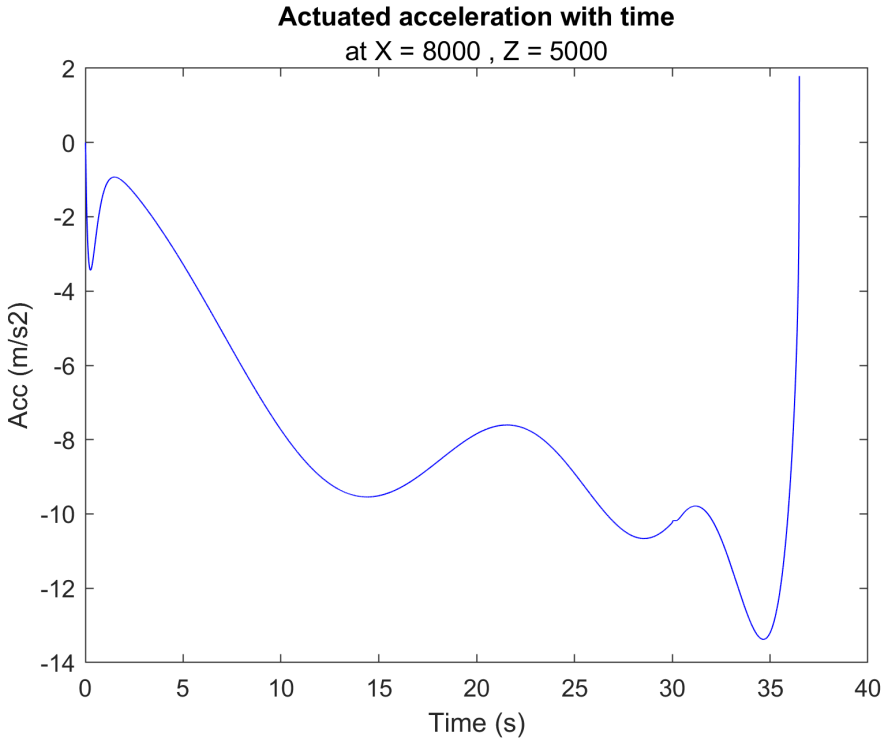


Figure 3.10 Actuated acceleration

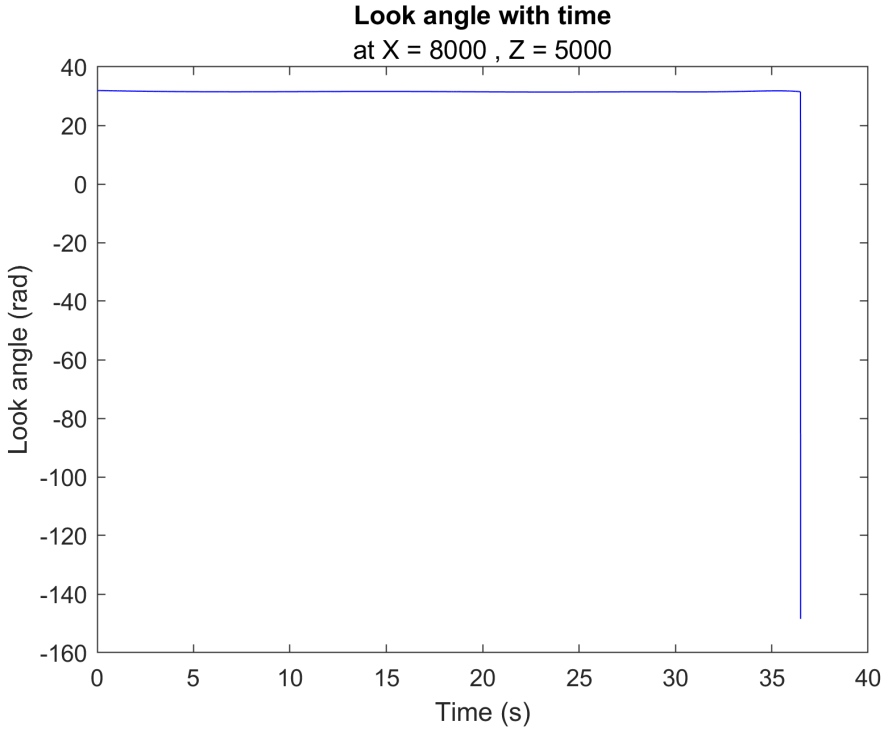


Figure 3.11 Loook angle

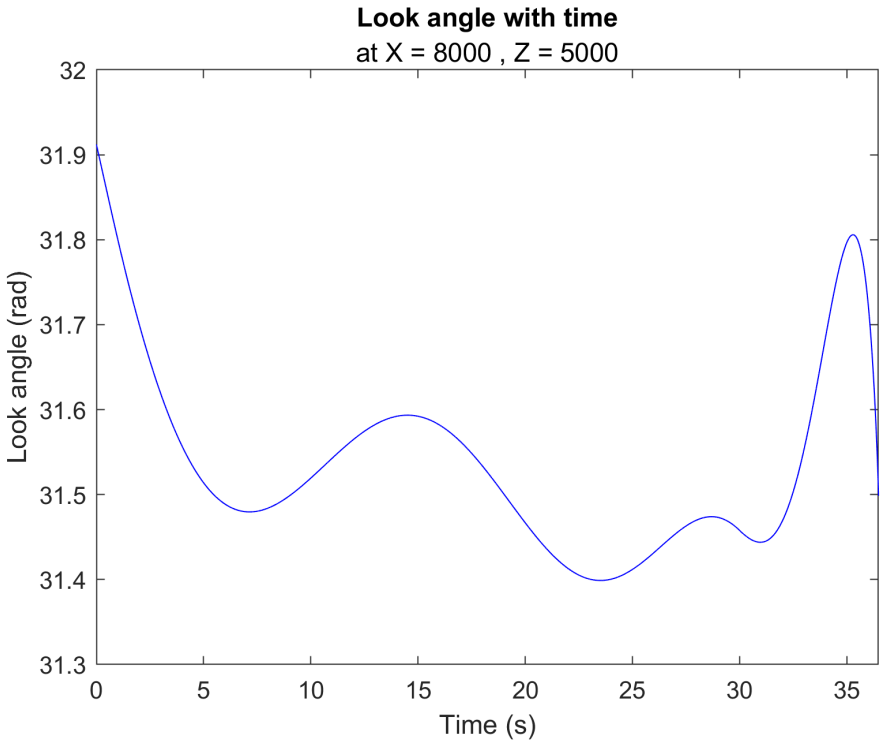


Figure 3.12 Detail of the look angle before the divergence

Chapter 4

Design Process and Team Organization

4.1 Design Process

The design process begins by establishing the final flight conditions of the first stage, which correspond to the initial conditions of the second stage, from which the final conditions of the latter are derived. Once the flight conditions for both stages are defined, the detailed design phase commences, divided into relevant technical areas: starting with guidance, moving through lethality, and continuing with propulsion, aerodynamics, and structures. Upon completing the analysis of the second stage, the design of the first stage is revisited, utilizing the results obtained from the various technological areas.

The report follows the logical flow of the design process, beginning with flight simulations that provide an overview of the objectives to be achieved. This enables the establishment of requirements through an iterative cycle, which in turn guides the design of the missile's two stages. Before delving into the analysis of the two stages, chapters on guidance, which initialize navigation and the theoretical bases used, and on lethality, covering all aspects related to the warhead, are addressed.

The analysis of the second stage is divided into interconnected technological areas. It starts with propulsion, defining the dimensions of the engine and velocity profiles, followed by aerodynamics, where the wings and nose are designed, accompanied by fluid dynamics and aerodynamic analyses, and concludes with structural design, which includes sizing the entire structure of the considered stage. Based on the design results of the second stage, the first stage is then designed, adapting all previous analyses and observations to its functionality. The process concludes with off-design conditions, providing a clearer understanding of the missile's design limits.

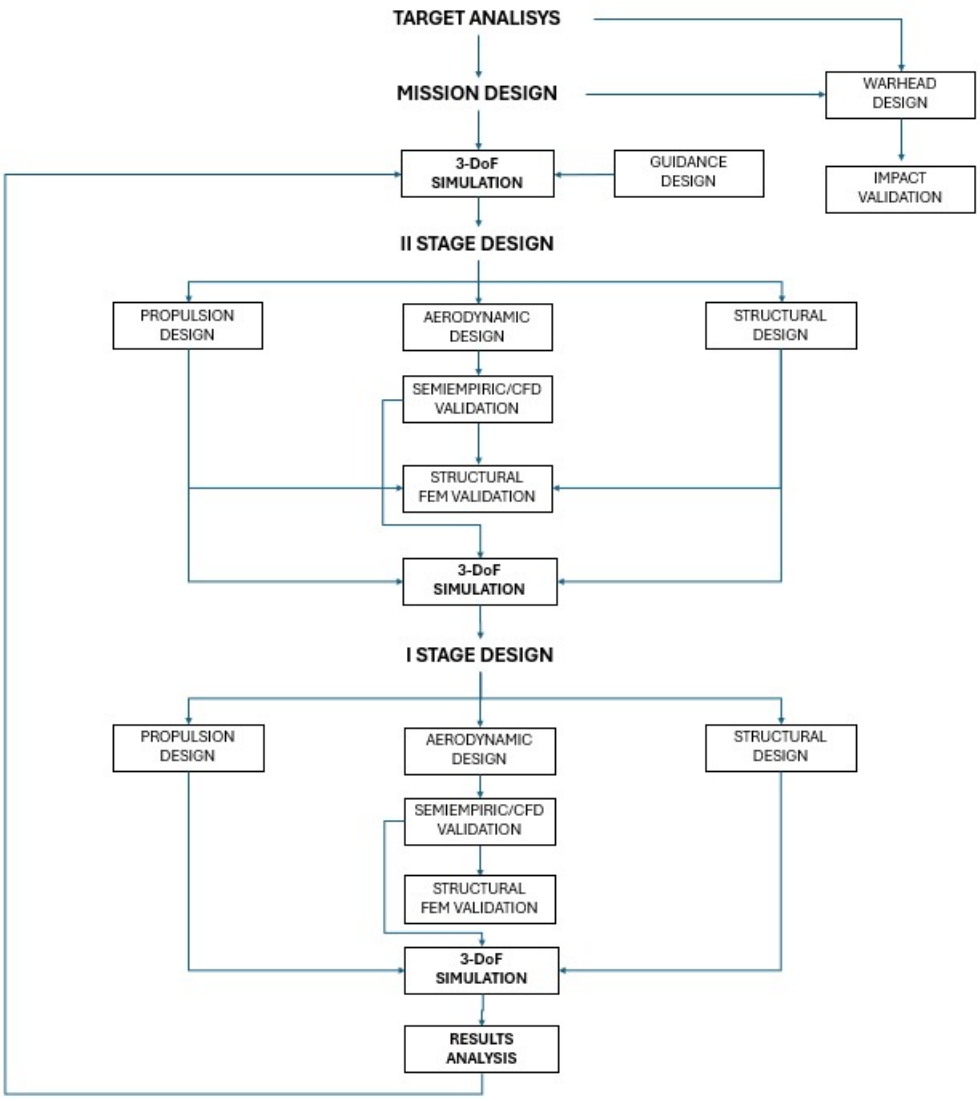


Figure 4.1 Design Flow chart

4.2 Team Organization

We are pleased to present ourselves as the founding and executing team of the **SAGITTARIUS** project.

- **Vincenzo Junior Di Rosa**, a Master's degree student in the Fluid Dynamics and Propulsion program at the University of Naples Federico II, serves as the project team leader and system engeneer, overseeing the missile's design organization.
- **Gabriele Foggiano**, a Master's degree student in the Fluid Dynamics and Propulsion program at the University of Naples Federico II, is responsible for aerodynamics, from the design of the fins to the nose, and collaborates in the propulsion area and the general organization of the team.
- **Francesco Gasperini**, a Master's degree in the Aeronautics program at the University of Naples Federico II, is responsible for the structures area and collaborates in the aerodynamics and propulsion areas, executing FEM analyses.
- **Matteo Giannitti**, a Master's degree student in the Space program at the University

of Naples Federico II, is responsible for lethality and collaborates on the guidance and control of the missile.

- **Carmine Marra**, a Master's degree in the Fluid Dynamics and Propulsion program at the University of Naples Federico II, co-responsible for the propulsion area, with a particular focus on the first stage, and collaborates in the aerodynamics area, as well as co-executing CFD analyses.
- **Antonio Niro**, a Master's degree in the Space program at the University of Naples Federico II, is responsible for guidance and control, CAD execution and mechanical architect , as well as collaborating in the general organization of the team.
- **Davide Sergio**, a Master's degree student in the Fluid Dynamics and Propulsion program at the University of Naples Federico II, co-responsible for the propulsion area, with a specific focus on the second stage, and collaborates in the aerodynamics area, as well as co-executing CFD analyses.

The synergy of the team and the interconnection between the various technological areas have made the realization of the missile, which will be presented in the next chapter, possible.

Chapter 5

Simulation

This chapter provides a comprehensive overview of a MATLAB and Simulink based simulation model designed to compute the trajectory, impact point, and probability of kill (P_k) of a missile.
The model uses various inputs such as thrust values, burn time, missile masses, moments of inertia, initial conditions, and more to facilitate missile design trade-offs.
The base model is taken from the MathWorks ®repository. [6]
Based on the chosen mission design scenario, which is to cover an area of radius 5 km and altitude 8 km, the thrust and burn time are chosen so that Sagittarius is able to fulfill the mission and intercept the target in the best possible way.
The values of thrust (T) and burn time (t_b) will be then passed to the propulsion department to properly size the missile.

5.1 Input parameters

The input data of the simulation model are taken from a MATLAB file and they are divided as follows:

- *Masses taken from the CAD model.*
  - "Full second stage mass". This is the mass of Sagittarius' second stage with the full propellant loaded. This correspond to $Fullmass = 11.224kg$.
  - "Grain mass" is the loaded propellant mass. It corresponds to $m_p = 2.524kg$.
  - "Empty second stage mass" equal to $Emptymass = 8.70kg$.
    It is calculated with the difference between the full second stage mass and the grain mass.

- *Moment of inertia* (I_{yy}) for both the full and empty configuration are also taken from the CAD model. Values are the following:
  - $I_{yy,FULL} = 1.034kg \cdot m^2$
  - $I_{yy,EMPTY} = 0.904kg \cdot m^2$
- *Missile's characteristic dimensions*, represented in this case by the circular cross-sectional area of the missile and its diameter.
  - $S_{REF} = 0,005m^2$
  - $D_{REF} = 0,08m$
- *Solid rocket motor data.*
  These are the key parameter for the simulation, modified at each iteration to reach in the optimal way the target in every scenario.
  The final values are:
  - $T = 150\ N$
  - $t_b = 30s$
- *Center of gravity that changes with time.*
  It starts from a value of $X_{CG} = 476.45mm$ up to $X_{CG} = 418.75mm$ when the propellant grain is exhaust.
- Center of pressure $X_{CP} = 0.6873m$
- Drag Coefficient C_D: modeled with 5 different values to have a dependance on Mach number.
- Aerodynamics coefficients:
  - $C_{N,\alpha} = 7.609\frac{1}{rad}$
  - $C_{X,fin} = 0$

  - $C_{Z,fin} = 5.61$
  - $C_{M,q} = -0.65$

- Actuators data
  - $\omega_n = 150 \frac{rad}{s}$
  - $\zeta = 0.707$
  - $fin_{max} = -fin_{min} = 35 deg$

- Sensors position with respect to Center of Gravity and maximum load factor n.
  - $l_{acc} = 0.07465 m$
  - $n = 6$

- The X and Z position where the defense systems have identified the target.

- Target velocity and trajectory.
  For the purpose of the simulation the following values are considered:
  - $V_{tgt} = 60 \frac{m}{s}$;
  - $\theta_{tgt} = 180$ deg.
    The value of θ_{tgt} means the target is moving along a straight line and towards the launch base.

- The initial position (X_i, Z_i) and initial attitude θ_i of the second stage.
  These values are computed depending on the considered scenario with an auto-aiming procedure (see next section) that integrates the equation of motion for the booster phase.

5.2 Auto-aiming

Sagittarius is equipped with a simple but effective auto-aiming function that ensures maximum accuracy in every operational scenario. Specifically, the auto-aiming system is designed to visualize the target and adjust the missile's trajectory before launch.

The underlying concept of the auto-aiming function is to select the correct pitch angle θ for launching the interceptor. To calculate this angle, a dedicated function has been developed, which relies on predicting the motion of both the missile and the target. To determine the value of θ, the motion equations of both the missile and the target are considered as if both were moving in uniform rectilinear motion. This assumption holds true for the target, but not for the missile. However, after launch, the navigation system will guide the missile until it intercepts the threat.

For the interceptor, within the aforementioned motion equation, a constant speed of 200 m/s is assumed, and it starts from the origin of the coordinate system. The target, on the other hand, moves as described in 3, but more generally, it is assumed to move in uniform rectilinear motion with a constant speed of 60 m/s and a zero pitch angle.

From the two motion equations, a system of two equations with two unknowns, namely the time t and the pitch angle θ of the interceptor's trajectory, is solved. This is achieved using the MATLAB function `fsolve`, which solves homogeneous systems of order n. However, since it is a trigonometric equation, it admits multiple solutions. Consequently, to find the correct solution for the auto-aiming purpose, proper initialization of the auto-aiming function is required.

Script 5.1 Aiming function

```
function F = AimingFun(x)

global v_aim x_ini_tgt h_ini_tgt v_tgt theta_tgt g

% along the y axis
F(1) = v_tgt*sin(theta_tgt)*x(2) + (h_ini_tgt) - v_aim*sin(x(1))*x(2) +
    0; % +0.5*g*x(2)^2;
F(2) = v_tgt*cos(theta_tgt)*x(2) + (x_ini_tgt) - v_aim*cos(x(1))*x(2) +
    0;

end
```

Script 5.2 Detail of the main calling the function

```
        % Aiming
        v_aim = 200;
        [sol_theta_ini,t_crash_preview]  = fsolve(@AimingFun, [0, 30]);
           % Initial Body Attitude [rad]
        theta_ini = min(sol_theta_ini);   % First solution
        theta_ini = 1.3*theta_ini + 0.008*(1/theta_ini);
        theta_ini_deg = theta_ini/d2r;
        look_angle_ini = atan((h_ini_tgt-h_ini)/(x_ini_tgt-x_ini));
```

Once the system of equations is solved, the smaller solution is selected due to the previously mentioned reasons regarding the multiplicity of the solution. Furthermore, the launch angle is calculated as follows:

$$\theta_{launch} = 1.3 * \theta + 0.008 * (1/\theta)$$

This is done so that the initial launch angle is slightly greater than the one calculated in the *Aiming Function*, in order to account for gravity effects. This adjustment allows the interceptor to perform better and reach greater distances.

5.3 MATLAB function for booster phase

The booster phase is the first phase of this two-stage mission, where the missile must reach the assigned speed in the shortest time possible.

To simulate this phase of the mission, due to its brevity, it is possible to assume that the missile can be approximated as a material point with variable mass, under the effect of aerodynamic force, gravity force, and thrust. The motion equations are solved in the missile's axis, and the forces are calculated as follows:

$$F = \dot{m} \cdot c \tag{5.1}$$

$$D = 0.5 \cdot \rho \cdot V^2 \tag{5.2}$$

Furthermore, it can be asserted that the approximation of the missile as a material point is justified not only by the brevity of the flight but also by the strong stability of the missile in the lift-off configuration, here the booster hasn't in which the booster has not yet separated from the second stage. The interaction with the aforementioned forces, makes the missile behave as a material point.

Some input data are required to simulate this part of the mission. Their values are displayed here, while the methods used to obtain them will be explained in chapter 9.

Charcteristic velocity c^*	1532 m/s
Thrust coefficient cf	1.507
Total initial mass $m0$	16.695 kg
Mass flow rate $\dot{m}$	1.57 kg/s
Drag coefficient C_D	0.37

Table 5.1 Characteristics of 7075 Aluminum

Script 5.3 Datail of the *ode*45 solving the booster motion equations

```
% derivatives
V_axial_dot = (mdot_boost*c)/(m0-mdot_boost*t) - 0.5*rho*S_ref*Cd*V_axial
    ^2/(m0-mdot_boost*t) - g*sin(theta);
V_normal_dot = g*cos(theta);
x_dot = V_axial*cos(theta) + V_normal*sin(theta);
z_dot = V_axial*sin(theta) - V_normal*cos(theta);
theta_dot = 0;

dStatedt = [V_axial_dot; V_normal_dot; x_dot; z_dot; theta_dot];
```

5.4 3-D.O.F Simulink model for sustain phase

The Simulink model used to perform the trade-off analysis is structured as shown in "figure 5.1 ": The smaller blocks perform simple operation, while the red, yellow and blue

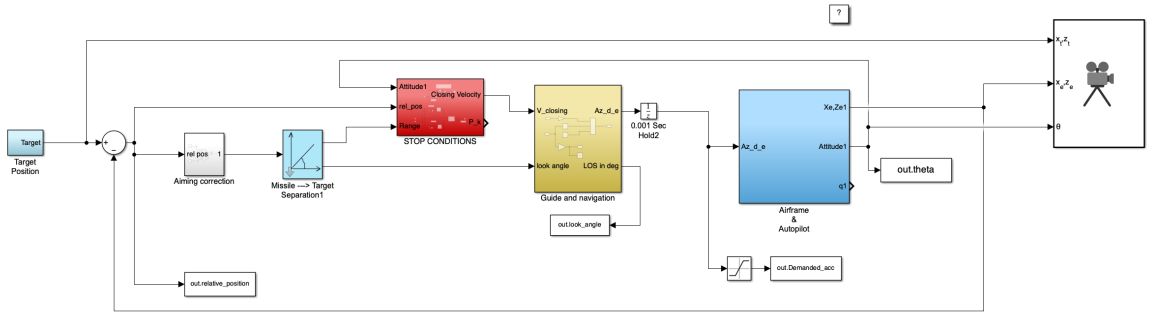


Figure 5.1 Simulink Model Overview

block will be covered with a deeper level of details.
Starting from the left, the block *Target position* calculates at every instant the position of the target based on its direction and speed.
The *Aiming correction* block performs an ***Aiming point selection*** to guide the missile so that the warhead explosion inflicts the maximum possible damage to the target according to the calculations described in Chapter 7
The *missile to target separation* block performs a transformation from Cartesian to Polar coordinates $[x, y] -> [r, \theta]$.

5.4.1 Stop Conditions block

Implemented stop conditions

The simulation stops when two conditions are met:

1. Closing velocity becomes greater than zero: $V_C = \dot{R} > 0$. That is because as the missile gets closer to target the range decreases and its first derivative has negative values. When the closing velocity is positive it means the missile is moving away from the target.
2. The range is $R >= 1$.

When these two conditions are met we are in a situation where the missile has surpassed the target by not more than 1 m.

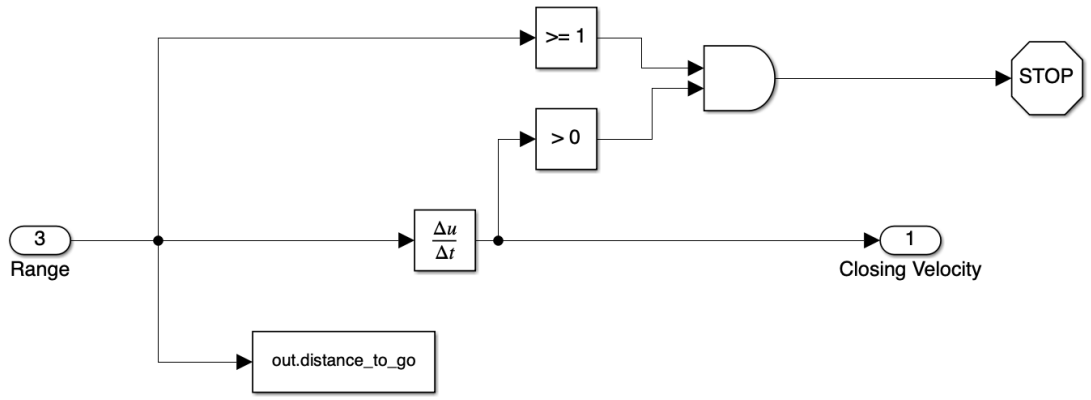


Figure 5.2 Stop Conditions

Probability to Kill (P_k) computation

This block calculates the P_k (Probability to Kill) at each timestep based on the varying attitude and the relative position between the target and the missile. It is divided into two sections to optimize the calculation of this probability for cases where the angle is greater than 45 deg and for cases where it is less than 45 deg. The calculations differ slightly between these two scenarios (see Chapter 7).
It processes the attitude and relative position in the following way:

- For angles greater than 45 deg, the upper section processes the inputs through a gain block, an inversion block, and a 3-D lookup table.
- For angles less than 45 deg, the lower section performs similar operations with a separate gain block, inversion, and 3-D lookup table.
- A conditional block then determines which of the two sections to use based on whether the angle exceeds 0.785398 rad (approximately 45 deg).

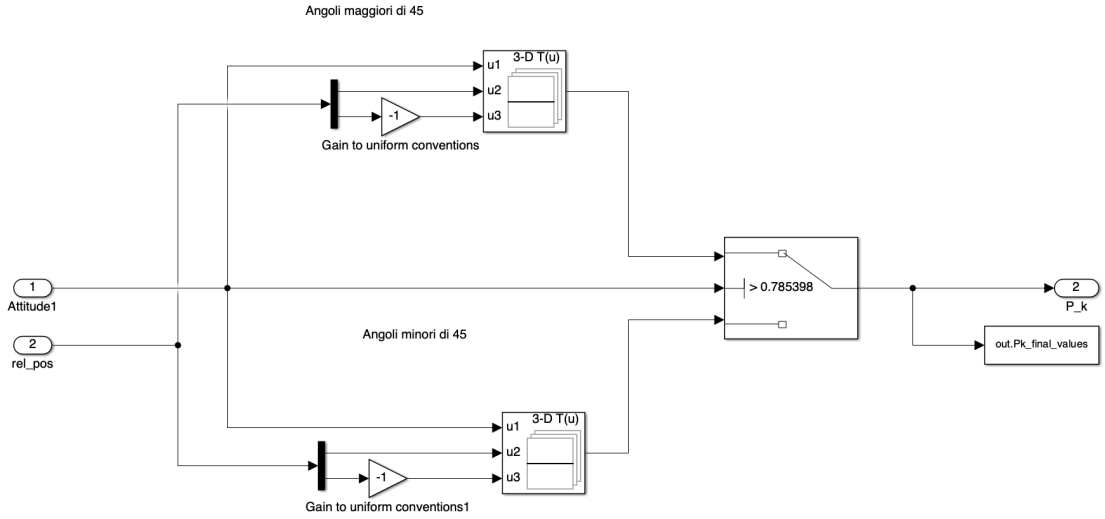


Figure 5.3 P_k calculation

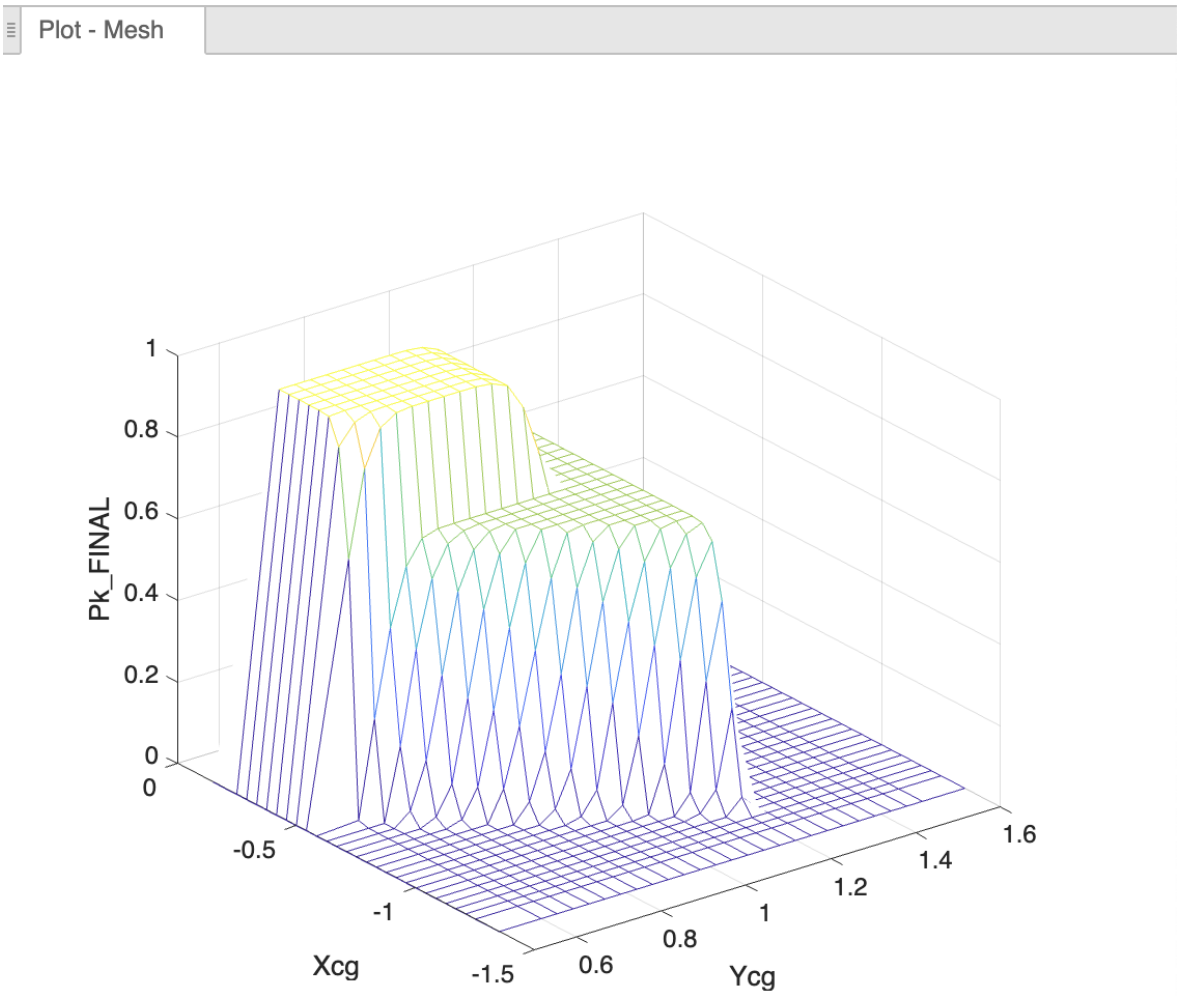


Figure 5.4 Example of P_k with fixed θ

5.4.2 Guide and Navigation block

This block takes in input the *Closing Velocity* V_C and the *Attitude* (α) to provide the required normal acceleration to actuate in order to reach the target.
It implements the *Proportional Navigation Law* described in Chapter 6.

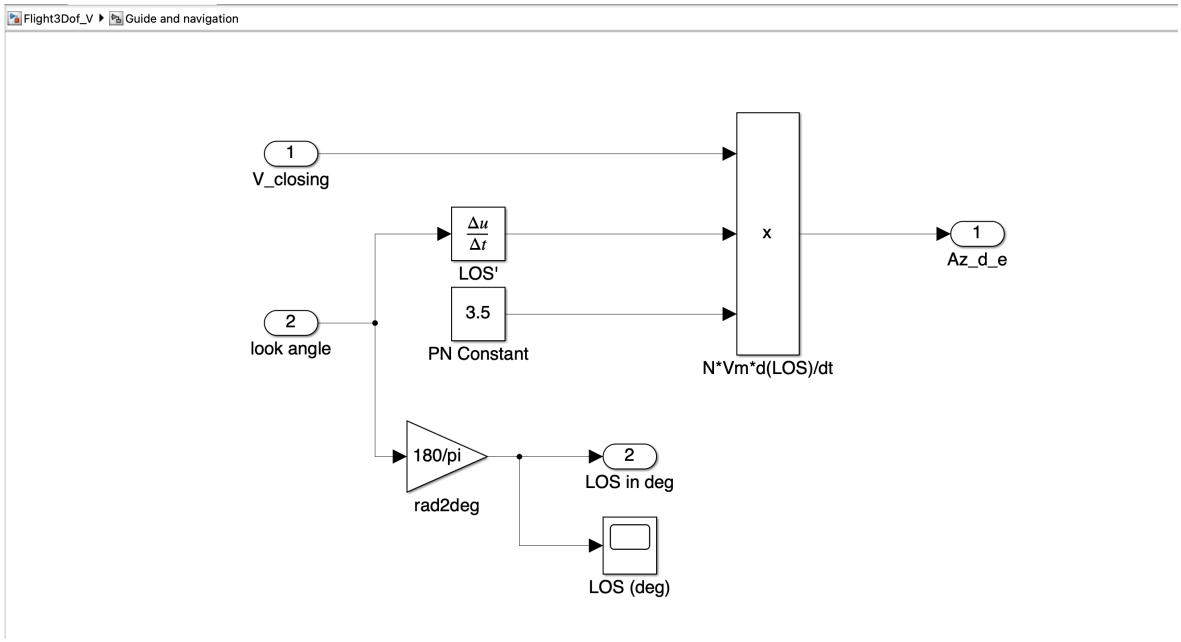


Figure 5.5 PN Block

5.4.3 Air frame and Autopilot block

This is one of the most important block of all the simulation. It contains the loop to calculate the attitude, the position and the pitch of Sagittarius for each time step.
All the block are described in detail in the following sections.

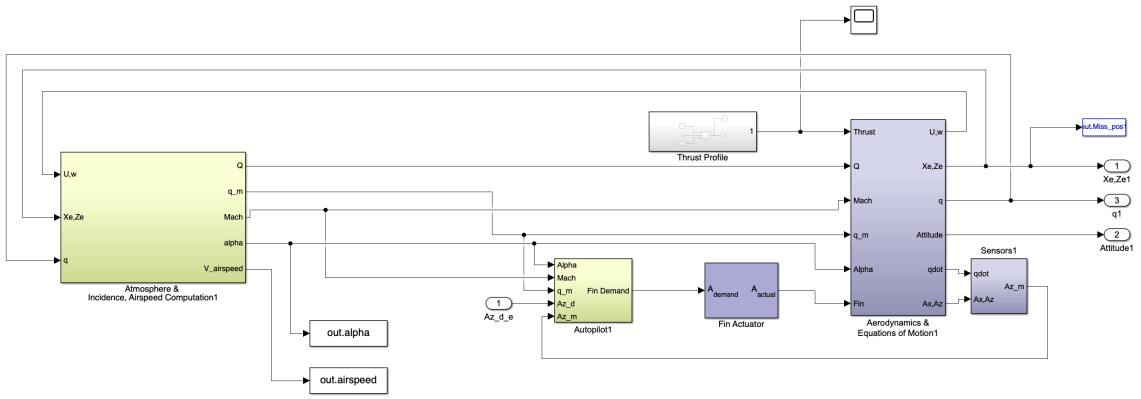


Figure 5.6 Air frame and Autopilot block

Atmosphere, Incidence and Airspeed block

The block dedicated to the calculation of attitude, velocity, and atmospheric pressure requires the input of the missile's instantaneous position X_e, Z_e .
Using this data, the *ISA Atmosphere Model* block from the Aerospace Block-set calculates the air density ρ and the speed of sound a at that specific altitude (h). These values are necessary for calculating the flight Mach number M and dynamic pressure Q.

For the calculation of the incidence Mach number, the components of the instantaneous velocity *(U, w)* are also required as inputs.
These two components, as well as the instantaneous position are obtained from the *Aerodynamics and Equations of Motion* block described below.

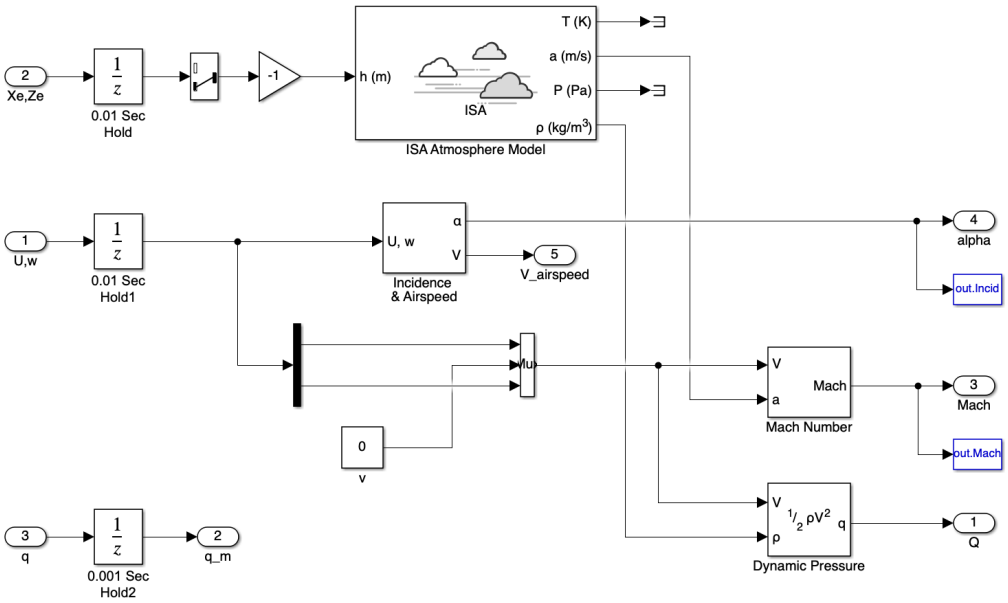


Figure 5.7 Atmosphere, Incidence and Airspeed block

Autopilot

The commanded normal acceleration, A_{Z_d} is passed to the missile's autopilot system. In this case, a dedicated autopilot with closed loops is designed as shown in the following figure.

The system begin with a *Gain scheduled coefficients* block which adapt the control gains

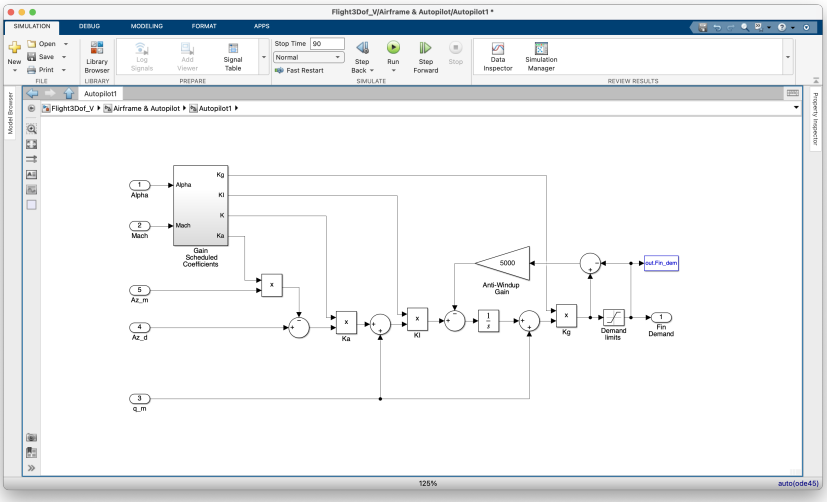


Figure 5.8 Autopilot Block

(K_a, K, K_I, K_g) based on two inputs, the *Mach* at which the missile is flying and its *angle of attack* (α).

The block, as seen in the figure 5.9 integrates four different 2D-lookup tables to extrapolate the correct value of the gain for the specific case.

The values of the gain coefficients are taken from the default ones brought by MathWorks ® for mach between 2 and 4 and scaled linearly to add values of M between 0 and 2. [5]

Initially, the normal acceleration demanded by the *PN guidance law* is compared with the actual acceleration measured A_{Z_m} by an accelerometer placed at a distance of $l = 0.07465m$ from the missile's center of gravity to calculate the error. More specifically the measured acceleration is multiplied for the K_a scheduled gain coefficient (which is always less than 1) and then subtracted from the measured acceleration.

The computed error is the input for the subsequent control action, performed by the following control loops:

- ***Error calculation loop***: this control multiplies the acceleration error by the gain K, providing an immediate corrective action proportional to the error magnitude.
- ***Attitude Control Loop***: this loop uses the output from the previous one to control the vehicle's attitude using the parameter q_m, passed from the aerodynamics computation. This signal is then scaled for the K_I coefficient and the integral of the error over time is computed using the $\frac{1}{s}$ block.
- ***Gain Adjustment***: finally the integrated signal is multiplied by the K_g coefficient

Lastly, an ***anti-windup*** gain mechanism with a gain of 5000 is implemented to prevent the integrator from accumulating excessive error when the actuator is saturated.

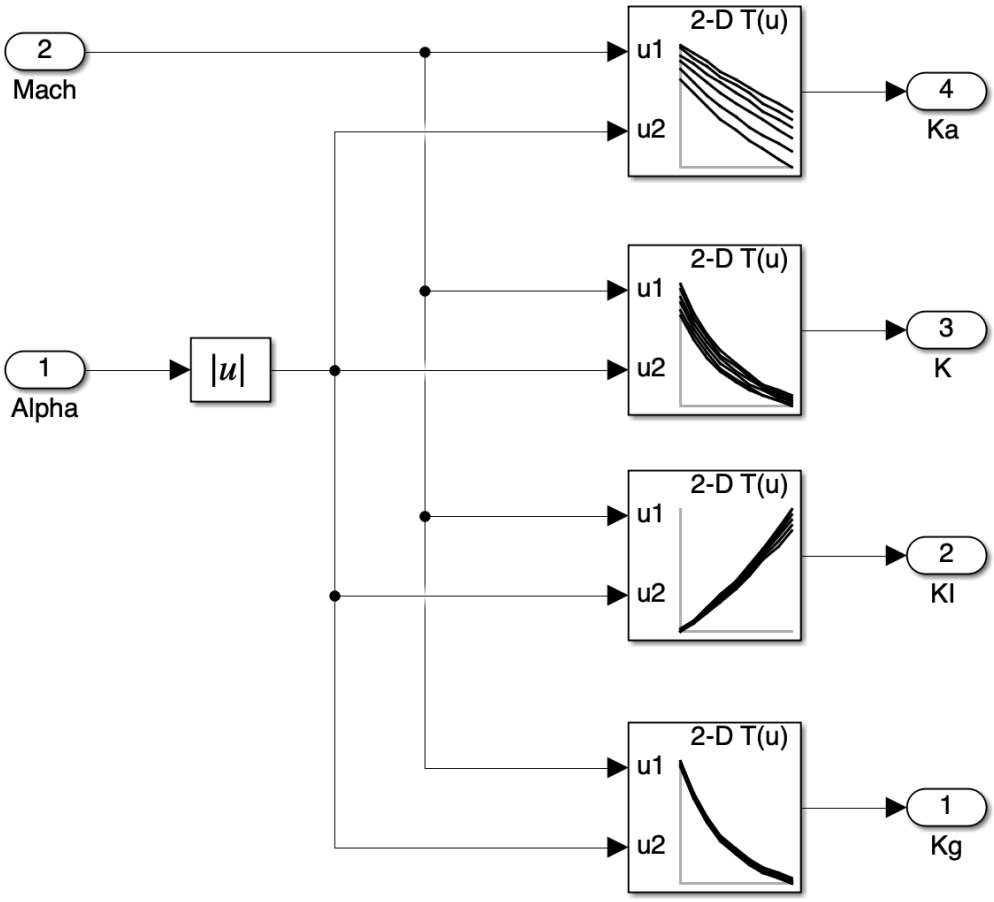


Figure 5.9 Scheduled Gain coefficients (K_a, K, K_I, K_g) 2D lookup table

This mechanism ensures that the integral action is effective even when the control commands are constrained by physical limits.
The adjusted control signal is passed through a series of operations and limits to produce the final fin demand.
First of all there is a saturation block that prevent the autopilot to demand deflections that are over the structural limits of the fins. For this specific case the limits are set with the parameters *fin max* and *fin min* respectively at +35 deg and - 35 deg, which are the structural and aerodynamic limits chosen for the fins of the Sagittarius missile, as reported in the Aerodynamics and Structures chapters.
The output represents the commanded fin position required to correct the missile's trajectory and align it with the PN guidance commands.

Non Linear Second Order Actuators

The *Fin demand* value is then passed to the actuator system. This model is been modeled in Simulink thanks to the Aerospace Block-set's *Non-Linear Second order actuator*. The mathematical base typically involves ordinary differential equations (*ODE*s) with nonlinear terms. These equations describe the relationship between the input and output of the actuator, incorporating nonlinear effects such as saturation, friction, backlash, and other non-ideal behaviors.
A general form of a second-order nonlinear differential equation that might describe an actuator is:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(u(t), x(t), \dot{x}(t))$$

that can be written as:

$$\ddot{x}(t) + \frac{b}{m}\dot{x}(t) + \frac{k}{m}x(t) = \frac{f(u(t), x(t), \dot{x}(t))}{m}$$

- m is the mass parameter
- b is the damping coefficient
- k is the stiffness coefficient
- $x(t)$ is the position of the missile at instantaneous time t, while $\dot{x(t)}$ and $\ddot{x(t)}$ are respectively the first and second derivative of the position, representing instantaneous speed and acceleration.
- $u(t)$ is a
- $f(u(t), x(t), \dot{x(t)})$ is a nonlinear function that represents the actuator's non-linearity.

And considering the following relationships:

- $\omega_n = \frac{k}{m}$, referring to the *natural frequency* of the system.
  For the purpose of the simulation this value is set at $\omega_n = 150\frac{rad}{s}$
- $\zeta = \frac{b}{2\sqrt{mk}}$, refers to the *damping ratio*.
  For the purpose of the simulation this value is set at $\zeta = 1/\sqrt{2} = 0.707$

The general equation is finally:

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2 x(t) = \frac{f(u(t), x(t), \dot{x}(t))}{m}$$

Implementation of *Non-Linear Second order actuator* in Simulink takes in input the demanded normal acceleration from the three-loop autopilot and actuates it into the actual acceleration A_{actual} used in the Aerodynamics equations. It can be seen in the following figures:

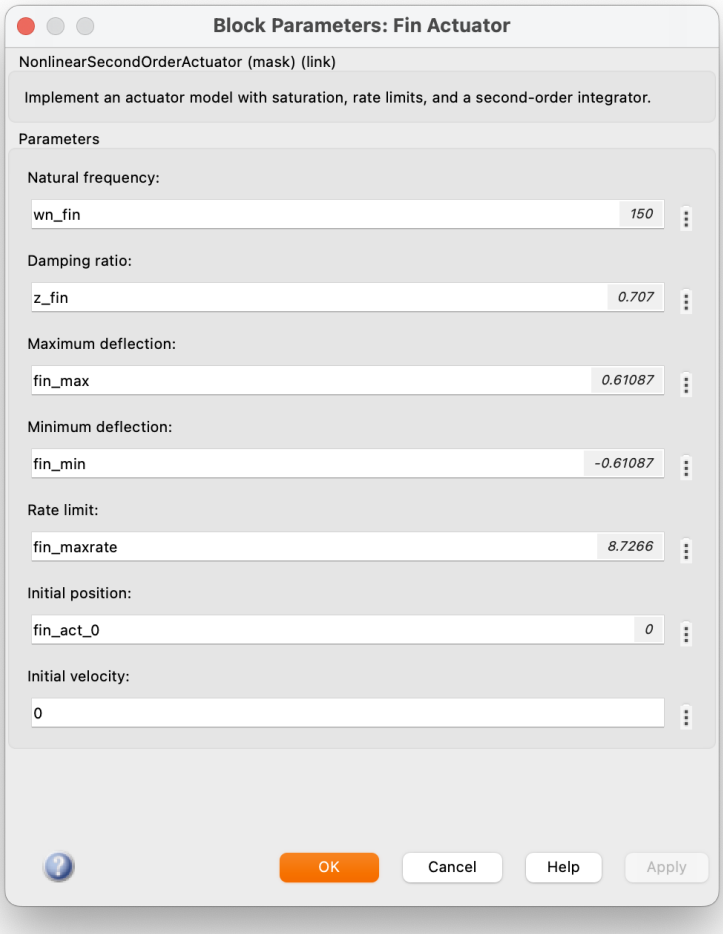


Figure 5.10 Fin Actuator Block with inputs

Thrust Profile

In the thrust profile block, a simple switch block is installed to provide thrust to the missile only if it is under the maximum burn time. The thrust values and burn time are given as inputs, as highlighted in the previous section.

Aerodynamics and Equations of Motion

The depicted block represents the dynamic model of the missile. Its configuration is shown in the figure 5.11 below:
It is divided in two different sections:

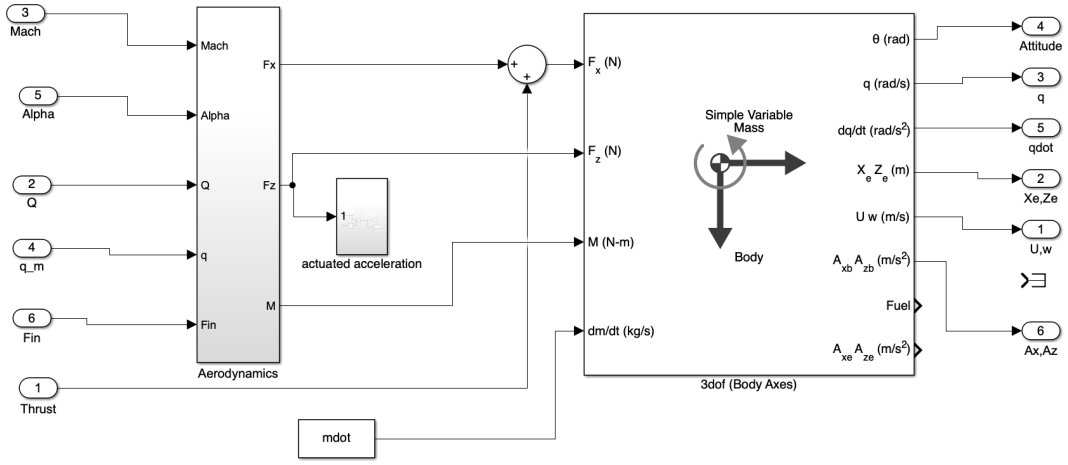


Figure 5.11 Aerodynamics and Equations of Motion Block

Aerodynamics:
It takes in input the following values:

- Mach
- Alpha
- Dynamic Pressure
- Pitch Rate
- Actuated Fin deflection

It gives in output the forces on X axis and on Z axis, as well as the momentum.
In the next figures it is possible to see how the aerodynamics is modeled:

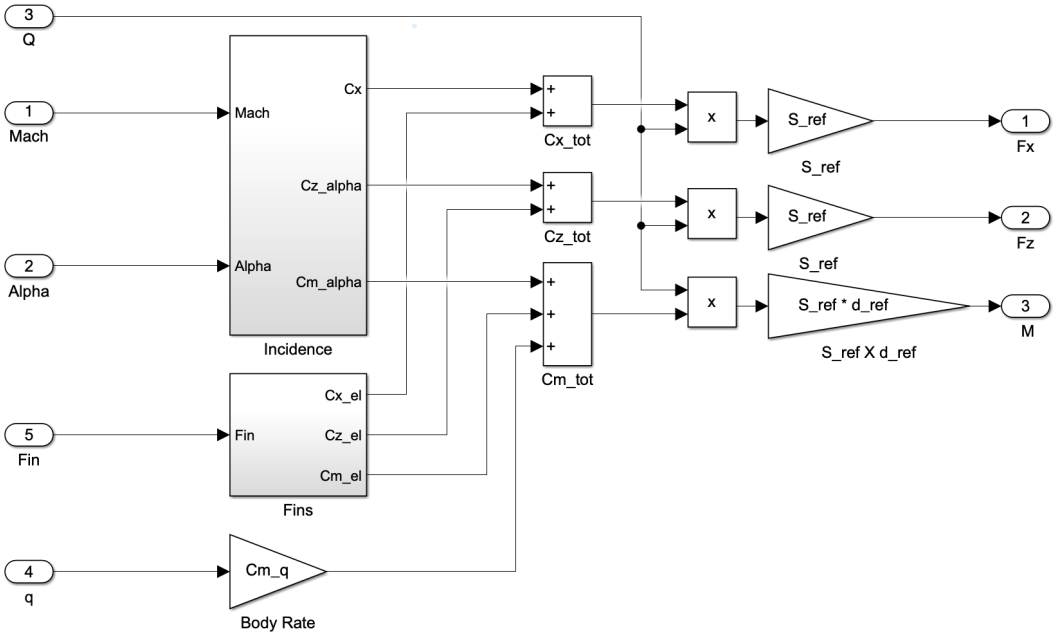


Figure 5.12 Aerodynamics block

Forces on X and Z axis are calculated with the following formula, also described in the Aerodynamics Section:

$$F_z = S_{ref} \cdot Q \cdot C_z$$

$$F_x = S_{ref} \cdot Q \cdot C_x$$

$$M = S_{ref} \cdot d_{ref} \cdot Q \cdot C_{m,tot}$$

where:

- $C_{z,tot} = C_{z,\alpha} + C_{z,fin}$
- $C_{x,tot} = C_{x,\alpha} + C_{x,fin}$
- $C_{m,tot} = C_{m,\alpha} + C_{m,fin} + C_{m,q}$

The above coefficients are calculated for each different situation in the following manner:

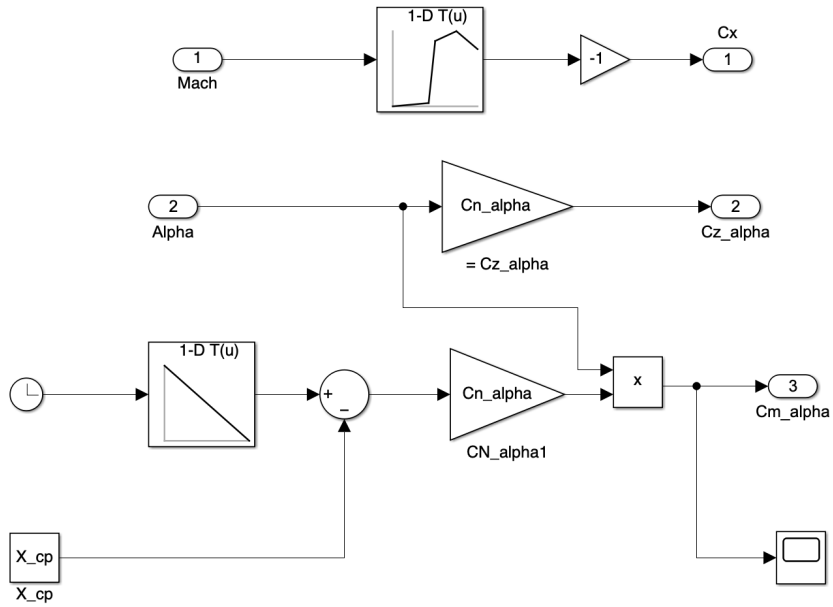


Figure 5.13 Incidence block

The C_x (drag coefficient) is modeled with data verified with the CFD simulation (8) and RASAero (8). It is possible to see the rapid increase in the drag coefficient following the achievement of the *Drag Divergence Mach number* (M_{DD}):

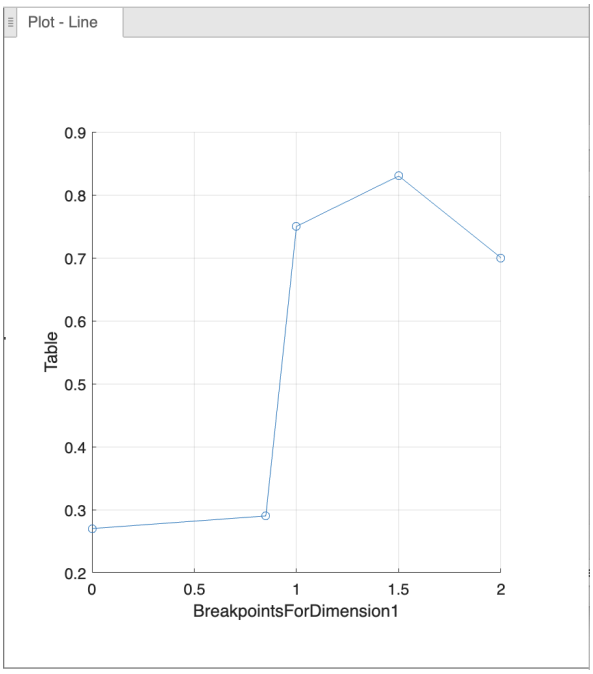


Figure 5.14 CD as function of Mach.

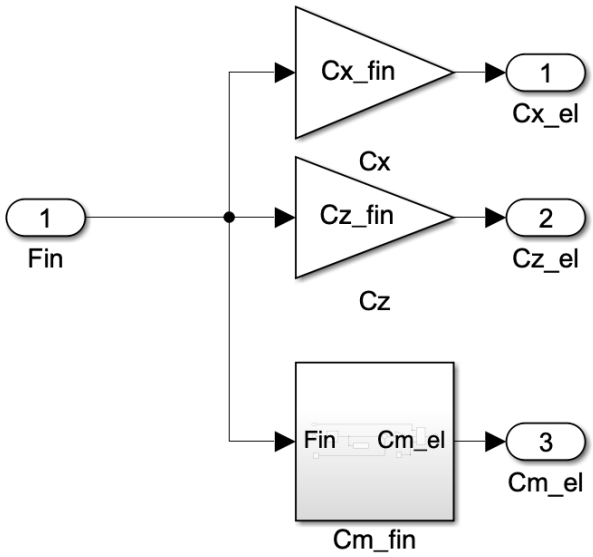


Figure 5.15 Fin coefficients

3 D-o-F (Body Axes):
This is a pre-existing block in the Aerospace Block-set that integrates the equations of motion in 3 degrees of freedom with respect to the body axes using the output from the Aerodynamics block and considering the mass variation linearly.
Indeed it takes in input the value of $\dot{m}$ *(mdot)*, which represents the rate of change of mass, corresponding to the propellant consumption rate calculated in Chapter 8.3 regarding the second stage propulsion, which is the subject of the simulation.
All the inputs required in the mask are taken from the MATLAB input file, as specified in Section 5.1.
The output of the block is composed by:

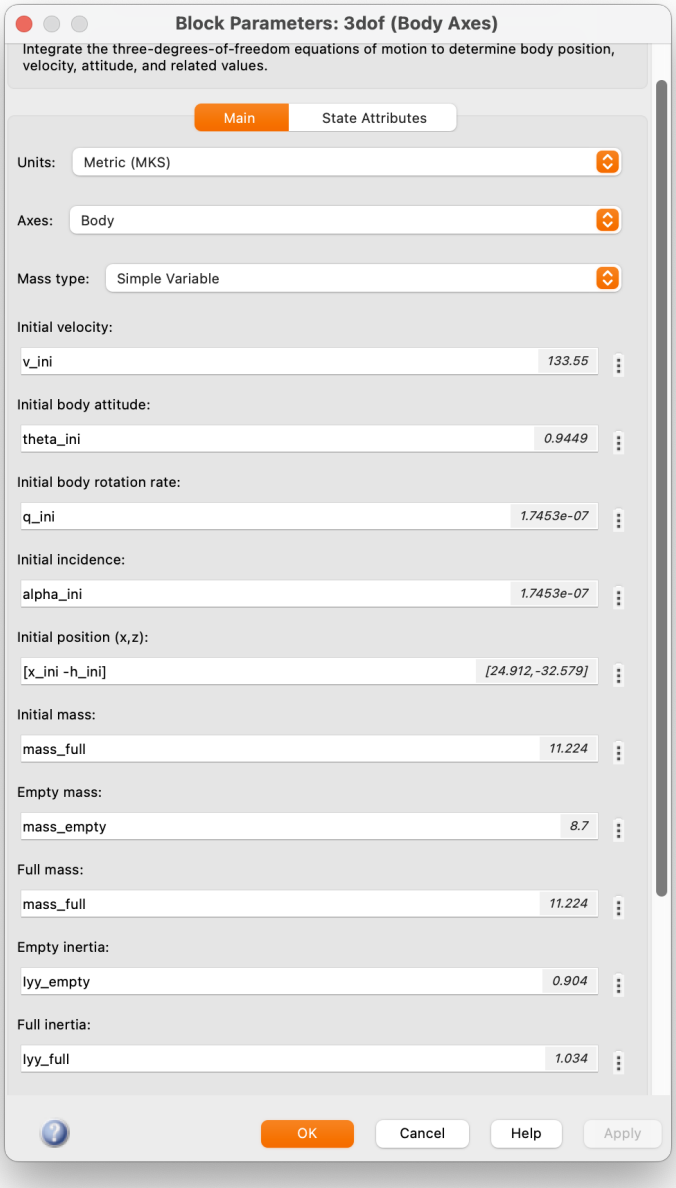


Figure 5.16 3Dof (Body Axes) block

- Attitude θ
- Pitch q and Pitch Rate $\dot{q}$

- Position on (X,Z) axis
- Speed on (X,Z) axis
- Acceleration on (X,Z) body axis
- Fuel consumption. This value has not been used in the simulation.

5.4.4 3 D-o-F Animation block

The block on the right represents an animation camera that tracks the relative position of the missile with respect to the target. This block receives three primary inputs:

- X_e, Z_e: The coordinates of the missile within the reference system (presumably either terrestrial or target-based).
- θ: The missile's elevation angle to correctly display its attitude.

The camera continuously updates its position and viewing angle based on these inputs, ensuring that the animation accurately depicts the missile's approach towards the target. This functionality is crucial for real-time visualization of the missile's trajectory and guidance accuracy.
The result of the animation is a representation of the trajectory of Sagittarius (in blue) and the trajectory of the target (in red).

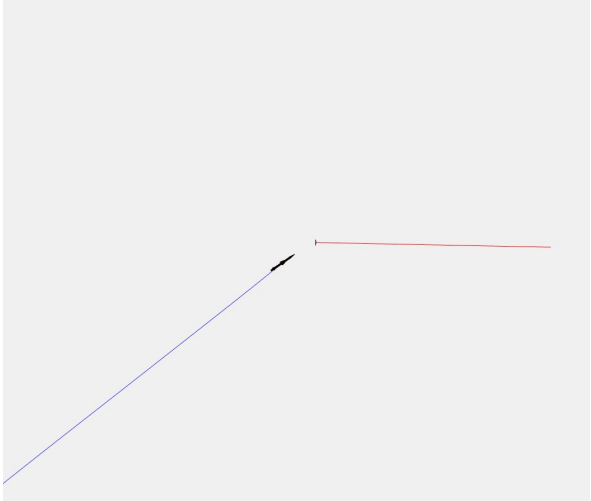


Figure 5.17 Trajectories result from the simulation

Chapter 6

Guidance

This chapter details the design of a guidance system for Sagittarius, a small ground-to-air missile, focusing on the integration of onboard sensors, navigation algorithms, and real-time data processing shows its practical implementation performed within the trade-off analysis conducted using Simulink.

6.1 Introduction

Missile guidance is a critical technology in modern warfare, encompassing the methods and systems used to direct missiles towards their targets with high precision.
There are several commonly used guidance techniques, each with its own advantages and applications. One widely used method is command guidance, where the missile receives commands from an external source, typically a ground-based radar or a control station, which tracks the target and directs the missile towards it. Another prevalent technique is inertial guidance, which relies on gyroscopes and accelerometers to measure the missile's movement and maintain its course without external inputs. This method is particularly useful for long-range missiles and those requiring stealth, as it minimizes the risk of detection and jamming.
Additionally, homing guidance is extensively utilized, involving the missile actively seeking out the target using onboard sensors. This can be further divided into active, semi-active, and passive homing. Active homing uses radar or sonar systems to emit signals and detect their reflections from the target, while semi-active homing relies on external illumination of the target, and the missile homes in on the reflected signals. Passive homing, on the other hand, detects energy emitted by the target itself, such as heat from an aircraft's engines, making it effective for heat-seeking missiles.
Sagittarius uses a semi active seeker to track the target in real time.

6.2 Proportional Navigation

6.2.1 Theoretical foundations of proportional navigation

Base Principles

Proportional Navigation (*PN*) is one of the fundamentals guidance law, mostly used for its simplicity.
The core principle of PN is that the missile's commanded acceleration is proportional to the rate of change of the line-of-sight (LOS) angle between the missile and the target. This method is favored due to its simplicity and effectiveness in guiding missiles to intercept moving targets.

In *PN law* the missile is guided in a way that its acceleration is always perpendicular to the line-of-sight.
The special feature of proportional navigation is not going to target the missile precisely, but rather the intended point of interception. Indeed the missile effectively leads the target by anticipating its future position based on the current motion, ensuring interception without requiring continuous course corrections.

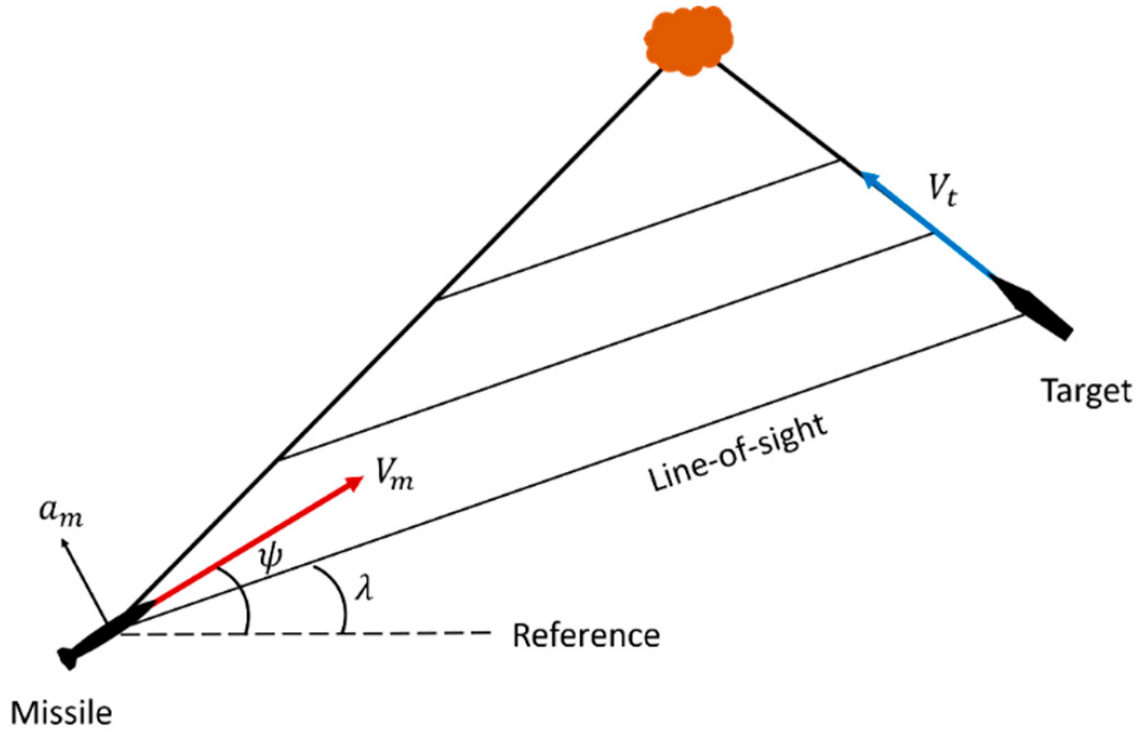


Figure 6.1 Conceptual diagram of proportional navigation guidance

Key parameters

Understanding the key parameters and equations involved in Proportional Navigation (PN) is essential for effectively implementing and analyzing this guidance technique. Here, we outline the primary variables and the mathematical framework that underpins PN.

- ***Line-of-Sight (LOS) angle*** (λ):
  The LOS angle is the angle between the missile's velocity vector and the line connecting the missile to the target.

- ***LOS Angular Rate*** ($\dot{\lambda}$):
  This is the rate at which the LOS angle changes over time. It is a crucial variable in determining the necessary acceleration commands in PN.

- ***Closing Velocity*** (V_C):
  Closing velocity is the rate at which the distance between the missile and the target decreases. It is given by the negative range derivative (R) between the missile and the target: $V_C = -\dot{R}$

- ***PN Navigation Constant (N):***
  It is a positive value that determines the strength of the guidance system.
  It typically ranges from 3 t o 5 for practical applications with higher values for more aggressive maneuvers. It is important to consider that a higher navigation gain can improve interception performance but may also increase the demand on the missiles's control system and propulsion capabilities.

- ***Commanded normal Acceleration*** A_C:
  This is the normal acceleration command generated by the PN law, which the missile must follow to stay on course to intercept the target. This value is then passed to the actuator systems.

Fundamental equations

The Proportional navigation law can be mathematically expressed using the following core equation:

$$A_{Z_d} = N \cdot V_c \cdot \dot{\lambda}$$

To better understand the relationship, consider the following equations involved:

- Closing Velocity:
$$V_C = -\frac{\delta R}{\delta t}$$
- LOS Angular rate
$$\dot{\lambda} = \frac{\delta \lambda}{\delta t}$$

This equation states that the normal acceleration command A_{Z_d} is directly proportional to the product of the closing velocity and the LOS angular rate, scaled by the navigation constant. It is important to take into account several considerations for practical implementation

- Sensors accuracy
  Accurate measurement of the LOS angle and its change rate is critical, as well as the real acceleration of the missile performed by the accelerometer.
- Actuators limits
  The guidance law gives the commanded acceleration. It has to be actuated by the missile's control surfaces
- Environmental factors:
  Wind, turbulence and atmospheric conditions must be considered while designing the guidance law ad they can affects the missile's flight path.

[12]

6.2.2 Proportional Navigation in Simulink

Proportional Navigation for Sagittarius is implemented using Simulink as shown in subsection 5.4.2.
More specifically a product block is used to multiply the closing velocity V_C with the rate of change of LOS and the guidance navigation constant.
For this specific case $N = 3.5$ is chosen to prevent too much commanded normal acceleration resulting in required maneuvers not withstood by the missile structure.
It is important to remember that the maximum load factor for which Sagittarius was designed is $n_{lim} = 6$.

Chapter 7

Warhead Design & Performance

The warhead is a crucial part of the missile, designed to inflict damage on the target. It can contain explosives, chemical, biological, nuclear, or conventional agents.

7.1 Warhead sizing

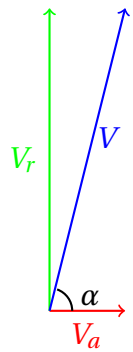
Cylindrical fragmentation warhead has been selected, which is a typical choice to shoot down a small aerial target. Given the cylindrical shape of the warhead, upon explosion, the fragments will be imparted with a radial velocity component, which can be estimated using the Gurney formula [10]:

$$V_r = \sqrt{2E}\sqrt{\frac{c/m}{1+0.5\frac{c}{m}}} \tag{7.1}$$

where:

- E is the energy, per unit mass of explosive, convertible to mechanical work;
- c/m is the ratio between the mass of the explosive and the mass of the fragments.

This component must be added to the axial velocity component (V_a), which is equal to the missile's velocity; so the fragment's vectorial velocity (V) is the sum of these two.



In the design, the effect of aerodynamic resistance was then considered, which is responsible for the aerodynamic deceleration of the fragment (in the direction of V) according to the law:

$$V_{new} = V_{old} \cdot e^{-C_D \rho A m_{sf} d} \tag{7.2}$$

where:

- V_{new} is the velocity before aerodynamic deceleration;
- V_{old} is the velocity after aerodynamic deceleration;
- C_D is the drag coefficient of the single fragment;
- ρ is the air density;
- A is the wetted surface of the single fragment;
- m_{sf} is the mass of the single fragment;
- d is the distance traveled by the single fragment.

The last aspect to consider concerned the penetration of the fragment. It was necessary to understand whether a given fragment, impacting a specific target with a certain velocity, was able to penetrate or not. To this end, it has been utilized the THOR formulas which assumes the target as a plate of thickness [9]:

$$V_{res} = V_{imp} - 0.3048 \cdot 10^{c_{11}} \cdot (61023.75 \cdot t \cdot A)^{c_{12}} \cdot (915432.1 \cdot m_{sf})^{c_{13}} \cdot (1/cos(\alpha))^{c_{14}} \cdot (3.28084)^{c_{15}} \cdot V_{imp}$$

$$M_{res} = m_{sf} - 6.48 \cdot 10^{c_{31}-5} \cdot (61023.75 \cdot t \cdot A)^{c_{32}} \cdot (915432.1 \cdot m_{sf})^{c_{33}} \cdot (1/cos(\alpha))^{c_{34}} \cdot (3.28084)^{c_{35}} \cdot V_{imp}$$

where:

- V_{imp} is the velocity of the fragment upon impact with the target;
- V_{res} is the residual velocity of the fragment after the penetration;
- M_{res} is the residual mass of the fragment after the penetration;
- A is the contact area of the fragment and the target;
- c_{ij} are the coefficient of materials for target.

Once the theoretical concepts have been presented, it is possible to discuss the choices made:

- composition B as explosive since it has an high value of $\sqrt{2E}$ ($\sqrt{2E} = 8800 ft/s$); [10]
- pre-cut spherical fragments are a common choice for similar applications; furthermore, they simplify the analysis due to their symmetry, and there is more information available regarding their drag coefficient C_D;
- radius of the spherical fragment equal to 5 mm;
- steel as the material for the fragments(thus $m_{sf} = \rho \cdot v = 4.1\ g$);
- c/m=0.5.

With the aforementioned data, it is possible to verify that, setting d = $\sqrt{5}$ m, [1] the aerodynamic deceleration is less than 2%, consequently, it has been considered negligible in subsequent analyses. By doing so, the initial velocity of the fragment was also equal to the impact velocity and they are:

$$V = V_{imp} = 1714.638\ \frac{m}{s}$$

Applying the THOR formula and considering the target as an aluminum plate with a thickness of 10 mm, [2] it has been obtained:

$$V_{res} = 763.965\ \frac{m}{s} \qquad M_{res} = 2.6\ g$$

Considering the dimensional requirements of the entire missile, it has been decided to assign the warhead a length of 150 mm (L_w) and a diameter of 77 mm (D_w). In this way it has been obtained an L/D ratio of approximately 2, which promotes efficient acceleration of the fragments and allows for a fairly accurate estimation of the velocity using the Gurney formula. In order to estimate the number of spherical fragments per ring, it has been set:

$$n_{fr} = \frac{\pi D_w}{\pi r_f} = 15$$

This number is always rounded down.

[1] according to the GNG system, $\sqrt{5}$ m is the maximum distance at which the study of P_k was conducted (see next section).

[2] it has been used slightly thicker plate than usual to be conservative

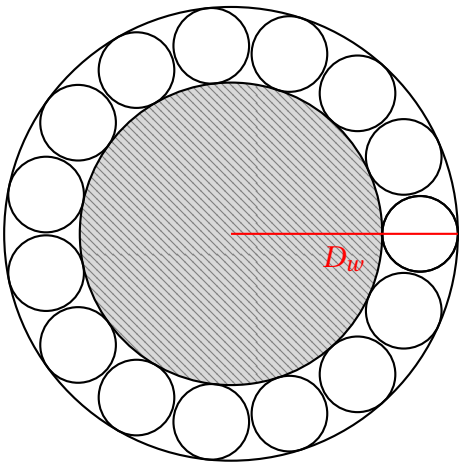


Figure 7.1 Section view of the warhead: fragments ring+explosive in the middle

The number of fragment rings placed along the length of the warhead was:

$$n_r = \frac{L_w}{2r_f} = 15$$

This number is always rounded down.

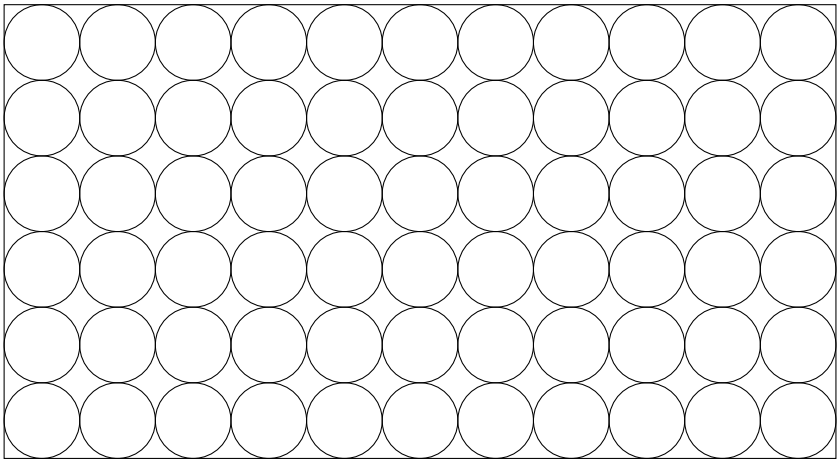


Figure 7.2 Lateral view of the warhead: fragments distributed along the length

Therefore it has been calculated the total number of fragments:

$$n_f = n_{fr} \cdot n_r = 225$$

A this point the total mass of the fragments was computed:

$$m_f = n_f \cdot m_{sf} = 0.9189\ kg$$

thus the mass of explosive:

$$m_e = \frac{c}{m} \cdot m_f = 0.4595\ kg$$

Since the mass of the explosive and the fragments represents 70% of the mass of the warhead (the remaining part is known as "dead mass"), it has been estimated the total mass of the warhead:

$$m_w = \frac{m_e + m_f}{0.7} = 1.9691\ kg$$

Knowing the density of Composition B ($\rho_e = 1650\ \frac{kg}{m^3}$) and the mass, then it has been calculated the diameter of the explosive:

$$d_e = 2\sqrt{\frac{m_e}{L_w \rho_e \pi}} = 0.0486\ m$$

7.2 Fragment's pentration simulation

An initial finite element model, employed to validate the assertions derived from Thor's formula, utilizes the classic plate fragment representation. This approach, while inherently simplified, is designed to simulate the behavior of an individual fragment upon impact with the target. Notably, the model excludes the explosion effects, focusing exclusively on the velocity imparted to the fragment. This allows for a more controlled analysis of the fragment's kinematics under the given conditions. A comprehensive illustration of the model is presented below:

- Fragment:
  - The fragment is a deformable sphere with a radius of 5 mm and a mass of 4.1 g;
  - Fragment's velocity is perpendicular to the target surface and is 1714.6 m/s;
  - The fragment is made of steel.
- Target:
  - The target is a plate with a width and length of 1 m and a thickness of 10 mm;
  - The target is made of aluminum.

Given the aforementioned premises, it is possible to translate the following data into Keyword format within the LS-PrePost environment.

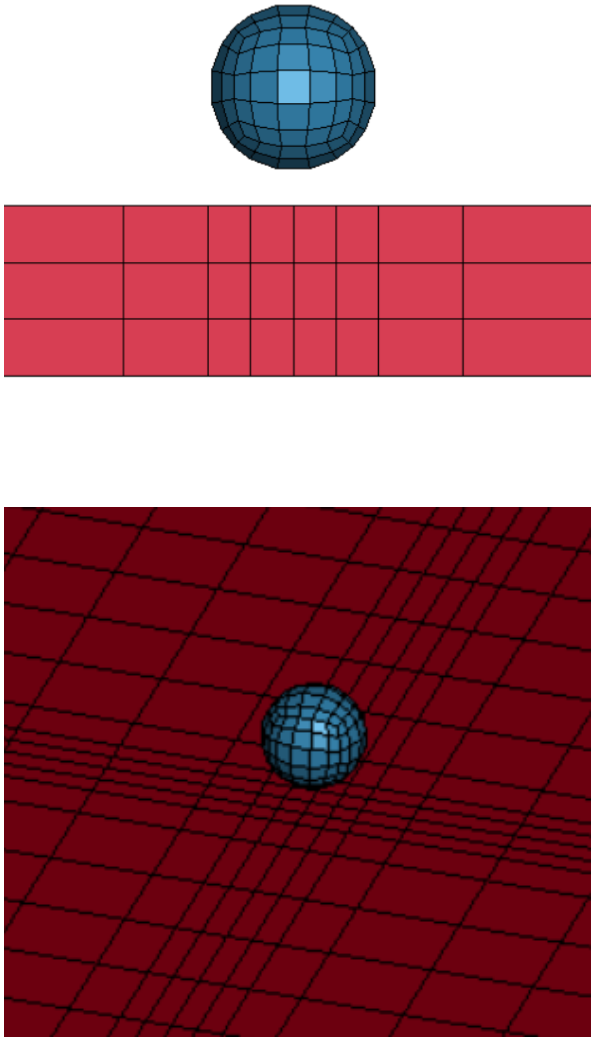


Figure 7.3 Model

Let us now examine the material Keywords selected for this simulation. As illustrated in figure 7.4, it is important to note that $MAT_003_PLASTIC_KINEMATIC$ has been assigned solely to the target, while $MAT_024_PIECEWISE_LINEAR_PLASTICITY$ has been selected for the fragment. The $MAT_003_PLASTIC_KINEMATIC$ model for the target provides accurate simulation of plastic deformation, an essential aspect for capturing realistic material behavior under impact. This material model incorporates kinematic hardening, enabling detailed analysis of stress distribution and strain localization as the target yields and deforms upon contact.

Indeed, in selecting the material attributes, particular attention was given to the tangent modulus, which was assigned as a zero value for the plate. This choice effectively eliminates the need to consider any potential plastic deformation of the plate, simplifying the model by focusing exclusively on its fracture behavior.
For the fragment, $MAT_024_PIECEWISE_LINEAR_PLASTICITY$ is chosen due to its suitability for high-strength materials, allowing a clear representation of both elastic and plastic behaviors under high-impact conditions. This choice ensures that the frag-

ment maintains rigidity under initial loading while accurately depicting energy transfer during impact. Moreover, *MAT*_024 enables control over parameters such as yield strength and strain-rate sensitivity, providing an effective model for simulating the fragment's behavior as it impacts and transfers kinetic energy to the target. In fact, a relatively high yield stress was specified, ensuring that its resistance to deformation is consistent with the nature of the impact phenomenon being studied, and adequately captures the material's response to the applied stresses.

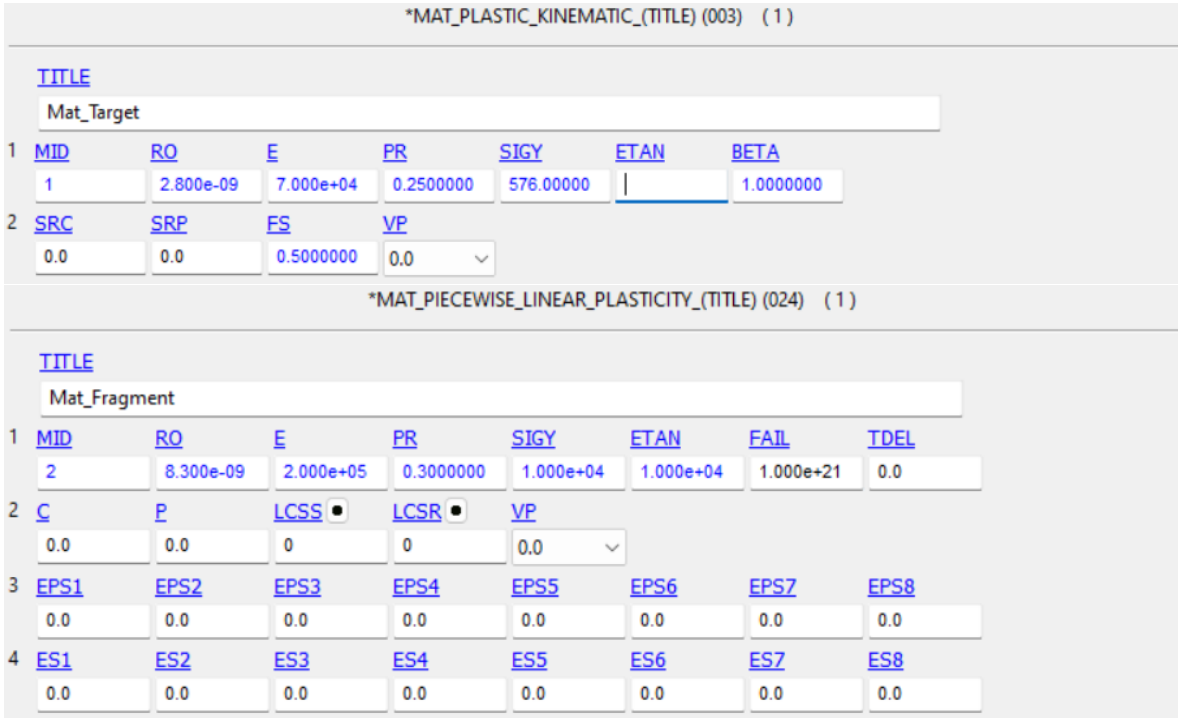


Figure 7.4 Material choice

Furthermore, it can be observed that for the target, a Poisson's ratio lower than the typical value of 0.3, commonly associated with metals, has been chosen. This decision is justified by the fact that a lower Poisson's ratio makes the plate less sensitive to lateral deformation when subjected to stress, which in turn could enhance its resistance to unintended deformations in transverse directions. By selecting this value, the model better reflects the desired mechanical behavior of the target under the specific loading conditions, ensuring that it is more resistant to distortions that could arise in the perpendicular direction to the applied forces.
To validate the numerical results and accurately simulate the impact phenomenon, skin friction values have been assigned to the material. These values are the outcome of several simulations aimed at obtaining the most realistic representation of the phenomenon. By fine-tuning the friction parameters, the model ensures a more precise and reliable simulation of the interaction between the fragments and the target, reflecting the actual physical behavior as closely as possible.

To simulate the contact interaction effectively, the *ERODING_CONTACT_SURFACE_TO_SURFACE* Keyword was chosen (as shown in figure 7.5). This option, suited to the simplicity of the contact scenario, accurately captures the interaction dynamics while

keeping computational demands low. In this configuration, the fragment is designated as the master surface due to its role as the primary source of impact and its higher density, while the plate is assigned as the slave surface. To preserve simplicity while optimizing initial contact, the parameters SFSA and SFSB were adjusted rather than left at default values.

This change enhances the immediate interaction between the master and slave surfaces, which is crucial for accurately simulating the impact dynamics. In the next chapter, we will refine these adjustments further to ensure robust adhesion between the shock wave and the dispersed array of fragments, allowing for an even more nuanced representation of contact behavior. For this initial phase, however, these adjusted values provide the necessary balance between precision and simplicity, with further developments to be explored subsequently.

Before presenting the simulation results, it is essential to outline the initial velocity con-

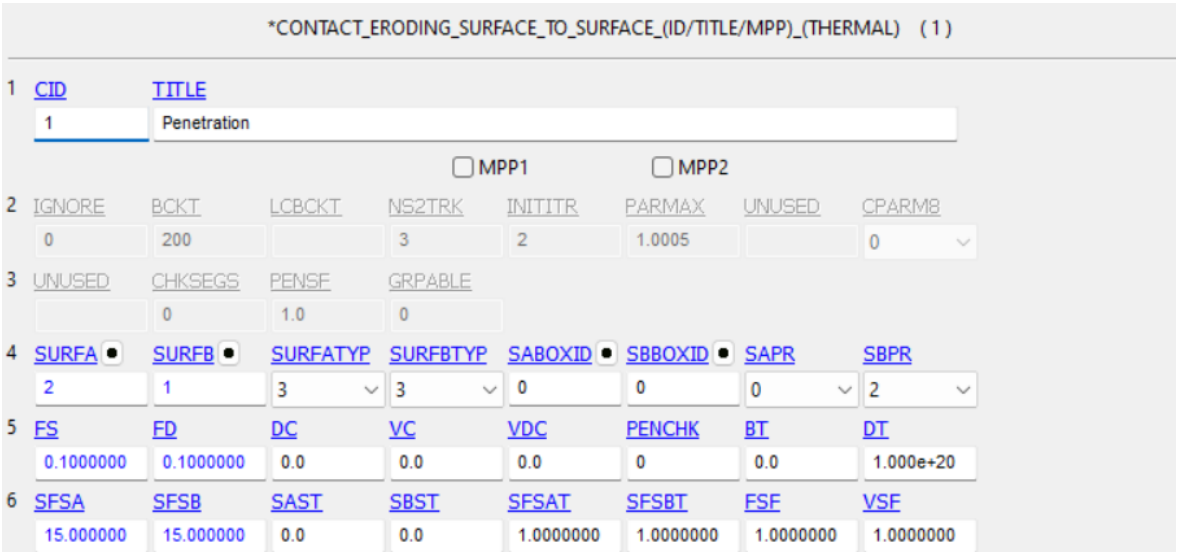


Figure 7.5 Contact

dition applied, shown in figure 7.6 providing a complete picture of the setup. This initial condition is intended to simulate the projectile's motion immediately following the explosion of the warhead. Specifically, the analysis focuses on a point in time just after the shock wave has passed, with the fragment already fully accelerated toward the plate. This approach allows us to isolate and examine the impact dynamics of the fragment on the plate, capturing the post-explosion phase without needing to model the shock wave itself.

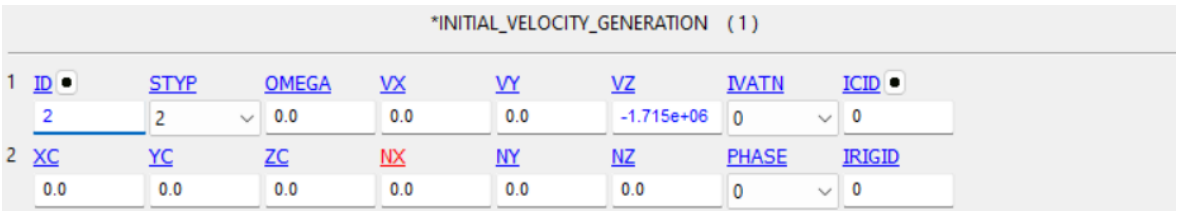


Figure 7.6 Initial Condition applied

The simulation is now initiated, and the results are examined. From a purely graphical perspective, it is evident that the penetration was successful, indicating that the tar-

get has been theoretically neutralized. Figure 7.7 presents a detailed view of the plate penetration and the corresponding deformation of the fragment upon impact. This visual evidence provides insight into the impact dynamics and material response, highlighting the interaction effects between the fragment and the target surface.

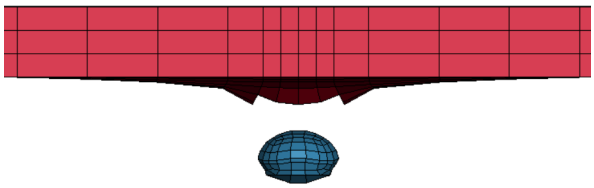


Figure 7.7 penetration

Attention must first be directed to the type of mesh applied in the model. For the plate, a finer mesh was implemented in the central region to improve contact precision, with this refinement extended into adjacent areas to ensure node alignment and a seamless connection across elements.
For the fragment, a standard mesh was selected, as it was considered sufficient for the simulation objectives. Previous experience has demonstrated that this level of discretization, when paired with the chosen contact model, accurately captures the problem dynamics. This balanced approach allowed for reliable simulation results without incurring excessive computational costs, maintaining both efficiency and fidelity in the analysis.

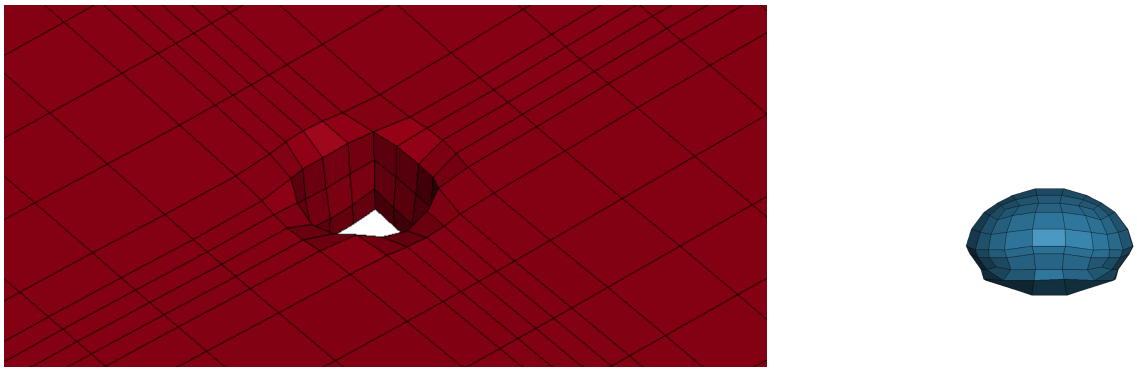


Figure 7.8 penetration focus

Figure 7.8 presents detailed views from the perspectives of both the plate and the fragment, illustrating the impact quality, as demonstrated by the complete perforation of the target. These images provide additional confirmation of the initial assumptions: the fragment displays noticeable plastic deformation, consistent with the materialâĂŹs response under high-stress impact conditions, while the plate shows only a clean penetration hole with no significant outward deformation. This outcome reflects the effec-

tiveness of the mesh refinement in capturing the localized impact zone on the plate, while avoiding unnecessary distortion beyond the target area.
AnalysisâĂŹs results are summarized in the plots illustrated in the figure 7.9:

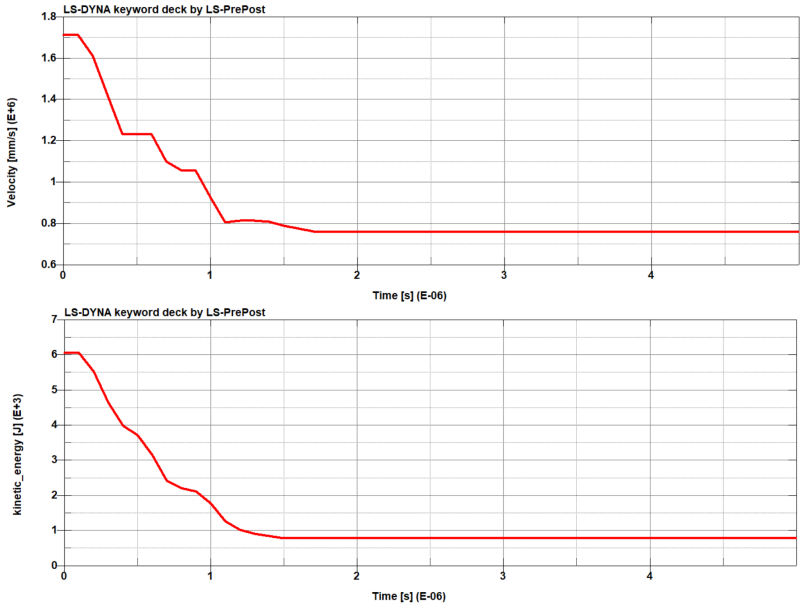


Figure 7.9 Simulation Plots

Then the results obtained with LS-DYNA and the ones obtained with the THOR formula were compared:

	LS-DYNA	**THOR**
residual kinetic energy	730 J	761 J
residual velocity	760 $\frac{m}{s}$	764 $\frac{m}{s}$
residual mass	2.6 g	2.6 g

Table 7.1 Comparison between the results of LS-DYNA and THOR

The results obtained align well with the theoretical framework applied. Specifically, minor discrepancies in the values can largely be attributed to the conservative assumptions embedded within the Thor formula. By further refining the simulation parameters, it would have been possible to narrow the gap even more between the results obtained through the Thor equation approach and those produced by the DYNA simulation. However, since the primary aim of this model was to provide an initial qualitative understanding of the impact dynamics, these results are considered satisfactory. They offer a solid foundation for further exploration and confirm that the model captures the essential characteristics of the phenomenon.

7.3 P_k estimation

To estimate the lethality of the warhead, it has been conducted a study on the kill probability based on the x and y distance between the target and the missile, and the missile's pitch angle θ. [3] For each set of values, the number of fragments hitting the target has been estimated and, assuming that each fragment hitting the target was a Bernoulli experiment with a success probability of 0.07, the P_k has been computed.

First of all, it's necessary to explain the conventions and references used:

- The target is considered as a rectangular prism with a width and length of 1 m, and a height of 0.3; [4]
- Given the top fragments belonging to the xy plane, the Y distance represents the vertical distance between the lowest point of the lowest fragment and the beginning of the target;[5]
- The X distance represents the horinzontal distance between the end of the rightmost point of top fragments belonging to the (x, y) plane, and the beginning of the target;
- At any given time, the target can move horizontally at any speed between v_{min} and v_{max};
- X_{cg} and Y_{cg} represent the horizontal and vertical distances, respectively, between the missile's and warhead's centers of mass. [6]

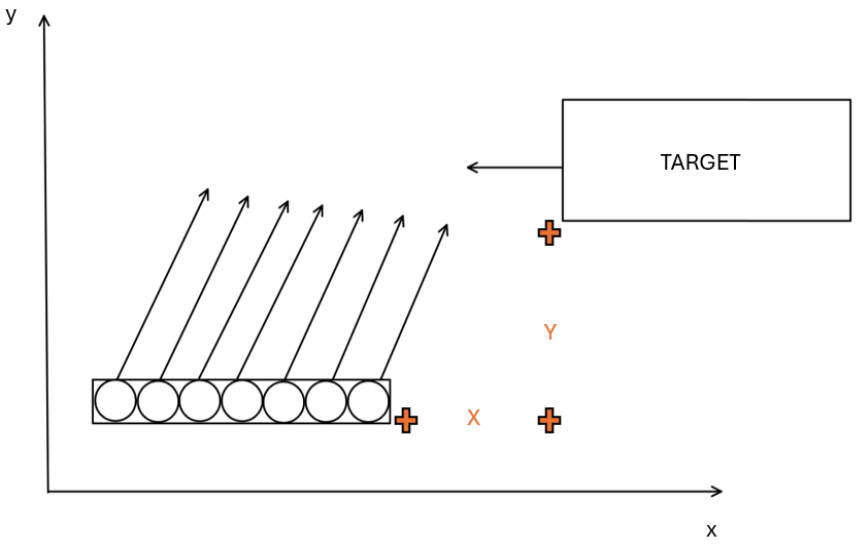


Figure 7.10 Example of scenario($\theta = 0$)

[3] the problem is $2D$, so the target and the missile lie on the same plane, thus only 3 degrees of freedom are involved

[4] in this way, the volume occupied by the target in our study is a fraction of the real one, allowing us to be conservative

[5] it is noted that in this discussion the y-axis indicates altitude, whereas in other chapters it has been represented by the z-axis

[6] these distances can be related to X and Y through appropriate algebraic sums

The idea was to propagate the top fragments belonging to the (x, y) plane to see which of them would end up on the target:

- A time step has been chosen for the propagation;
- The axial and radial velocities of the fragments, as well as the pitch angle, were known so, using this information, V_x and V_y were computed;
- The target's maximum and minimum horizontal speeds were known, consequentially, by exploiting the relative distances X and Y (or X_{cg} and Y_{cg}) at each time step, the region of the plane where a portion of the target was located could be identified (this region was determined as the intersection of the area obtained by propagating using V_{max} and with the one obtained by propagating using V_{min});
- During each time step, the coordinates of each fragment were compared to the region encompassing the target's location. If the fragment's coordinates were found within this target region, it was deemed to have impacted the target and was consequently included in the count. Otherwise, the propagation process for that fragment continued until the distance it had traveled surpassed a predefined threshold. Upon reaching this threshold, the propagation was halted, and the fragment was not considered to have struck the target. [7]

By following these instructions, it was possible to calculate how many rings out of n_{fr} hit the target. Then, since the fragments were deployed on a cylindrical warhead, it was necessary to add a $3D$ reasoning to understand how many fragments hit the target for each individual ring. To this end, it was necessary to define the angle β, which depends on the radial distance (itself depending on X, Y, and θ) and on the geometric characteristics of the target and warhead:

[7] the threshold serves not only as a distance beyond which the fragment can be presumed to have flew away without striking the target, but also as a boundary beyond which aerodynamic drag will have induced a substantial error, rendering the model invalid.

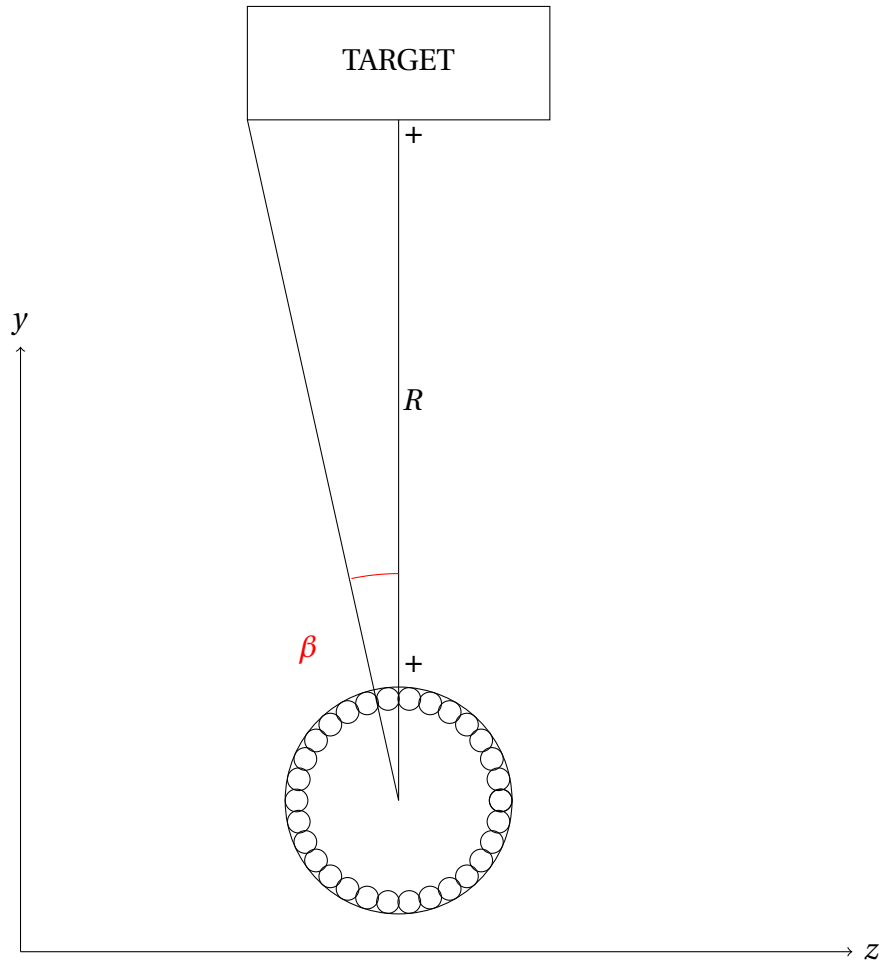


Figure 7.11 Beta angle

In the MATLAB code, for each ring that hit the target, it has been calculated the radial distance R, and subsequently β. By summing all these fragments, the total number of fragments hitting the target (n_{tot}) has been derived. At this point, it was possible to calculate P_k imposing:

$$P_k = 1 - (1 - p)^{n_{tot}} \tag{7.3}$$

Subsequently the P_k associated with each triplet of values (X, Y, θ) of interest was calculated. After explaining the underlying reasoning, it is possible to present the MATLAB codes implemented:

Script 7.1 MATLAB script for P_k estimation(case $0 < \theta \leq \pi/4$)

```
load("warhead.mat")
dmiss=0.12; %missile's diameter [m]
d_cg_w=0.06875; %distance between the missile's center of gravity and the
    start of the warhead[m]
Lt=1;%target's length [m]
ht=0.3; %target's height in [m]
lat=1; %target's width [m]
vtmax=60; %maximum target velocity [m/s]
vtmin=0; %minimum target velocity [m/s]
lnose=0.2; %nose length [m]
N=25;
thetamax=pi/4;
thetavec=linspace(0,thetamax,N);
```

```
Xvec=linspace(-2,2,N);
Yvec=linspace(0.5,1,N);
Xcgfix=Xvec;
Ycgfix=Yvec;
result_matrix = zeros(N^3, 5);
counter=1;
p=0.07;
jj=0;
for i1=1:N
    theta=thetavec(i1);
    for i2=1:N
    X=Xvec(i2);
        for i3=1:N
        jj=jj+1;
        Y=Yvec(i3);
        vyf=v0*cos(theta)+vm*sin(theta);
        vxf=vm*cos(theta)-v0*sin(theta);
        ntot=0;
              for j=1:nc
                  if j==1
                  y_fi_pv(j)=d_cg_w*sin(theta);
                  else
                   y_fi_pv(j)=d_cg_w*sin(theta)+y_ff_pv(j-1)-2*rf*cos(
   theta);
                   end
              y_ff_pv(j)=y_fi_pv(j)+2*rf;
                  if j==1
                  x_fi_pv(j)=d_cg_w*cos(theta);
                  else
                  x_fi_pv(j)=d_cg_w*cos(theta)+2*rf*(j-1)-2*rf*(j-1)*sin(
   theta);
                  end
              x_ff_pv(j)=x_fi_pv(j)+2*rf;
              x_fc_pv(j)=x_fi_pv(j)+rf;
              y_fc_pv(j)=y_fi_pv(j)+rf;
              t=0;
                 while true
                 t=t+0.0000001;
                 x_fc_ac=x_ff_pv(j)+vxf*t;
                 y_fc_ac=y_fc_pv(j)+vyf*t;
                 x_ti=X-Lt/2+t*vtmax;
                 x_tf=X+Lt/2+t*vtmin;
                 y_ti=Y-ht/2;
                 y_tf=y_ti+ht;
                   if x_fc_ac >= x_ti && x_fc_ac <= x_tf && y_fc_ac >=
   y_ti && y_fc_ac <=y_tf
                   flag=1;
                   break;
                   end
                 dist=sqrt(x_fc_ac^2+y_fc_ac^2);
                   if dist >5
                   flag=0;
                   break;
                   end
                 end
                 if flag == 1
```

```
                R=-sin(theta)*(x_fc_ac-x_fc_pv(j))+cos(theta)*(y_fc_ac-
   y_fc_pv(j));
                betamax=atan(0.5*lat/(R+dmiss/2));
                Lc=betamax*d;
                k=floor((Lc-rf*pi)/(2*rf*pi));
                n=2*k+1;
                ntot=ntot+n;
                else
                ntot=ntot+0;
                end
            end
        Pk=1-(1-p)^ntot;
        result_matrix(counter, :)=[rad2deg(theta), X, Y, ntot, Pk];
 Ycg(i3), ntot, Pk];
          Pk_matrix(i1,i2,i3) = Pk;
          counter=counter+1;
        end

    end
end
```

Script 7.2 MATLAB script for P_k estimation(case $\pi/4 < \theta < \pi/2$)

```
load("warhead.mat")
dmiss=0.12; %missile's diameter [m]
d_cg_w=0.06875; %distance between the missile's center of gravity and the
    start of the warhead[m]
Lt=1;%target's length [m]
ht=0.3; %target's height in [m]
lat=1; %target's width [m]
vtmax=60; %maximum target velocity [m/s]
vtmin=0; %minimum target velocity [m/s]
lnose=0.2; %nose length [m]
N=25;
thetamax=pi/2;
thetavec=linspace(pi/4,thetamax,N);
Xvec=linspace(-2,-0.75,N);
Yvec=linspace(0,1,N);
Xcgfix=linspace(-1.25,-0.25,N);
Ycgfix=linspace(0.6,1.5,N);
result_matrix = zeros(N^3, 5);
counter=1;
p=0.07;
jj=0;
for i1=1:N
    theta=thetavec(i1);
    for i2=1:N
    X=Xvec(i2);
        for i3=1:N
        jj=jj+1;
        Y=Yvec(i3);
        vyf=v0*cos(theta)+vm*sin(theta);
        vxf=vm*cos(theta)-v0*sin(theta);
        ntot=0;
              for j=1:nc
                  if j==1
```

```
                y_fi_pv(j)=0;
                else
                 y_fi_pv(j)=y_ff_pv(j-1)-2*rf*cos(theta);
                 end
            y_ff_pv(j)=y_fi_pv(j)+2*rf;
                if j==1
                x_fi_pv(j)=0;
                else
                x_fi_pv(j)=2*rf*(j-1)-2*rf*(j-1)*sin(theta);
                end
            x_ff_pv(j)=x_fi_pv(j)+2*rf;
            x_fc_pv(j)=x_fi_pv(j)+rf;
            y_fc_pv(j)=y_fi_pv(j)+rf;
            t=0;
               while true
               t=t+0.0000001;
               x_fc_ac=x_ff_pv(j)+vxf*t;
               y_fc_ac=y_fc_pv(j)+vyf*t;
               x_ti=Lw*cos(theta)+X-t*vtmax;
               x_tf=Lw*cos(theta)+X-Lt-t*vtmin;
               y_ti=Y;
               y_tf=y_ti+ht;
                 if x_fc_ac >= x_tf && x_fc_ac <= x_ti && y_fc_ac >=
y_ti && y_fc_ac <=y_tf
                 flag=1;
                 break;
                 end
               dist=sqrt(x_fc_ac^2+y_fc_ac^2);
                 if dist >5
                 flag=0;
                 break;
                 end
               end
               if flag == 1
               R=-sin(theta)*(x_fc_ac-x_fc_pv(j))+cos(theta)*(y_fc_ac-
y_fc_pv(j));
               betamax=atan(0.5*lat/(R+dmiss/2));
               Lc=betamax*d;
               k=floor((Lc-rf*pi)/(2*rf*pi));
               n=2*k+1;
               ntot=ntot+n;
               else
               ntot=ntot+0;
               end
         end
     Pk=1-(1-p)^ntot;
     result_matrix(counter, :) = [rad2deg(theta), X, Y, ntot, Pk];
     Xcg(i2)=X+Lt/2+Lw*cos(theta)+d_cg_w*cos(theta);
     Ycg(i3)=Y+d_cg_w*sin(theta)+ht/2;
     result_matrixguida(counter, :) = [rad2deg(theta), Xcg(i2), Ycg(i3
), ntot, Pk];
     Pk_matrix(i1,i2,i3) = Pk;
     counter=counter+1;
     Pk_y_vec(i3) = Pk;
     Y_value(i1,i3) = Ycg(i3);
     end
```

```
        X_value(i1,i2) = Xcg(i2);
        Pk_x_mat(i2,:) =  interp1(Ycg, Pk_y_vec, Ycgfix);
    end
    [XCG, YCGFIX1] = meshgrid(Xcg, Ycgfix);
    x_new = linspace(-1.25, -0.25, N); y_new = linspace(0.6, 1.5, N);
    [XCGFIX, YCGFIX2] = meshgrid(Xcgfix, Ycgfix);
    Pk_FINAL(i1,:,:) = interp2(XCG, YCGFIX1, Pk_x_mat, XCGFIX, YCGFIX2, '
   linear');
end
```

It can be noted that two codes have been created: one for lower pitch angle values and another for higher values. This is because depending on this angle's value, it made sense to study certain relative positions (X, Y) rather than others. It is noted that in the code for θ greater than $\pi/2$, an interpolation was performed to calculate P_k relative to fixed X_{CG} and Y_{CG} positions, which do not depend on the value of θ.

Assuming that the probability of a single fragment hitting the target is $p = 0.07$, and hypothesizing that this probability p varies with residual kinetic energy as a ramp function, the following plot has been obtained:

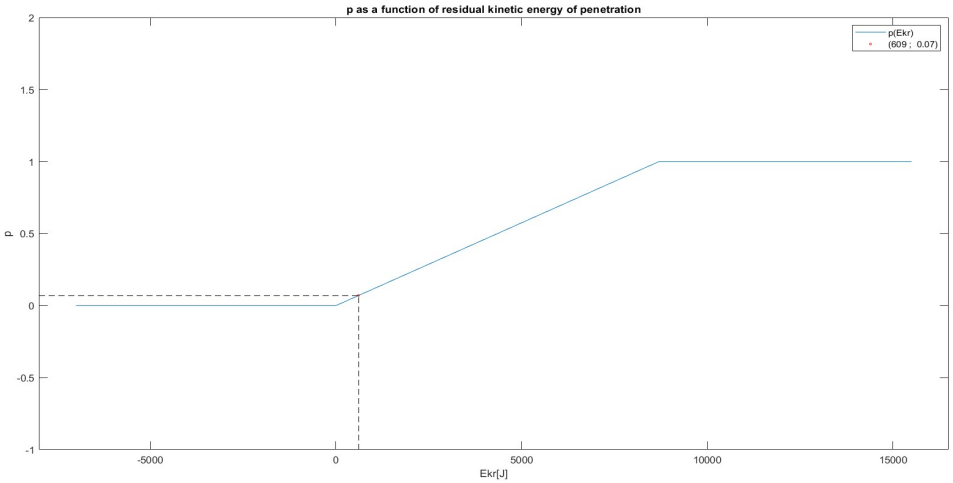


Figure 7.12 p as a function of residual kinetic energy of penetration

Chapter 8

II Stage

In this chapter, we will focus on the the second stage of the missile useful for the "Sustain phase". The second stage, allows the missile to maintain the desired cruising speed and follow the required trajectory until reaching and hopefully neutralizing the target. The operational life for this stage will be on the order of tens of seconds. This stage is designed to ensure precise and stable navigation, enabling the missile to follow the predefined trajectory with great accuracy.

8.1 II-stage Overview

Below is a general overview of the second stage, which is responsible for maintaining the predefined speed for the set flight time. Below are the main dimensions and technical specifications:

- **Total length**: 976 mm;
- **Caliber**: 80.6 mm;
- **Number of fins**: 4;
- **Mass with loaded propellant**: 11.224 kg;
- **Mass without propellant**: 8.7 kg;
- C_G **on X asis with loaded propellant**: 474.9 mm;
- C_G **on X asis without propellant**: 419.1 mm;
- **Moment of Inertia** I_{YY} **with respect to** C_G **with loaded propellant**: $1.033 kg \cdot m^2$;
- **Moment of Inertia** I_{YY} **with respect to** C_G **without propellant**: $0.904 kg \cdot m^2$;
- C_p **on X asis** : 687.3 mm;
- **Maximum load factor n**= 6;
- **Accelerometer position with respect to minum** C_G **(L_acc)**: 74.65 mm.

These data are the result of the combined outputs provided by various branches involved in the overall design, such as propulsion, aerodynamics, structural analysis, and 3-D.o.F. simulation.

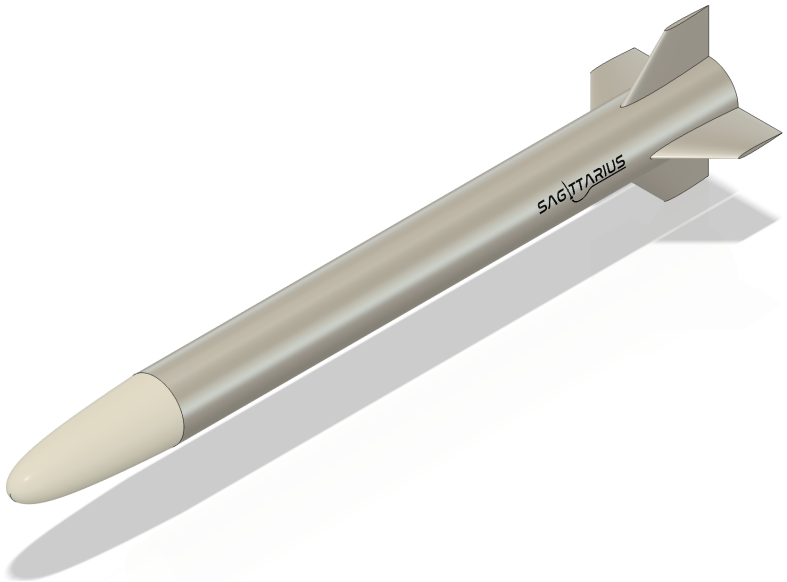


Figure 8.1 Second Stage 3/4 View

8.2 Mechanical Architecture

This paragraph provides a summary of the components of the second stage and explains the design choices related to them.

8.2.1 Design Description

The entire design of the second stage for the sustain phase revolves around the main payload, which is a 10 kg warhead (see chapter 7).[11]
The following figure shows the internal components of stage II:

Figure 8.2 Sagittarius Second Stage Internal Components

The warhead will be the first functional component immediately after the nose. Specifically, the nose has been designed using the elliptic parameters, which minimize drag for a given volume. It has among its aerodynamics roles (described in **??**) the task of housing the semi-active seeker, essential for the guidance system.
The selection of the seeker has not been thoroughly detailed due to project complexity. However it has been assumed to be a truncated cone shape, rounded at the front to accommodate the laser, with a weight of 1.925 Kg. It occupies almost the entire nose section, with a length of 200 mm on the x-axis.
Following there is the warhead, described in the CAD model as a cylinder with a length of 150 mm and a lateral surface area of 4778 mm^2. It has a weight of 1.969 Kg . The position of the warhead's center of gravity in relation to the overall center of gravity of the missile is of particular importance as specified in 7.
After that, there is a space reserved to every electrical component, such as batteries, motherboard and CPU. These components are necessary as they power and control the various sensors onboard the missile, such as the Seeker, accelerometer, and temperature sensors.
Additionally, the fins of the second stage are adjustable through actuators that must be in contact with the central computer.
There are four small actuators connected to the four fins of the second stage which are useful to deflect them up to ±35 deg . Each of them has an assumed weight of 0.05387 Kg.
It is important to note that the actuators have been placed in the space between the steel, that covers the nozzle, and the aluminum outer shell. Due to space constraints, it should be emphasized that the fins are not hinged at the optimal point (i.e., c/4), but at 46 mm

to fin's leading edge.
The fins were designed using aerodynamic software to ensure the best compromise between stability and maneuverability for the missile (see **??**.

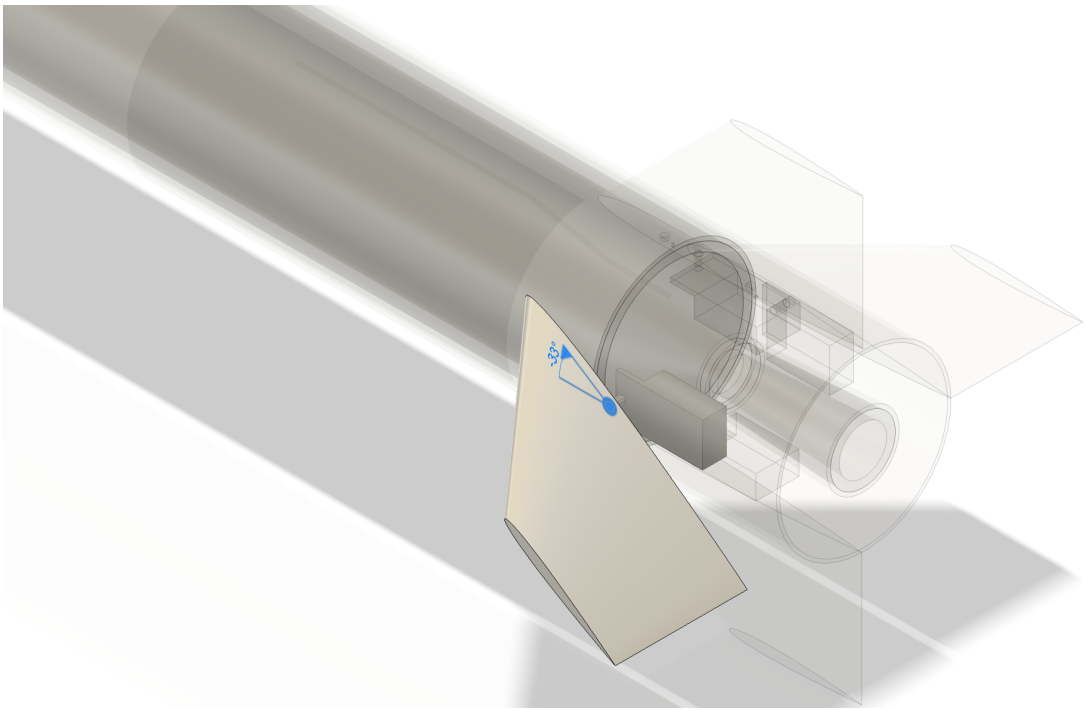


Figure 8.3 Lateral Fin deflected by an actuator

After the space dedicated to electrical components there is the propulsion section, which is composed by the propellant grain, its case (Inner shell), thermal liner and nozzle.
The propellant grain has been designed using custom Matlab codes, based on inputs provided by 3-D.o.F. (see 5). Given the requirement for constant thrust, a cylindrical grain shape was selected. The nozzle is frustoconical and designed to optimize axial outflow and avoid heating problems.
The liner, made of porous graphite, surrounds the grain and the nozzle, providing adequate thermal insulation, and has been designed starting from the thermally critical area, the nozzle throat.
The inner shell is a layer of steel surrounding the grain, providing the first layer of structural coating.

The outer shell is made of aluminum and encases the second stage in its entirety, except for the Radome. The outer shell is the part in contact with the external environment.
The outer shell and the inner shell are separated by a 1 mm thick circular section, intended for the passage of cables necessary to connect the electronic components to the actuators. The structural connection between the inner and outer shells is made through eight special graphite thermal breakers, as specified in the structural section.

The following figures show the Second Stage front, back and lateral view.

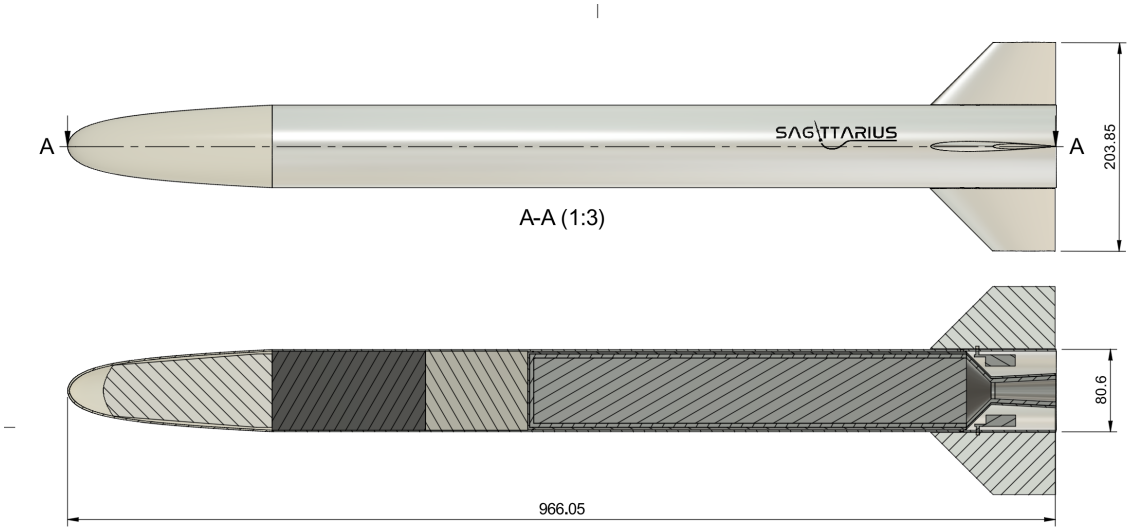


Figure 8.4 Lateral View and Section

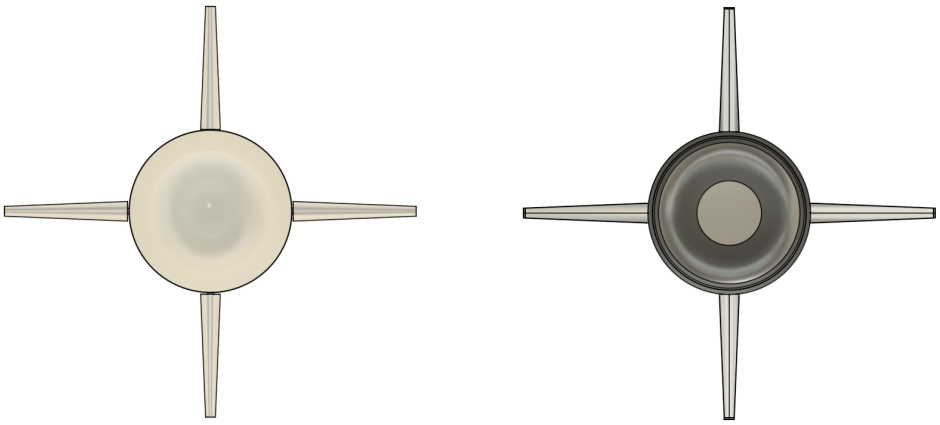


Figure 8.5 Front and Back View

8.3 Sustain Propulsion

In this section the propulsion requirements and solutions for the thrust to be provided in the sustain phase, from the detachment from the booster until interception, will be enquired.

8.3.1 Thrust profile

For the design of the second stage, we started with data provided by 3-D.o.F. (see chapter5). This analysis showed the necessity of having a constant thrust (also known as "step thrust") of 150 N for a duration of 30 seconds.

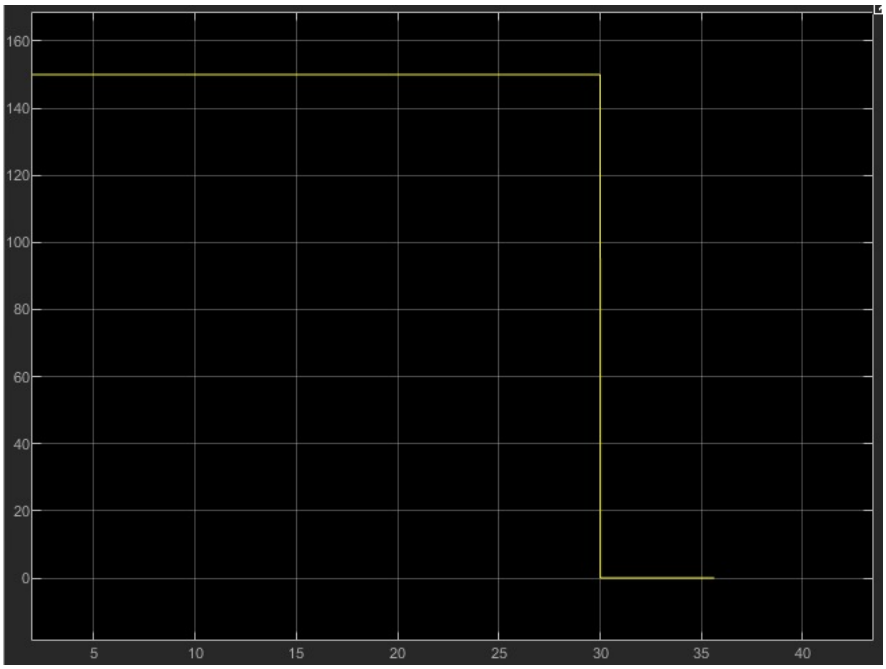


Figure 8.6 Thrust Profile

Based on these requirements, the choice of grain shape was narrowed down to two options: a cylindrical "cigarette" grain or a "star" grain, both characterized by a constant thrust profile. The "star" grain option had the advantage of keeping the combustion away from the walls for most of the burn time, allowing for a reduction in the thickness of the thermal insulation layer. However, despite this advantage, after various analyses, the choice fell on the cylindrical "cigarette" grain. With a Loading Fraction of $\epsilon_l = 1$ (the grain occupies the entire available volume), this option allows for a significant reduction in the size and weight of the entire stage, even considering the thickening of the insulating layer.

8.3.2 Rocket motor design

In the design of the propulsion system for the second stage, numerous factors must be considered, as they can lead to significantly different performance outcomes. The choice of propellant and the geometric configuration of the grain are key elements that directly influence the performance of the booster.

Choice of Propellant

The choice of propellant will be a solid propellant as it is particularly suitable for providing high performance with a short firing time, on the order of tens of seconds, making it ideal for the assigned operational scenario. This type of propellant offers an excellent balance between power and ease of use, making it ideal for military applications where rapid response is crucial, solid propellants also provides a 10-20 years of duration without loosing their properties.

Initially, a mixture of Nitroglycerin (NG) and Nitrocellulose (NC) was chosen due to its low cost and the absence of visible exhaust fumes, a useful characteristic in military contexts. However, this choice was later discarded due to its lower performance compared to more modern propellants.

HTPB-1912 falls into the category of "Composite Propellants", in which the oxidizer is dispersed in the form of crystals within a reducing matrix made of polymers. Aluminum powder is added to the matrix to increase the heat of reaction and specific impulse.

The chosen oxidizer, Ammonium Perchlorate, is not sensitive to impacts and leaves no residues. These features are enhanced by the rubbery matrix, making it extremely beneficial for military applications.

The final choice was HTPB-1912, which is a mixture composed of 19% Aluminum powder and 69% Ammonium Perchlorate, mixed in a base of 12% Hydroxyl-terminated Polybutadiene (HTPB). This propellant has a density (ρ) of 1854.55 kg/m^3 and a flame temperature of 3440 K. The Specific Impulse typically provided by this mixture is 260-265 seconds.

Other necessary parameters for stage sizing are:

- $a = 6.475 \times 10^{-5}$, Propellant Coefficient [8]
- $n = 0.4$, Pressure Exponent (Sutton) [8]

These parameters allow for the calculation of the grain regression rate:

$$\dot{r} = a \cdot P_0^n$$

Problems with Alternative Propellant

As previously mentioned, the initial choice of propellant was a mixture of Nitroglycerin and Nitrocellulose, specifically the so-called "Italian Balistite" studied by Alfred Nobel. This propellant falls into the category of "Homogeneous Propellants" (or "Double Base"), which contain both oxidizing and reducing agents within the same molecule. When appropriately ignited, these propellants decompose exothermically, providing high energy and behaving like explosives. In propellants made by Nitroglycerin and Nitrocellulose, the Nitroglycerin behaves as the oxidizer, while the Nitrocellulose as the fuel, the

optimum O/F ratio is set to be around 8, thus lower O/F ratios are usually used since the Nitroglycerin has very poor mechanical characteristic, while the Nitrocellulose are better[8]. In the "Italian Balistite" the O/F is set to 1.

The strengths of these propellants are numerous:

- Absence of fumes and toxic exhaust products;
- Good control over the regression rate;
- Low density, resulting in reduced weight;
- Low costs.

However, this choice was overcome for both performance and safety reasons. The "Italian Balistite" mixture has a Heat of Reaction (2550 K) and a Specific Impulse (210 s), significantly lower than HTPB-1912. More critically, it is very unstable and extremely sensitive to impacts, as both Nitroglycerin and Nitrocellulose may have an explosive behaviour if hit by bullets.

Below is a comparison between two grains designed using the two different propellant mixtures:

$Propellant$	$HTPB-1912$	$NG-NC$
I_{sp}	182.09 s	162.16 s
l	0.423 m	0.2126 m
d	0.064 m	0.1026 m
c^*	1666 m/s	1483 m/s
c_f	1.071	1.071
m_p	2.52 kg	2.83 kg

We can see that the grain designed with HTPB-1912 has smaller dimensions (particularly the diameter, which affects aerodynamic resistance), lower weight, and higher performance. The weight issue is further emphasized when considering the outer structure that encases the grain, which would also increase in size, leading to a further increase in weight.

Grain Geometry

The choice of geometry is fundamental to obtaining the thrust profile consistent with 3-D.o.F. (see5). The required thrust profile is a constant thrust profile since the second stage must overcome aerodynamic resistance to maintain a constant velocity profile over time and, at the same time, allow for any necessary trajectory corrections to neutralize the target.

For this reason, a cylindrical “cigarette” grain was chosen, which, thanks to its constant burning area, allows for constant thrust throughout the operational life.

Figure 8.7 Grain with a cigarette geometry [Sutton]

Input from C.E.A.

Using the C.E.A. software, we derived the parameters for the chosen propellant, HTPB-1912, relevant to our specific application.

Several assumptions were made for the use of C.E.A.:

- $\gamma = 1.1$, a value compatible with similar cases in the literature (Sutton);
- $R = 287\,\mathrm{J/(kg \cdot K)}$, Universal Gas Constant ;
- $c_p = 1005\,\mathrm{J/(kg \cdot K)}$, Specific Heat at Constant Pressure (Sutton);
- $P_0 = 7$ bar, Chamber Pressure (Design Choice).
- $P_0/P_exit = 7$.

Given these input values, the C.E.A. provides an equivalent exhaust velocity value of 1784.5 m/s. The C.E.A. also outputs the Expansion Ratio, which is the ratio between the Exit Area and the Throat Area. The Expansion Ratio value is 1.9.

Nozzle Design Choices

Before proceeding with the actual sizing of the nozzle, it was decided to have a nozzle adapted to sea level. This decision was made because the operational altitudes of the missile will be relatively low and to avoid potential problems related to flow separation within the nozzle's divergent section.

After various iterations, it was determined that the optimal exit diameter for our nozzle is 0.0022 m.

We will return to the actual sizing of the nozzle later.

Calculation of $\dot{m}$ and I_{sp}

The first objective was to calculate the mass flow rate of burned propellant per second. This operation is straightforward knowing the Thrust and Equivalent Exhaust Velocity:

$$\dot{m} = \frac{T}{c}$$

Resulting in: $\dot{m} = 0.084\,\mathrm{kg/s}$

Next, we proceed with the calculation of the Specific Impulse:

$$I_{sp} = \frac{T}{\dot{m} \cdot g}$$

Resulting in: $I_{sp} = 182.1$ sec

It can be observed that this value is slightly lower than that reported by Sutton, primarily due to the lower Thrust requirement, which necessitated a lower value of c_f, a parameter that affects nozzle efficiency.

Script 8.1 Matlab code to calculate $\dot{m}$ and I_{sp}

```
%% mdot and Isp calculation
mdot = Thrust/c;
Isp=Thrust/(mdot*g) %Specific Impulse [sec]
```

Grain Sizing

It is possible to calculate the change in volume of the propellant grain over time:

$$\dot{V}_{\text{grain}} = \frac{\dot{m}}{\rho_p}$$

The calculated value is: $\dot{V}_{\text{grain}} = 4.53 \times 10^{-5}\,\text{m}^3$

The next step is the calculation of the cross-sectional area of the grain:

$$A_{\text{grain}} = \frac{\dot{V}_{\text{grain}}}{\dot{r}}$$

Resulting in: $A_{\text{grain}} = 0.0032\,\text{m}^2$

At this point, the diameter and length of the cylindrical grain have been calculated:

$$d_{\text{grain}} = 2\sqrt{\frac{A_{\text{grain}}}{\pi}}$$

Resulting in: $d_{\text{grain}} = 0.064\,\text{m}$

To determine the length of the grain:

$$l_{\text{grain}} = \dot{r} \cdot t_{\text{fin}}$$

Resulting in: $l_{\text{grain}} = 0.4231\,\text{m}$

Finally, to complete the sizing, we calculate the volume and mass of the cylindrical grain:

$$V_{\text{grain}} = l_{\text{grain}} \cdot A_{\text{grain}}$$

Resulting in: $V_{\text{grain}} = 0.0014\text{m}^3$

$$m_p = V_{\text{grain}} \cdot \rho_p$$

Resulting in: $m_p = 2.52\,\text{kg}$

Script 8.2 Matlab code to grain dimentions

```
%% Sizing
Vol_dot = mdot/rhop;              %Grain volume variation  [m^3/sec]
A_grain = Vol_dot/l_dot;          %Grain section [m^2]
d_sustain = 2*sqrt(A_grain/pi)    %Grain diameter [m]
l_sustain = l_dot*t_fin_sustain   %Grain length [m]

Vol = l_sustain * A_grain;        %Grain volume [m^3]
mp = rhop * Vol                   %Propellant mass [kg]
save("propellant_mass","mp");
```

Calculation of c_f and c_{star}

The coefficients c_f and c_{star} allow evaluating the design of the nozzle and the combustion chamber, respectively. With the previous data at hand, it is possible to compute these two coefficients:

$$c_f = \frac{T}{P_0 \cdot A_{cr}}$$

$$c_{star} = \frac{P_0 \cdot A_{cr}}{\dot{m}}$$

Resulting in: $c_f = 1.071$, $c_{star} = 1666.1$ m/s.

As mentioned earlier, the value of c_f is slightly lower compared to those found in Sutton, due to the lower Thrust required to maintain constant velocity.

The value of c_{star} is perfectly in line with literature values.

Script 8.3 Matlab code to calculate c_f and c_{star}

```
%% cf and cstar calculation
cf=Thrust/(P0*Acr)
cstar=P0*Acr/mdot
```

8.3.3 Nozzle Sizing

Now it is possible proceed with the sizing of the nozzle, for which a truncated conical geometry has been chosen, ensuring good performance, design simplicity, and reduced manufacturing costs.

Starting with the known data:

- Inlet Diameter: 0.064 m(equal to grain diameter);
- Exit Diameter: 0.022 m;
- Expansion Ratio: 1.9 (from CEA).

First, the Exit Area is obtained, then using the Expansion Ratio, the Throat Area (Critical Area), and finally, the Throat Diameter:

$$A_e = \pi \left(\frac{d_e}{2}\right)^2$$

$$A_{cr} = \frac{A_e}{r_{exp}}$$

$$d_{cr} = \sqrt{\frac{4A_{cr}}{\pi}}$$

Resulting in: Throat Diameter $d_{cr} = 0.016$ m.

Next, the values for the convergent and divergent section angles are chosen. The convergent section angle (θ_{conv}) was chosen as 45 deg, as indicated in the literature [8] to avoid flow separation and excessive wall heating. After several trials, the divergent section angle (θ_{div}) was chosen as 3 deg. This angle minimizes thrust losses by favoring axial outflow and prevents excessive elongation of the nozzle dimensions, benefiting from its compact size.

Now, it is possible to compute the length of the convergent and divergent sections of the nozzle:

$$l_{\text{conv}} = \left(\frac{d_{\text{grain}}}{2} - \frac{d_{\text{cr}}}{2}\right) \tan\left(\frac{\pi}{2} - \theta_{\text{conv}}\right)$$

$$l_{\text{div}} = \frac{d_{cr}\left(\sqrt{\frac{A_{\text{exit}}}{A_{cr}}} - 1\right)}{2\tan(\theta_{\text{div}})}$$

Resulting in: $l_{\text{conv}} = 0.024$ m, $l_{\text{div}} = 0.0582$ m.

Script 8.4 Matlab code to calculate l_{conv} and l_{div}

```
%% Length Calculations

L_conv=((d_sustain/2)-(d_cr/2))*tan(pi/2-theta_conv);  %Convergent
    section length [m]
L_div=d_cr*(sqrt(A_exit/A_cr)-1)/(2*tan(theta_div));   %Divergent section
     length [m]
```

Another critical aspect of the design was the sizing of the Throat. This was not assumed to be infinitely small but rather 4 mm long. This choice was necessary to provide space for the servo-actuators responsible for controlling the fins.

By adding the Throat length to the lengths of the Convergent and Divergent sections, the total length of the nozzle is obtained, which is 0.0862 m.

Finally, it possible to evaluate the efficiency of the designed nozzle using an additional parameter compared to c_f, namely λ, which represents the "Percent Axial Thrust":

$$\lambda = \frac{1 + \cos(\theta_{\text{div}})}{2}$$

Resulting in: $\lambda = 0.9993$.

In other words, the percentage of Thrust lost due to the inclination of the Divergent section is less than 0.07%.

Script 8.5 Matlab code to calculate l_{conv} and l_{div}

```
%% Axial Thrust Loss Calculation

lambda=0.5*(1+cos(theta_div));  %Axial thrust percentage
T_loss_percentage=1-lambda;     %Thrust loss percentage
```

8.4 Aerodynamics

This chapter elucidates the wing design process, which begins with Wings.3, a software employed to define the wing geometry of the missile. Through a series of deliberate evolutions, grounded in rigorous preliminary structural studies, it enables precise exploration of how geometric modifications impact the aerodynamic coefficients crucial for the missile 's stable and efficient flight. Subsequently, the implementation of these results into the semiempiric aerodynamic validation software *RASAERO* program, has yielded trends across all aerodynamic coefficients and facilitated a stability analysis of the stage under consideration. Following this, an analysis of the aerodynamics of the nose is conducted. After completing the design phase, a Computational Fluid Dynamics (CFD) analysis was performed using Ansys Fluent software. This analysis aimed to verify, by examining the fluid dynamic behavior of the entire stage, whether the system conforms to the assumptions initially made.

8.4.1 Wings

To ensure maximum maneuverability of the missile, a skid-to-turn control system has been employed, allowing the use of control surfaces to generate lateral forces. These surfaces deflect the airflow, thereby producing a force that acts laterally on the missile. Given the initial parameters, the mass is set at 11.224 kg, increased by a safety factor of 30%, the diameter at 0.08 m, and the reference area at 0.005 m^2, all pertaining to the "sustain" phase of the missile. Subsequently, a load factor of n=6 has been assigned, which subjects the structure to a lateral force

$$W = 14.59 \cdot 9.81 = 143.140N \tag{8.1}$$

$$F = W \cdot n = 858.401N \tag{8.2}$$

This load factor objective is significantly higher than standard requirements (three times the target's load factor). To accommodate this, for the purpose of wing design, a well-structured iterative process has been implemented in the Wings.3 program, which aims to derive the wing geometry and the normal force coefficient relative to the missile axis as a function of the angle of attack $C_{z,\alpha}$, this coefficient is then utilized to determine the total normal force.

$$Fz = C_{z,\alpha} \cdot \alpha \cdot 0.5 \cdot \rho \cdot (v_{inf}^2) \cdot S_{ref} \tag{8.3}$$

If the load factor is below 6, an increment in geometric dimensions is initiated, continuing the cycle until the desired value is achieved, all of this at a fixed angle of attack α of 35 degrees. During development, considerations were made regarding the parameters for infinite flight, ensuring that the wing would not encounter supersonic Mach numbers. Specifically, an infinite velocity v_{inf} of 250 m/s was selected, corresponding to a Mach number M_{inf} of 0.73. The profile under consideration is a NACA 0008, chosen for its symmetry and thus its ability to adapt to various flight conditions. With an angle of attack of 5 degrees, it does not enter the drag divergence zone, avoiding shock wave formation. However, reducing the thickness further increases the significance of stall phenomena at increasingly smaller angles of attack.

Table 8.1 Aerodynamic Performance of Various NACA Profiles

Profile	C_L	C_D	$E_{max}\ (\alpha)$	α_{Stall}
NACA 0015	0.6431	0.00921	77.9 ($\alpha = 9$)	16.75Âř
NACA 0012	0.6940	0.00972	75.6 ($\alpha = 7.5$)	15.5Âř
NACA 0010	0.6336	0.01033	70.2 ($\alpha = 9.75$)	14.5Âř
NACA 0008	0.6671	0.01020	68.3 ($\alpha = 6.75$)	12Âř
NACA 0006	0.6462	0.01479	56.6 ($\alpha = 4.5$)	9Âř
NACA 63015A	0.6411	0.00968	72 ($\alpha = 4.75$)	16.25Âř
NACA 63012A	0.6391	0.01033	64 ($\alpha = 7$)	13.25Âř
NACA 63A010	0.6733	0.01131	59.7 ($\alpha = 5.75$)	10.5Âř
NACA 64008A	0.6403	0.01590	49.9 ($\alpha = 4.25$)	7Âř

Wings.3

The program begins with the precise setting of the initial wing dimensions, establishing an optimal geometric ratio of 1 to 2 between the root chord (Cr=0.08) and the tip chord (Ct=0.04), which remains constant, complemented by the determination of the wingspan (S=0.04), chosen to be equal to the Ct. These proportions are selected to maximize aerodynamic efficiency and improve the load distribution along the wing during flight.

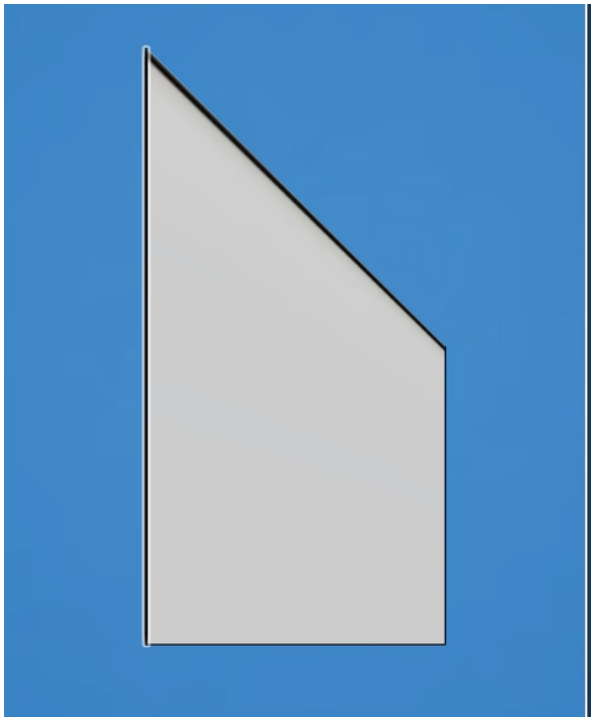


Figure 8.8 Wing for geometrical considerations

Calibrated increments for Cr, Ct, and S are introduced, orchestrated through a "while" loop that continues until a load factor of n=6 is achieved. It is observed that by increasing the span, the sweep angle, which will be discussed subsequently, is preserved. This cycle allows for the experimentation of various scenarios, calculating for each wing geometry the corresponding load factor and lateral force, thereby progressively refining the design

based on the results obtained at each iteration.

```
Cr: 0.0800 m, Ct: 0.0400 m, S: 0.0400 m, d: 0.0800 m, n: 3.83
Cr: 0.0820 m, Ct: 0.0410 m, S: 0.0410 m, d: 0.0800 m, n: 3.94
Cr: 0.0840 m, Ct: 0.0420 m, S: 0.0420 m, d: 0.0800 m, n: 4.04
Cr: 0.0860 m, Ct: 0.0430 m, S: 0.0430 m, d: 0.0800 m, n: 4.15
Cr: 0.0880 m, Ct: 0.0440 m, S: 0.0440 m, d: 0.0800 m, n: 4.25
Cr: 0.0900 m, Ct: 0.0450 m, S: 0.0450 m, d: 0.0800 m, n: 4.36
Cr: 0.0920 m, Ct: 0.0460 m, S: 0.0460 m, d: 0.0800 m, n: 4.47
Cr: 0.0940 m, Ct: 0.0470 m, S: 0.0470 m, d: 0.0800 m, n: 4.59
Cr: 0.0960 m, Ct: 0.0480 m, S: 0.0480 m, d: 0.0800 m, n: 4.70
Cr: 0.0980 m, Ct: 0.0490 m, S: 0.0490 m, d: 0.0800 m, n: 4.82
Cr: 0.1000 m, Ct: 0.0500 m, S: 0.0500 m, d: 0.0800 m, n: 4.94
Cr: 0.1020 m, Ct: 0.0510 m, S: 0.0510 m, d: 0.0800 m, n: 5.06
Cr: 0.1040 m, Ct: 0.0520 m, S: 0.0520 m, d: 0.0800 m, n: 5.19
Cr: 0.1060 m, Ct: 0.0530 m, S: 0.0530 m, d: 0.0800 m, n: 5.31
Cr: 0.1080 m, Ct: 0.0540 m, S: 0.0540 m, d: 0.0800 m, n: 5.44
Cr: 0.1100 m, Ct: 0.0550 m, S: 0.0550 m, d: 0.0800 m, n: 5.57
Cr: 0.1120 m, Ct: 0.0560 m, S: 0.0560 m, d: 0.0800 m, n: 5.70
Cr: 0.1140 m, Ct: 0.0570 m, S: 0.0570 m, d: 0.0800 m, n: 5.83
Cr: 0.1160 m, Ct: 0.0580 m, S: 0.0580 m, d: 0.0800 m, n: 5.97
Cr: 0.1180 m, Ct: 0.0590 m, S: 0.0590 m, d: 0.0800 m, n: 6.11
Final Cr: 0.1200 m, Final Ct: 0.0600 m, Final S: 0.0600 m, Final d: 0.0800 m, Final n: 6.11
```

Figure 8.9 Geometry increasing

The core of the geometric analysis is the determination of the sweep angle at the leading edge and at one-quarter of the chord

$$\Gamma = \tan^{-1}\left(\frac{Cr - Ct}{S}\right) = 0.78rad \tag{8.4}$$

$$\Gamma_{\text{deg}} = \frac{\Gamma}{\pi} \cdot 180deg = 45deg \tag{8.5}$$

$$\Gamma_{c/4} = \tan^{-1}\left(\frac{Cr - Ct + 0.25 \cdot Ct - 0.25 \cdot Cr}{S}\right) = 0.64rad \tag{8.6}$$

$$\Gamma_{c/4,\text{deg}} = \frac{\Gamma_{c/4}}{\pi} \cdot 180deg = 36.87deg \tag{8.7}$$

These angles were then utilized to ascertain the relative velocity to the wing profile and the corresponding Mach number

$$v_{\text{airfoil}} = v_{\text{inf}} \cdot \cos(\Gamma_{c_4}) = 200m/s \tag{8.8}$$

$$M_{\text{airfoil}} = \frac{v_{\text{airfoil}}}{a} = 0.58 \tag{8.9}$$

Once the 1 to 2 ratio of the chords is established, the sweep angle at the leading edge and at one-quarter of the chord will remain fixed, and given that the infinity velocity is also set, these velocity values will not change with variations in the wing geometries implemented during the cycle.

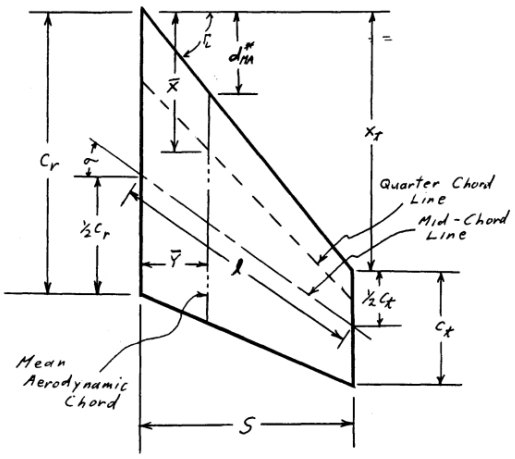


Figure 8.10 Wings geometry

In conclusion, the variation of the normal force coefficient relative to the missile axis as a function of the angle of attack is derived. This variation encapsulates the aerodynamic contributions from the wings and the wing-body interactions, which are evaluated with a K_{T_B} factor of 1.4 as suggested by Borrowmann [3], alongside the consistent contribution from the missile's nose, as Borrowmann's stipulations.

$$Cz_{\alpha_{\text{fin}}} = 16\left(\frac{S}{d}\right)^2 \div \left(1 + \sqrt{1 + \left(\frac{2l}{Cr + Ct}\right)^2}\right) \tag{8.10}$$

$$K_{T_B} = 1 + \frac{d}{2} \div (S + \frac{d}{2}) \tag{8.11}$$

$$Cz_{\alpha_{\text{fin_body}}} = K_{T_B} \cdot Cz_{\alpha_{\text{fin}}} \tag{8.12}$$

$$Cz_{\alpha_{\text{nose}}} = 2 \tag{8.13}$$

$$Cz_{\alpha_{\text{total}}} = Cz_{\alpha_{\text{nose}}} + Cz_{\alpha_{\text{fin_body}}} \tag{8.14}$$

The values obtained at the conclusion of this cycle are delineated below:

Geometric Parameters:

- Root chord (Cr) = 0.12 m
- Tip chord (Ct) = 0.06 m
- Wingspan (S) = 0.06 m

Variation of the Normal Force Coefficient Relative to the Missile Axis with Respect to the Angle of Attack:

- Total normal force coefficient $(CZ_\alpha) = 7.43\ \frac{1}{\text{rad}}\ (0.130\ \frac{1}{\text{deg}})$

- Normal force $(Fz) = 874.204$ N

These values are critical for assessing the aerodynamic performance and structural integrity of the missile under various flight conditions. The program also conducts final evaluations such as determining the mean aerodynamic chord, and from this, the wing thickness,providing essential data for the validation of the optimized design.

$$\lambda = \frac{Ct}{Cr} \tag{8.15}$$

$$MAC = \left(\frac{2}{3}\right)\left(\frac{1+\lambda+\lambda^2}{1+\lambda}\right)Cr = 0.093m \tag{8.16}$$

$$t = 0.08 \cdot MAC = 0.0075m \tag{8.17}$$

8.4.2 Nose

Considering the aerodynamic requirements, an elliptical shape was selected for the nose to maximize the volume.

The major semi-axis measures 20 cm, while the minor semi-axis measures 4 cm. Operating in a subsonic regime, where shock waves are absent, negates the need for specialized geometries. The contribution to the lift coefficient ($C_{z,\alpha,nose}$) remains constant due to its shape, as detailed in Borrowman[3].

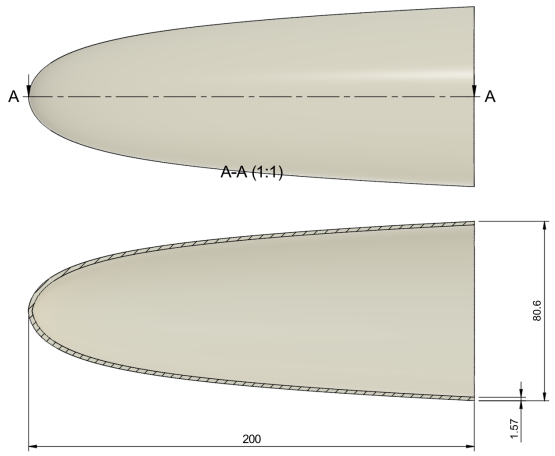


Figure 8.11 Nose geometry

8.4.3 Rasaero Analysis

To verify the accuracy of the data obtained in the 8.4.1 section, the RASAERO software was used. Geometric data of the II stage and the fins were inputted into the program, resulting in values for $C_{z,\alpha}$ that were then compared with the previously obtained values. The values were found to be consistent.

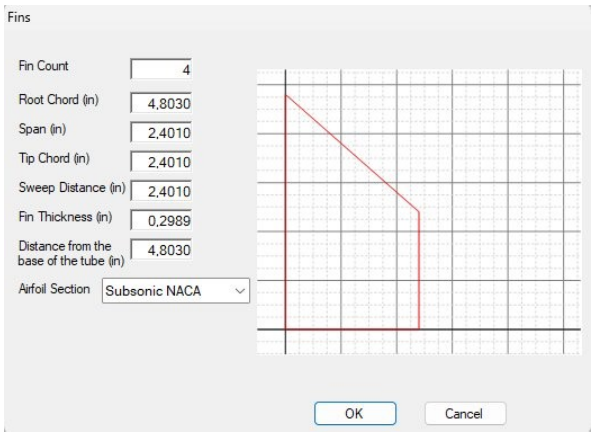


Figure 8.12 Rasaero input

In light of this verification, the remaining aerodynamic coefficients as a function of the Mach number were derived from RASAERO:

$$C_{D,\alpha}\,(\text{subsonic}) = 0.29 \tag{8.18}$$

$$C_{Z,\alpha} = 7.609\,\text{rad}^{-1}\,(0.1328\,\text{deg}^{-1}) \tag{8.19}$$

$$C_{M,\alpha}\,(\text{initial}) = C_{Z,\alpha} \times \frac{(C_{g,\text{position}} - C_{p,\text{position}})}{d} = -19.8\,\text{rad}^{-1}\,(-0.345\,\text{deg}^{-1}) \tag{8.20}$$

where d is the outer diameter of the missile.

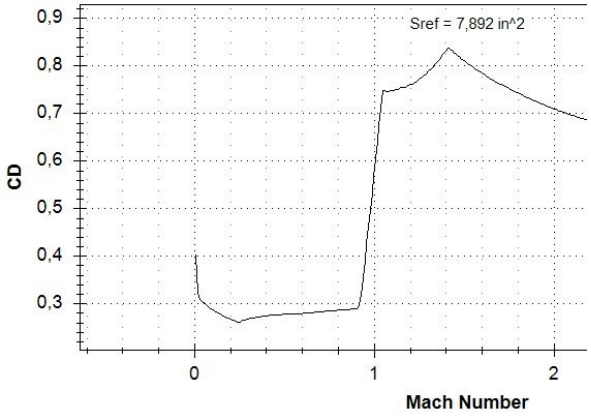


Figure 8.13 C_D vs Mach

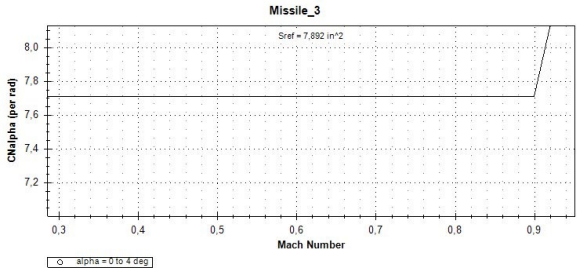


Figure 8.14 $C_{Z,\alpha}$ vs Mach

The aerodynamic moment coefficient $C_{M,\alpha}$, changes over time as the positions of the center of gravity (C_G) vary. This variation directly affects the missile's maneuverability

over time, as illustrated in the 5. The RASAERO software also allows for a stability analysis, ensuring that the center of pressure is positioned as far aft as possible relative to the center of gravity under the worst flight conditions (i.e., at launch). This configuration aims to enhance the stability of the missile. However, such a setup might typically compromise maneuverability. Nonetheless, through this analysis, an optimal configuration has been identified that ensures both superior stability and maneuverability.

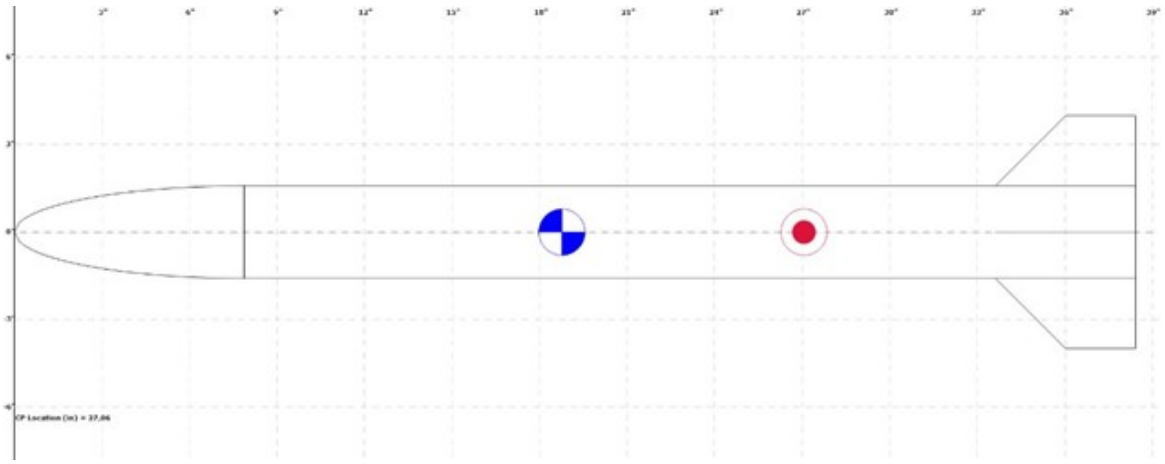


Figure 8.15 RASAERO

8.4.4 CFD Analysis

Following the geometric and aerodynamic design phases detailed in 8.4.1, 8.4.2 and 8.4.3, a fluid dynamic analysis of the entire sustain phase was conducted. This section will outline the methodologies employed and present the results obtained from this analysis.

Methodologies

The initial step involved importing the geometry to be analyzed from the CAD drawing. This geometry was then enclosed within a rectangular parallelepiped with dimensions of 3m x 2m x 2m. Utilizing the native Ansys software "Meshing," a computational mesh was generated for the analysis.

The subsequent phase involved performing the actual Computational Fluid Dynamics (CFD) analysis, where various setting parameters were defined. The analysis was conducted under steady-state conditions, employing the SST k-ω turbulence model. Given the regime in question, where compressibility effects are significant, it was assumed that air behaves as an ideal gas.

The flow velocity impacting the body, corresponding to the cruise speed, was determined during the booster design phase and set at 250 m/s. This velocity was established as a boundary condition at the inlet section of the mesh.

The computation process comprised 500 iterations, during which convergence was observed to occur significantly earlier.

Results

The main results that were focused on and are reported subsequently are:

$$D = 75N \tag{8.21}$$

$$C_D = 0.4 \tag{8.22}$$

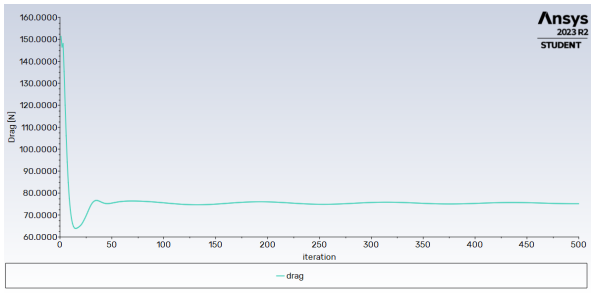


Figure 8.16 Drag

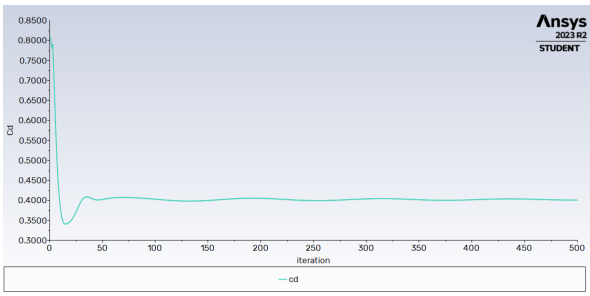


Figure 8.17 C_D

- The drag behavior is consistent, stabilizing around a value slightly above 75 N, which aligns with the drag reported by the 3-DoF analysis. It has to be noted that in the CFD analysis it has not been considered any gravitational effect, so the only force acting on the body is the Drag one.
- The velocity contour shows a deceleration at the nose section, followed by an expansion along the missile's profile. At the tail, a separation zone is visible where the velocities are significantly lower.

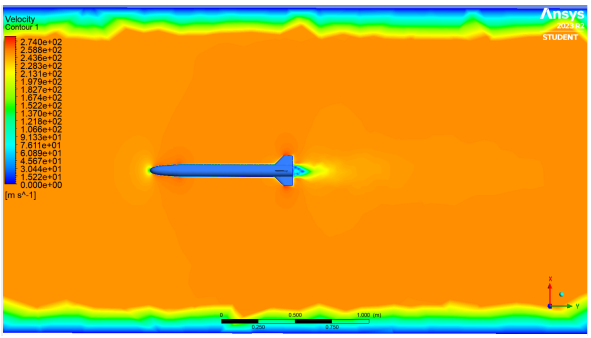


Figure 8.18 Contour velocity

- The pressure contour is consistent with the velocity contour, showing a noticeable increase at the front section of the nose, where it peaks. At the tail, in the separation zone, there is a region of strong pressure drop.

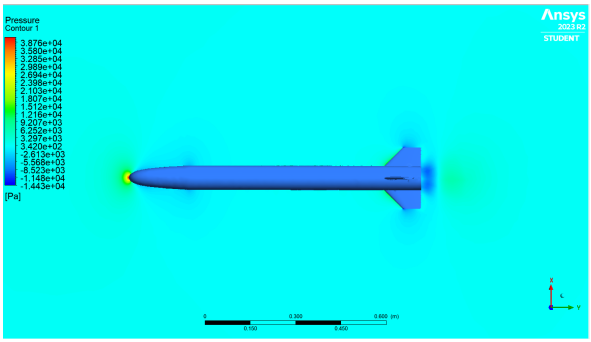


Figure 8.19 Contour pressure

8.4.5 Observations

- The coefficients related to the fins alone, obtained using the Wings.3 program, were extrapolated and utilized in the Simulink program;

- The drag coefficient (C_D) obtained from RASAERO is constant with respect to the angle of attack, but changes with Mach;
- The lift coefficient ($C_{Z,\alpha}$) from the Wings.3 program does not account for any compressibility factors, which RASAERO does; however, the values are similar, so this variation is not considered significant;
- Due to spatial constraints, the fins of the sustain stage are not hinged at a quarter of the chord, as would be optimal to nullify the moment, but rather at 46 mm from the leading edge also this variation is not considered significant;
- It should be noted that the CFD analysis was conducted considering only the sustain stage, as this stage is involved throughout the entire mission duration, whereas the booster is only present during the initial phases. Therefore, its aerodynamic contribution to the mission is negligible. Furthermore, including the booster in the analysis would significantly increase the computational cost, and thus it was not included in the fluid dynamic analysis;
- The C_D from the CFD analysis is higher than that obtained from RASAERO. This discrepancy is attributed to the fact that the CFD analysis did not consider the behavior of the nozzle, which may lead to separation at the final corner, slightly increasing the observed drag. This effect is not accounted for in the RASAERO model.

8.5 Structural Analysis

The structural analysis of the sustain encompasses three fundamental components:

- Outer shell: this is the aluminum structure that encases all the elements of the Sagittarius system, providing essential protection and support.
- Case grain: constructed from AISI 4340 steel, this structure houses the grain and is the core site of the combustion process, ensuring structural integrity and efficiency during operation.
- Nozzle: also made from AISI 4340 steel, the nozzle is designed to direct the exhaust flow effectively, playing a crucial role in the propulsion system's overall performance.

8.5.1 Outer Shell

The selection of materials employed in the missile is intricately linked to its mission requirements. The decision to utilize 7075 Aluminum for the outer shell is based on the specific forces and moments at play. Although these forces are not excessively high and the moments are significant, they remain manageable due to the compact dimensions of the Sagittarius. Moreover, given the short duration of the mission, fatigue behavior is not a primary concern. However, this aspect was still considered in our analysis to ensure a comprehensive and critical evaluation. In addition, 7075 Aluminum, which is also used for the dorsal wing panels, demonstrates superior resistance to compression and bending moments when compared to steel of equivalent weight. This characteristic makes it the material of choice for these components. It is also noteworthy that the missile is not subjected to forces that would generate significant torsional moments, thus rendering such moments negligible in our considerations. Ultimately, the careful selection of 7075 Aluminum is a reflection of its advantageous properties in terms of weight, strength, and durability, all of which align perfectly with the mission profile and operational demands of the Sagittarius missile. This meticulous approach ensures optimal performance and reliability throughout the missile's mission. Having introduced the material employed,

Young Modulus E	70 GPa
Density w	2800 Kg/m^3
Poisson Modulus ν	0.3
Conducivity k	200 $W/(mK)$

Table 8.2 Characteristics of 7075 Aluminum

now it is possible to proceed to the detailed sizing of the outer shell. In this context, several key[7] considerations have been taken into account for its design:

- Compression: The sizing for compression involves considering a critical Eulerian buckling load. This approach allows us to focus on the deflection behavior of the beam under compressive forces, ensuring that the structure can withstand the expected loads without experiencing instability.

- Shear: Shear forces are applied at the shear center, which is crucial for accurate analysis. In this way, the influence of the torsional moment is effectively eliminated, as it is considered negligible in this scenario and does not significantly impact the missile's overall dynamics.

- Bending Moment due to Shear: For bending moments induced by shear, we specifically considered straight bending. In this case, the neutral axis is aligned with one of the principal central inertia axes. This alignment simplifies the analysis and ensures that the structural response is accurately captured.

Furthermore, the forces acting on the structure will be detailed. This comprehensive analysis is essential for understanding the load distribution and ensuring the integrity of the design under operational conditions.
From a geometrical perspective, the outer shell features a hollow circular section. The geometric properties of this section are described by the following formulas:

- Area: The cross-sectional area is calculated to determine the material's capacity to withstand internal forces:
$$A = \frac{\pi}{2}(D^2 - d^2) \tag{8.23}$$
In this context, D is designated to represent the outer diameter of the section, while d signifies the inner diameter. This distinction between D and d allows for a clear differentiation between the external and internal dimensions, which is crucial for understanding the overall design and ensuring the integrity and performance of the outer shell under various load conditions.

- Polar Moment of Inertia: This property is crucial for assessing the resistance to torsional loading, even though torsion is considered negligible in this specific analysis:
$$I_p = \frac{\pi}{32}(D^4 - d^4) \tag{8.24}$$
Given the double symmetry of the section, the polar moment of inertia coincides with the trace of the inertia tensor. This symmetry simplifies the mathematical treatment and ensures that the moments of inertia about the principal axes are equal. Therefore:
$$I_p = I_{11} + I_{22} \tag{8.25}$$

- Moment of Inertia: For the doubly symmetric section, the moments of inertia about the principal axes are identical, which is a direct consequence of the geometric symmetry. This allows to express the moment of inertia of the cross-section succinctly and accurately:
$$I_p = 2I \tag{8.26}$$
Therefore:
$$I = \frac{I_p}{2} = \frac{\pi}{64}(D^4 - d^4) \tag{8.27}$$

- Minimum Radius of Gyration: The minimum radius of gyration is derived to understand the distribution of the cross-sectional area relative to the centroid, providing insight into the stability and buckling resistance of the structure. Furthermore, this value enables a thorough evaluation of the slenderness ratio of the alu-

minum tube. Such an evaluation is essential for determining the appropriate method for assessing the critical load the tube can withstand. Specifically, it allows to decide whether to rely on the Euler critical load, which is traditionally used for long, slender columns under axial compression, or to employ the more nuanced Euler-Johnson curves. The Euler-Johnson curves offer a refined analysis by incorporating the transition between the elastic and inelastic buckling behavior of the beam. Meticulously understanding the slenderness and corresponding buckling characteristics ensures the structural stability and integrity of the outer shell under various loading conditions, thereby enhancing the reliability and safety of the overall design. Therefore:

$$\rho = \sqrt{\frac{I}{A}} \tag{8.28}$$

In the comprehensive analysis, the assumption of a beam-like behavior characterized by a low slenderness ratio has been adopted [1]. This approach permits to utilize the Euler critical load as the primary indicator of instability without necessitating the consideration of buckling behavior through the more complex Euler-Johnson curves. To substantiate this assumption, a meticulous calculation for the slenderness ratio has been implemented within the program. The results of this calculation revealed that the slenderness ratio is significantly below the critical threshold, which is approximately equal to 100. This critical threshold is a well-recognized benchmark in structural analysis, below which the simpler Eulerian approach is deemed sufficient for predicting instability loads. By ensuring that slenderness ratio remains below this threshold, it is possible to apply the Euler critical load in the assessments. This not only simplifies the analysis but also maintains accuracy and reliability in predicting the structural behavior under various loading conditions. This methodical and detailed approach reinforces the validity of the assumptions and enhances the overall robustness of the structural analysis. Therefore:

$$\lambda_0 = \frac{l_0}{\rho} \tag{8.29}$$

Where:

- λ_0 represents the slenderness of the structure
- l_0 It represents the effective length of free deflection of the beam, a value that is dependent on the examined constraint coefficient. This effective length is crucial in structural analysis as it directly influences the stability and buckling behavior of the beam under axial loads. The constraint coefficient, which varies based on the type of supports and boundary conditions applied to the beam, determines how freely the beam can deflect or rotate at its ends. Therefore:

$$l_0 = \mu l \tag{8.30}$$

  Where μ is the constraint coefficient and l the length of the beam.

In the comprehensive analysis, the assumption of a cantilever beam configuration has been adopted, characterized by one end being fixed and the other end free. This beam is subjected to compression by the reaction force generated by the thrust and experiences a shear force that varies based on the maneuverability of the missile. Within this

context, the constraint coefficient, which plays a pivotal role in determining the beam's stability, is assigned a value of 2. To ensure a thorough and accurate design, the subsequent program executes three simultaneous sizing calculations. Each of these calculations determines a specific thickness value, considering different loading and boundary conditions. These thickness values are then consolidated into a single vector, allowing for a comprehensive comparison. The program evaluates these values and identifies the maximum thickness among them. This maximum value is crucial as it represents the final thickness of our cylindrical structure, ensuring that it meets all stability and performance requirements under the given loading conditions. By employing this detailed and methodical approach, the structural integrity of the missile's cylindrical shell is optimized for both strength and durability, thereby enhancing overall performance and reliability in various operational scenarios. For the sizing of the outer shell, it must have the following dimensions:

- Outer Diameter $D = 80.6mm$
- Length $l = 1.4m$
- Arm Length $b = 0.7360m$

These dimensions are determined by operational considerations, such as the length of the grain for the first and second stages, and the size of the warhead. It is important to note that this structure is responsible for housing all components of the Sagittarius missile and protecting them from the aerodynamic forces encountered during operation. The outer diameter must be sufficient to accommodate the internal components while maintaining structural integrity. The length is dictated by the need to contain the stages of propulsion and the warhead effectively. The arm length, or the distance from the central axis to the outer shell, ensures the stability and balance of the missile during flight. In essence, the outer shell is not just a protective casing but a critical structural element that ensures the overall functionality and effectiveness of the missile by providing the necessary support and protection for all internal components against external forces.

Compression Sizing

In assessing the compression behavior, it's possible to conceptualize a thin-walled tube as consisting of interconnected 1D elements along its circumference [7]. This approach lets to treat the structural response as influenced by individual elements, even under axially symmetric loading conditions where contributions can be segregated for primary structural elements (as known from the principle of superposition of effects). By considering each element's behavior, it's possible to effectively simplify the overall analysis to a one-dimensional problem (refers to figure 8.20), assuming an elastic foundation. This perspective acknowledges that the collective response of the tube can be understood through the characterization of these interconnected elements.
By enforcing radial equilibrium across a unit axial length:

$$KuRd\theta = \frac{2\sigma_c s d\theta}{2} \tag{8.31}$$

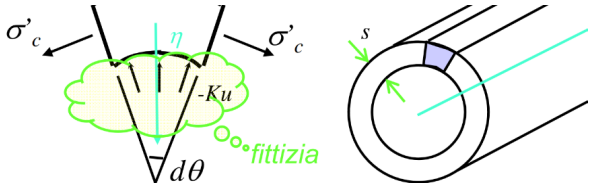


Figure 8.20 pipe segment

Where the circumferential tension as a function of u is equal to:

$$\sigma_c = E\varepsilon_c = E\frac{2\pi\,(R-u)-2\pi R}{2\pi R} = -\frac{E}{R}u \tag{8.32}$$

Substituting this value it's possible to determine the value of K, now considered per unit area:

$$K = \frac{Es}{R^2} \rightarrow k = 2\pi RK = 2\pi E\frac{s}{R} \tag{8.33}$$

This represents the stiffness of the element. Consequently, the outer shell, when subjected to axisymmetric loading, can be analyzed as if it were a beam resting on an elastic foundation, as shown in figure 8.21. This approach simplifies the analysis of the structure's behavior under such loading conditions.

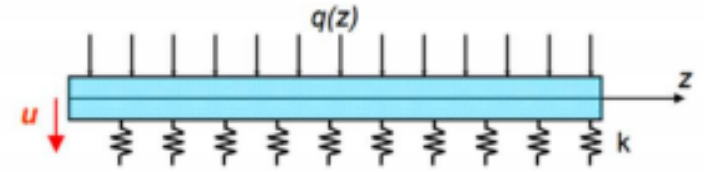


Figure 8.21 elastic Behavior

It is possible to study the behavior of a beam segment on an elastic foundation subjected to axial load S. It's possible to refer to figure 8.22

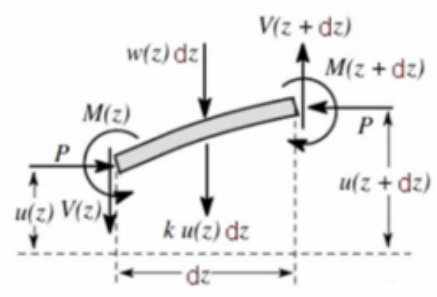


Figure 8.22 1D beam element

$$\begin{cases} \frac{dV}{dz} = w + ku \\ \frac{dM}{dz} = P\frac{du}{dz} + V \\ M = -E'I\frac{d^2u}{dz^2} \end{cases} \tag{8.34}$$

Where, for the sake of simplicity,it's possible to express:

$$E' = \frac{E}{1-\nu^2} \tag{8.35}$$

Where, with ν indicated the Poisson's Ratio equal to 0.3 for isotropic material.
With this premise established, let us assume the absence of any distributed load, so $w = 0$, therefore:

$$\frac{d^4u}{dz^4} + \frac{P}{E'I}\frac{d^2u}{dz^2} + \frac{ku}{E'I} = 0 \tag{8.36}$$

The solution, expressed as $u = Ae^{az}$, is derived by identifying the roots of the algebraic equation:

$$a^4 + \frac{P}{E'I}a^2 + \frac{k}{E'I} = 0 \rightarrow a^2 = \frac{1}{2}\left(-\frac{P}{E'I} \pm \sqrt{\left(\frac{P}{E'I}\right)^2 - \frac{4k}{E'I}}\right) \tag{8.37}$$

In the derived solution, the exponential decay term vanishes when the preceding root becomes zero; therefore:

$$P_{cr} = P_0 = \sqrt{4kE'I} \tag{8.38}$$

Moreover, it is possible to evaluate the critical axial stress:

$$\sigma_{cr} = \frac{P_0}{2\pi Rs} = \frac{E}{\sqrt{3\,(1-\nu^2)}}\left(\frac{s}{R}\right) \tag{8.39}$$

Formulate explicitly with respect to the thickness:

$$s = \sigma_{cr} R E^{-1}\sqrt{3(1-\nu^2)} \tag{8.40}$$

A formula has been derived to evaluate the thickness, where the only unknown, $P_0 = P_{cr}$, varies with the thickness.

$$P_{cr} = \frac{\pi^2 EI}{(l_0 MS)^2} \tag{8.41}$$

Therefore, it is possible to numerically assess the thickness to convergence, evaluating the Eulerian critical load corresponding to the first instability load for the treatment undertaken thus far, for each thickness value. Throughout each iteration, both the thickness and its corresponding Eulerian load are calculated, continuing until the difference between successive iterations is on the order of 10^{-5}. In the following analysis, we have taken into account a safety margin of $\sqrt{5}$.
Upon outlining the reasoning for the sizing, the following fundamental values are acknowledged:

- $F_c = 3626N$, which represents the compressive force that the outer shell must withstand, symbolizes the propulsive thrust of the missile;
- $\lambda = 100$, which represents the point of maximum slenderness, beyond which it becomes crucial to consider the instability phenomenon described by the Euler-Johnson curves, as previously mentioned.

Script 8.6 MATLAB code to outer shell: Compression Sizing

```
%% Compression Sizing
```

```
sigma_cr = 70 * 10^6;

for i = 1:1000000
    v = 0.3;
    tc = sigma_cr*(sqrt(3*(1-v^2)))*R/E;
    r = R - tc;
    d = 2*r;
    A = 0.25*pi*(D^2-(D-2*tc)^2);

    Ip = pi/32 *(D^4-(D-2*tc)^4);
    I = 0.5*Ip;

    ro = sqrt(I/A);
    P_E = (pi^2*E*I)/((2*h)^2*4);

    sigma_cr = P_E/A;
end
```

The following MATLAB script exemplifies the concepts discussed, assuming an initial condition with a $\sigma_{cr} = 70MPa$. The following analysis yields a thickness measurement of $s = 0.13mm$; evidently, this factor is not determinative in the discussion.
An additional insightful result stemming from the preceding analysis is the determination of the ultimate $\sigma_{cr} = 70.380MPa$. This finding not only aids in assessing subsequent behaviors but also serves as a significant indicator of the material under consideration. It is notable that this value remains closely aligned with the initial value utilized.

Shear Sizing

To assess the shear behavior, the formula developed by Jourasky [7] is used:

$$\tau = -\frac{TS}{Is} \tag{8.42}$$

Here, T represents the applied shear force, and S is the static moment of the section, defined by a vector of anomalies θ across the entire section, calculated as:

$$S = -R_{med}^2 s \sin\theta \tag{8.43}$$

Where:

$$R_{med} = \frac{R+r}{2} \tag{8.44}$$

Furthermore, with the Henry Von Mises's criterion it's possible to write:

$$\tau_{cr} = \frac{1}{\sqrt{3}}\sigma_{cr} \tag{8.45}$$

Thanks to considering the PSE, that allow to analyze contributions separately, thereby focusing solely on the shear stress as a result; in fact:

$$\sigma_{eq}^2 = 3\tau_{cr}^2 \rightarrow \tau = \frac{1}{\sqrt{3}}\sigma \tag{8.46}$$

In this scenario, a while loop within the MATLAB environment has been created. Its purpose is to iteratively adjust the thickness until the maximum shear stress falls below the maximum threshold that the structure can withstand; so until $\tau_{max} < \tau_{cr}$. Where the τ_{max} is the shear stress. The maximum shear stress occurs when the static moment is maximized, associated with a 90 degree anomaly.

Script 8.7 MATLAB code to outer shell: Shear Sizing

```
%% Shear Sizing
tt = tc;
while true
    r = R - tt;
    d = 2*r;
    A = 0.25*pi*(D^2-(D-2*tt)^2);
    Ip = pi/32 *(D^4-(D-2*tt)^4);
    I = 0.5*Ip;
    T = n*Wt;
    theta = linspace(0,2*pi, 100);
    Rmed = (R+r)/2;
    S = -(R^2*tt*sin(theta));
    tao = - ((T*S)/(I*tt));
    tao_max = max(tao);
    tao_cr = 1/sqrt(3) *sigma_cr;
    %stop condition
    if tao_max < tao_cr
       break ;
    end
     tt = tt+0.0000001;
end
```

The initial condition is set to the thickness determined in the compression design phase, as evaluating a greater thickness would be unnecessary. Furthermore, the shear acts at its shear center that coincides with the centroid due to the section's symmetry, thereby simplifying the analysis by negating any torsional moments.
Regarding the shear force, it is assumed to be $T = nW_t$ according to project mission, where:

- W_t represents the total structure's weight. Thanks to final CAD data, it is approximately $W_t = 12$ Kg;
- n represents the maneuverability of the missile and must be 3 times the target's one. For the mission that's established, where the target is a drone, with a maneuverability $n_{target} = 2$; so the $n = 3 \cdot n_{target} = 6$

The result is an estimated force $T = 705.6$N. The sizing process determines a thickness of $s = 0.15$ mm. Furthermore, as previously articulated by the Von Mises criterion, the critical shear stress is known $\tau_{cr} = 40$ MPa. This concludes the shear sizing analysis, and it is evident that this is not the critical factor for the problem under consideration.

Bending Moment Sizing

For the assessment of Flexural Moment behavior, the Navier formula for straight bending has been used [7]:

$$\sigma = \frac{M_f}{I} \cdot x \tag{8.47}$$

Where, x symbolizes the distance from the neutral axis, perpendicular to the axis of loading. It is worth noting that the maximal distance aligns precisely with the radius. Furthermore, the value of M_f is defined as:

$$M_f = T \cdot b \tag{8.48}$$

Hence, in light of the discussion, the stress σ_{max} equates to:

$$\sigma_{max} = \frac{M_f}{w_f} = \frac{M_f}{\frac{\pi\left(D^4 - d^4\right)}{32D}} \cdot MS \tag{8.49}$$

The parameter w_f represents the flexural resistance modulus; it is worth noting that also in this scenario, a stability margin $MS = \sqrt{5}$ should be duly considered. In assessing the thickness, a methodology akin to that utilized in shear analysis was employed.

A while loop was established to iteratively evaluate the σ_{max} value until it descends below the critical threshold, accounting for the stability margin. The thickness is systematically increased with each iteration.

Script 8.8 MATLAB code to outer shell: bending Moment Sizing

```
%% Bending Moment Sizing
tm = tt;
while true
    r = R - tm;
    d = 2*r;
    A = 0.25*pi*(D^2-(D-2*tm)^2);
    Ip = pi/32 *(D^4-(D-2*tm)^4);
    I = 0.5*Ip;
    ro = sqrt(I/A);
    M_f = T * b;

    wf = pi * (D^4-d^4)/(32*D)/MS;
    sigma_M_max = M_f/wf ;
    if sigma_M_max < sigma_cr
        break;
    end
       tm = tm+0.0001;
end
```

It is worth highlighting that an initial condition equivalent to the outcome of the prior sizing calculation has been opted for. The thickness obtained from the MATLAB code is 1.5 mm, thus determining the dimensions for the entire structure.

Outer Shell sizing final step

Upon the conclusion of the three-dimensional analyses, the derived thickness values are consolidated into a unified vector, with the maximum value denoting the thickness of the aluminum shell, equals to $t = 1.5$ mm:

Script 8.9 MATLAB code true thickness

```
t = [tc tt tm];
t = max(t);
```

Subsequently, an evaluation of the section's geometric properties and the accompanying structural stresses ensues:

Script 8.10 MATLAB code Geometry Characteristics

```
r = R - t;
d = 2*r;
A = 0.25*pi*(D^2-(D-2*t)^2);
Ip = pi/32 *(D^4-(D-2*t)^4);
I = 0.5*Ip;
ro = sqrt(I/A);
sigma_N = F_c/A;
Rmed = (R+r)/2;
theta = linspace(0,2*pi, 100);
S = -(R^2*t*sin(theta));
tao = - ((T*S)/(I*t));
tao_max = max(tao);
wf = pi * (D^4-d^4)/(32*D);
sigma_M_max = M_f/wf ;
```

This sets the stage for a detailed analysis of fatigue behavior using the Von Mises criterion, highlighting the prevalence of stresses generated by bending moments.

Script 8.11 MATLAB code Von Mises Stress

```
sigma_VM = sqrt(sigma_M_max^2+sigma_N^2+3*tao_max^2);
lamda = 2*h/ro;                          %Beam slenderness
```

As deduced from the results, the σ_{eq} equivalent stress closely aligns with that arising from the imposed bending moment; therefore:

$$\sigma_{eq} = \sqrt{\sigma_N^2 + 3\tau_S^2 + \sigma_{max,BM}^2} = 18MPa << 100MPa \tag{8.50}$$

The assessment of the equivalent stress not only elucidates the primary factors at play but also confirms the structure's resistance to fatigue, with stress levels remaining below 100 MPa, so this behavior can be deemed negligible.

Additionally, in order to avoid oversimplification errors, the slenderness ratio was reevaluated to verify its compliance with the prescribed limit; the resulting value of $\lambda = 51$ confirms its adherence to the upper threshold. The following depiction showcases the structure under consideration.
The visual representation, shown in figure 8.23, is rendered within the FUSION environment.

Figure 8.23 Outer Shell Sagittarius

FEM Analysis

In this section focused on the Finite Element Analysis (FEM) results examination, meticulous attention is dedicated to ensure the coherence and consistency with previously acquired data.
The construction of the analyzed model prioritizes the use of original Computer-Aided Design (CAD) specifications of the external components sourced from a proprietary project. This selection is intended to specifically highlight the behavior and features of the structure previously studied.

Additionally, a comprehensive illustration of the structure is provided, encompassing essential elements such as the nose, separation ring, and wings. These components will be elaborated upon in subsequent sections to discuss their individual functions and material compositions.

Material considerations involve a thoughtful analysis of each part's composition:

- The nose, constituted by radome [11], is crafted from a specialized pyroceramic material with a precise thickness of 1.5 mm.

  The radome, short for radar dome, acts as a protective shell enclosing and safeguarding radar equipment, antennas, or electronic sensors, allowing them to operate efficiently. Typically composed of materials transparent to radio waves, such as specialized plastics or composite materials, radomes maintain the performance of the internal devices. Their primary function is to shield electronic components from adverse environmental conditions like harsh weather, while also minimizing signal interference and preserving the accuracy and effectiveness of radar systems. For the pyroceramic characteristics refers to table 8.3;

Young Modulus E	120 GPa
Poisson Modulus ν	0,3
Compression limit stress σ_N	235 MPa
Shear limit stress τ_{max}	136 MPa

Table 8.3 Pyroceramic characteristics

- The separation ring is composed of sturdy AISI 4340 [11] with a matching thickness of 1.5 mm. For the steel characteristics refers to table 8.4;

Young Modulus E	175 GPa
Poisson Modulus ν	0,3
Compression limit stress σ_N	240 MPa
Shear limit stress τ_{max}	156 MPa
Expansion Coefficient a	$6,6 \cdot 10^{-6} C^{-1}$
Conducivity k	$17 W/mK$
Specific Heat c_p	$38,64 J/(K g K)$

Table 8.4 AISI 4340 characteristics

- An interim assumption designates the wings to be made of aluminum 7075-T651 [11] with an original thickness of 4.8 mm; For the alloy characteristics refers to table 8.5

Young Modulus E	73 GPa
Poisson Modulus ν	0,33
Compression limit stress σ_N	67 MPa
Shear limit stress τ_{max}	43 MPa
Expansion Coefficient a	$1,23 \cdot 10^{-5} C^{-1}$
Conducivity k	$200 W/mK$
Specific Heat c_p	$77,28 J/(K g K)$

Table 8.5 Alloy 7075-T6 characteristics

With meticulous attention to detail and a systematic evaluation of material properties and structural characteristics, the presented Finite Element Model offers deeper insights and analytical depth into the design and performance of the scrutinized structure.
In figure 8.24, Sagittarius has been discretized using the finite element method. To accentuate the distinct components, color differentiation has been applied to clearly delineate the considered elements.

Specifically, the pyroceramic material is represented in red, AISI 4340 in blue, and the alloy in brown.

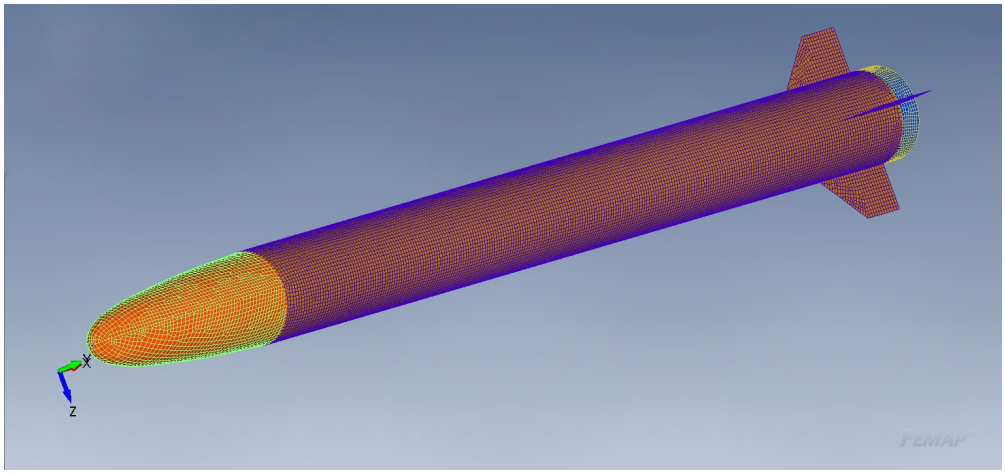


Figure 8.24 Sagittarius Discretized

Illustrating the discretized missile and the constituent materials allows to progress to the problem analysis. In order to simulate a plausible flight behavior of the missile through a static analysis, it has been modeled as a classical example of a simply supported beam. Specifically, two of the four steering fins have been designated as the supports upon which Sagittarius is positioned. This configuration enables the assessment of the maximum structural deformation and any potential fatigue phenomena. The central objective of the finite element problem is to determine the structural integrity under the applied loads. Following the establishment of constraint conditions, the forces are considered. As previously referenced during the dimensional analysis:

- A shear force of 705.6 N, exerted at the centroid;
- A compressive force, symbolizing the propulsive thrust, equaling 3626N, applied along the longitudinal axis of symmetry in close proximity to the ring.

The ensuing graphical representation depicts the outcome achieved in this analysis, shedding light on the structural response under the specified loading conditions:

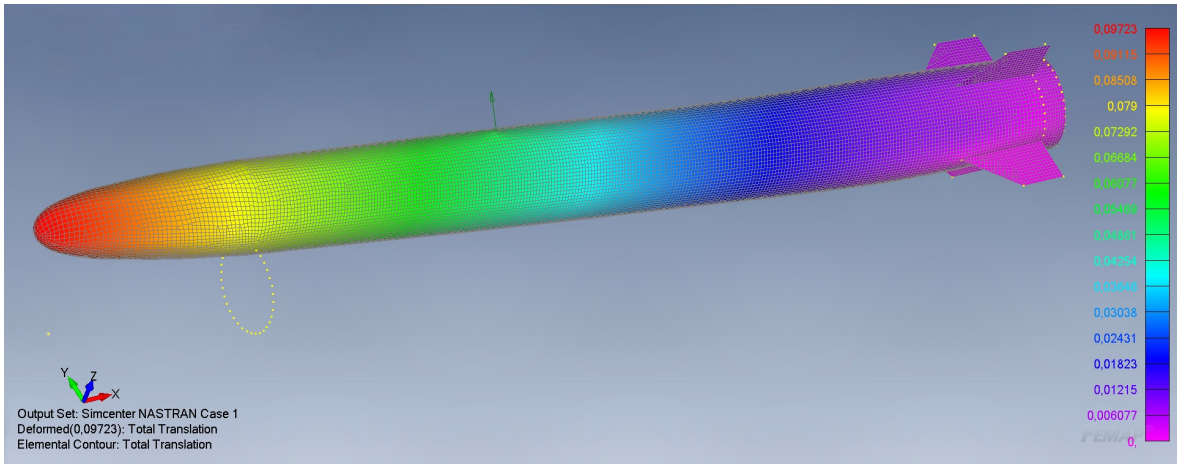


Figure 8.25 Outer Shell FEM

The area of maximum deformation in close proximity to the nose corresponds to the location where the bending moment reaches its peak, consequently leading to the highest stress levels. The analysis results emphasize the structural integrity of the outer shell designed for the mission, given the maximal $0,097mm$ deformation, which is deemed acceptable considering its brief duration. In terms of the Von Mises stress analysis findings, they exhibit consistency with the initial project assessments of 18 MPa, confirming both the absence of potential fatigue concerns and the structural adequacy to fulfill the mission's stability requirements. The results are shown in figure 8.25

8.5.2 Case grain

According to the propulsion section specifications, the sizing of the sustain grain case involves meticulous consideration of a notable 7 bar pressure and an elevated temperature of 3500K within the combustion chamber. This thermal extremity significantly surpasses the thermal tolerance of any natural material, signifying a critical challenge in maintaining mechanical resilience. To address this thermal predicament, a dedicated thermal liner has been meticulously engineered to effectively reduce the thermal gradient impacting the integrity of the steel structure. In the preliminary phases, it is postulated that the thermal liner implementation successfully moderates the inner surface temperature of the case grain to a manageable 800K, thereby enabling a streamlined sizing analysis founded on a discernible temperature differential of approximately $\Delta T = 600$ between the opposing surfaces of the case structure.
In extending the thermal scrutiny to the nozzle configuration, a comprehensive strategy integrating a thermal liner and a focused heat dissipation mechanism, particularly within the vital throat region, is being meticulously devised.
The calculation of the grain diameter, thus necessitating the determination of the case dimensions, was meticulously conducted during the propulsion phase, as elaborated upon in the Grain Sizing subsection. This rigorous analysis culminated in a definitive diameter measurement of 64 mm.
Furthermore, the subsequent stages of the structural design warrant stringent contemplation and application, ensuring that the overarching casing can effectively endure the substantial 7 bar pressure sustained within the combustion confines, while effectively navigating an intricate thermal gradient spanning $\Delta T = 600$. As alluded to in the foundational discourse, the ergonomically selected AISI 4340 steel alloy materialization stands as the optimal choice, strategically offsetting material limitations observed in aluminum, which exhibits mechanical inconsistencies beyond the critical 450 K threshold, thus affirming AISI 4340's operational resilience within the stipulated operational spectrum.

Sizing

In the process of sizing the Case grain, the Barlow equation [1] has been employed for calculations. Currently, the focus lies on the combustion chamber shell, as previously indicated in the preliminary phase, where due consideration must be given to any thermal stresses arising from the defined design conditions. Therefore, for the purpose of conducting this dimensioning, the following equation will be utilized:

$$\frac{\sigma_y}{MS} = \frac{p \cdot r_c}{s_c} + E \cdot \alpha \Delta T \rightarrow s_c = p \cdot r_c \left(\frac{\sigma_y}{MS} - E \cdot \alpha \Delta T \right) \tag{8.51}$$

Where:

- σ_y indicates the yield stress of the material. For the specific application at hand, a value of 350MPa has been chosen. This selection aligns with the targeted yield stress criteria pertinent to the material properties under consideration, ensuring a tailored approach to assessing the structural resilience of the component in the given operating conditions;

- MS denotes the safety margin, for which a value of $MS = \sqrt{5}$ has been assumed in this instance. This choice reflects the designated safety factor considered in the analysis;
- p represents the internal pressure to which the structure is subjected. For design hypothesis purposes, a value of 7 bar has been assumed in the combustion chamber. This specified pressure condition serves as a foundational parameter in the structural assessment, guiding the dimensioning process and ensuring the component's integrity under operational scenarios;
- s_c represents the thickness of the structure under examination;
- The variable E represents the Young's modulus of the material. This intrinsic material property signifies its stiffness and elasticity, playing a vital role in calculating structural deformations and stress distributions. It's assumed a value of $175GPa$, as reported in 8.4;
- α denotes the material's coefficient of thermal expansion. This crucial parameter influences the material's dimensional changes in response to temperature variations. The specific value for alpha can be found in 8.4, providing essential data for assessing the material's thermal behavior and structural stability under varying operating conditions;
- r_c denotes the radius of the grain and hence of the case; this specific value was established during the propulsion phase, measuring at 32 mm. The accurate determination of this parameter is crucial for the precise engineering and operational considerations related to the propulsion system design;
- ΔT represents the thermal gradient to which the structure is subjected; this value was established during the preliminary design phase and is set at $\Delta T = 600$. Accurately determining this thermal gradient parameter is critical for assessing the thermal response and stability of the structure under specified operating conditions.

This equation incorporates the effects of both internal pressure and thermal stresses on the tube, providing a comprehensive approach to ensuring structural integrity. By accounting for these factors, the design process aims to optimize the performance and reliability of the Case grain component within the broader system configuration.
In the formula, the dividing factor of 2 has not been taken into account, as the focus lies on assessing the radial pressure load rather than the axial load. This deliberate omission of the division by 2 reflects the specific intent to analyze the radial pressure on the shell, underscoring the targeted nature of the stress evaluation approach. The distinction in consideration aligns with the nuanced mechanics of radial versus axial pressure effects on the structural integrity of the case under examination. Illustrating the previously stated information in the Matlab environment, code strings related to parameter definitions are omitted for the purpose of elegance and conciseness.

Script 8.12 MATLAB code Case thickness

```
delta_Ta = 600;
sigma_Ta = Ec*alfaa*delta_Ta;
```

```
t_c = (p*r_c)/(sigma_ya/1.5-sigma_Ta);
```

Utilizing the equation provided below,thanks to the MATLAB code, has yielded a thickness of 1.8 mm, thereby defining the specific design of the case as shown in figure 8.26:

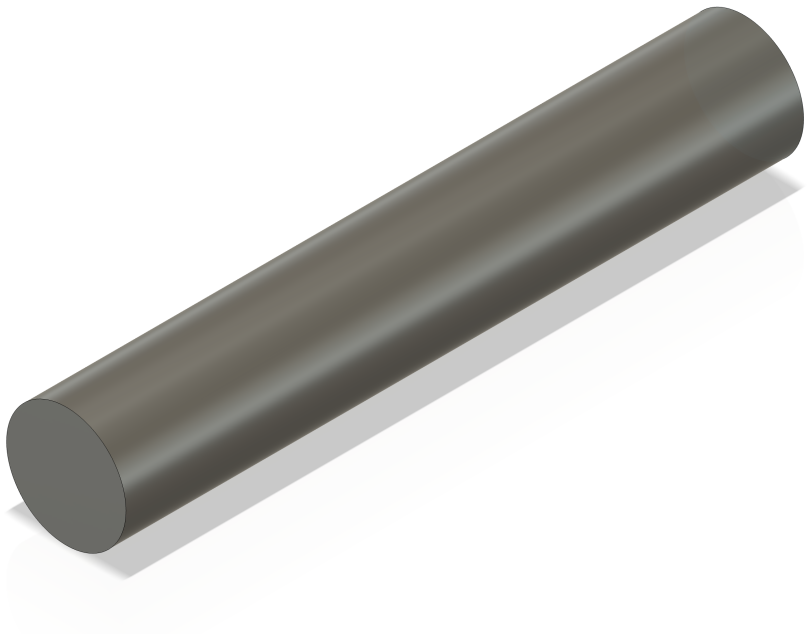


Figure 8.26 Case Grain

FEM

Within the context of the Outer Shell discussion, a thorough finite element analysis was carried out for this structure using the FEMAP environment. Two distinct types of analyses were undertaken: The initial analysis focused on evaluating the case's ability to withstand the pressure resulting from the combustion process within the chamber. Subsequently, a thermal analysis was conducted to examine the temperature variation across the material thickness. This comprehensive approach provides valuable insights into the structural integrity and thermal behavior of the component under operational conditions. Below, the finite element discretized structure is presented, prior to the application of forces.

The discretized model depicted in 8.27 serves as a crucial representation for analyzing the structural behavior under operational conditions. The subsequent analysis focuses on the case's ability to endure the pressure resulting from combustion, highlighting its pivotal role in the system.

It is noteworthy that the thermal liner plays a significant role in mitigating thermal stresses, underscoring the importance of its design and material properties in ensuring the overall integrity of the structure.

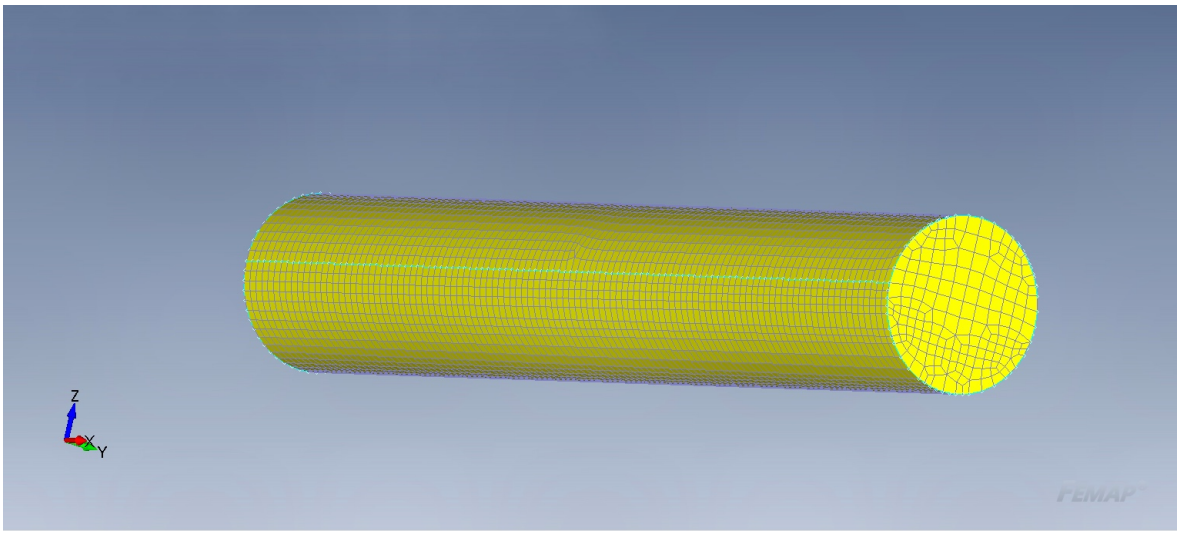


Figure 8.27 Case discretized

The results lead to a surface translation of approximately 0.1 mm, a finding very similar to that obtained in the Outer Shell analysis. Consequently, as asserted in the subsection concerning the respective FEM, this deformation is considered acceptable, thus ensuring that the sizing conducted is coherent with the specific problem and the parameters encountered thus far. The above observations are highlighted in 8.28.
Regarding the thermal gradient effect, the finite element model has provided insights into the thickness variation across the material, highlighting the structural response to temperature changes. This detailed analysis is visually represented in Figure 8.29.

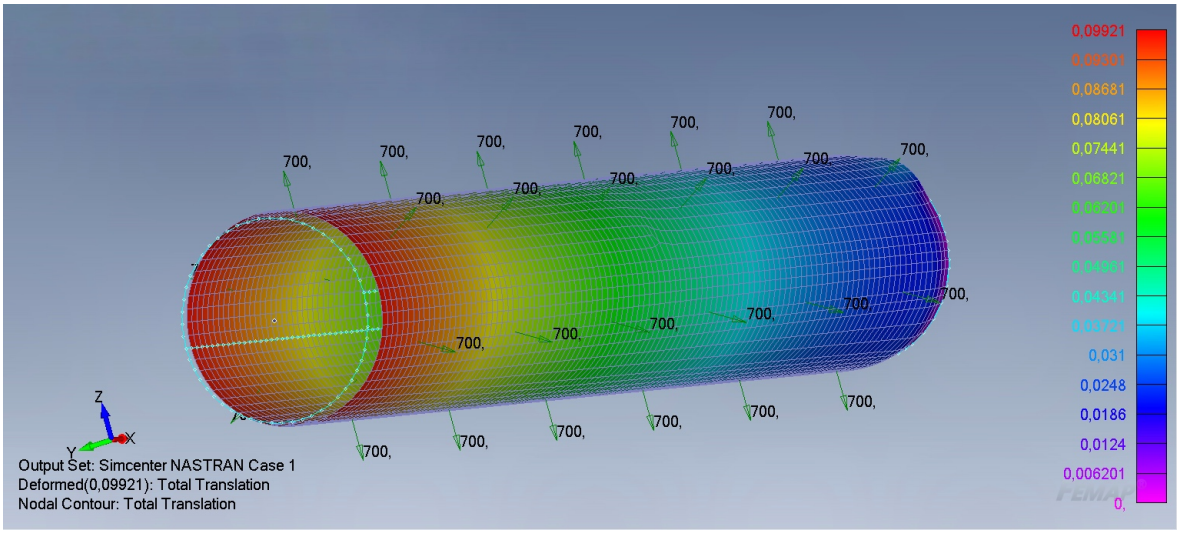


Figure 8.28 Case FEM

It is noted that a temperature of $255K$ has been assigned to the outer face of the case, in contact with the Outer Shell. This temperature value aligns with the adiabatic wall temperature of air.

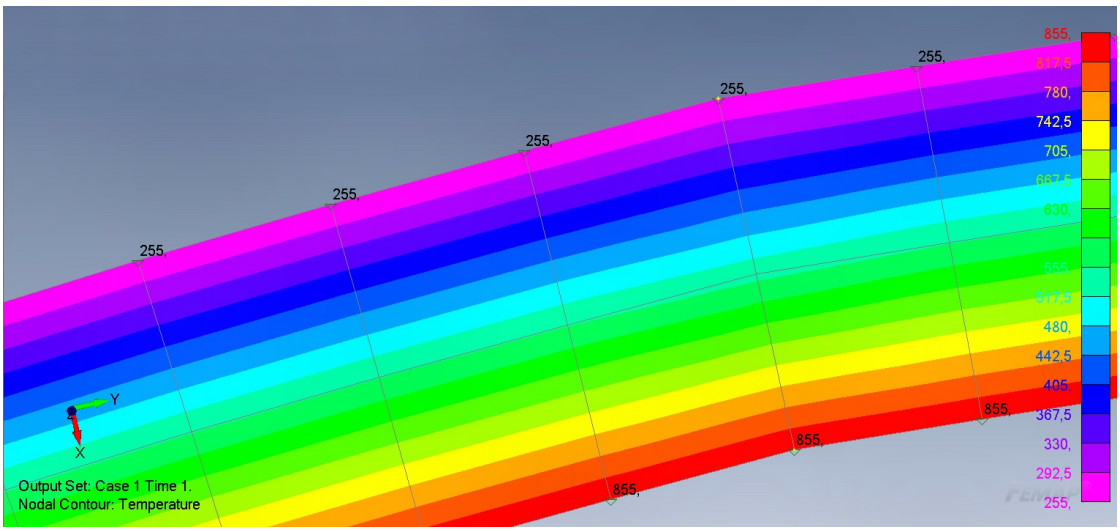


Figure 8.29 Case FEM

This selection was based on the exclusion of aluminum's mechanical behavior in the analysis, as it tends to degrade at elevated temperatures. Therefore, the finite element analysis overlooks this component, although its minimal impact has been considered in the thermal liner sizing to enhance precision in the analysis. Additionally, it is essential to consider that there is a 1 mm space between the Outer Shell and the Case through which air circulates; this space is utilized for the placement of wires related to the electronic components. Consequently, the previous statement regarding the removal of the aluminum layer and, hence, the gap where the wires are located is further justified, as considering the cooling process for these two layers could lead to the overheating of the wires and a subsequent loss of efficiency in conducting electrical current. In conclusion, it is assumed that the Alloy, Gap, as well as the outer wall of the casing are at the adiabatic wall temperature.[8] This temperature is evaluates:

$$T_{aw} = T_{air}\left(1 + \frac{r(\gamma_{air} - 1)}{2}M^2\right) = 255K \tag{8.52}$$

Where:

- r expresses the recovery factor; it is determined based on the turbulent nature of the flow experienced by the Sagittarius missile during its mission. Given the Prandtl number for air, which is approximately $Pr = 0.73$, the recovery factor in turbulent flow is calculated as:

  $$r = \sqrt[3]{Pr} \tag{8.53}$$

  Therefore, for the Sagittarius missile, the recovery factor (r) is approximately 0.90;

- γ_{air} is the parameter "gamma" for air refers to the specific heat ratio or adiabatic index, denoted by the symbol. It represents the ratio of specific heats, specifically

the ratio of the heat capacity at constant pressure (c_p to the heat capacity at constant volume (c_v) of a gas. It signifies the thermodynamic properties of air and is a crucial factor in determining its behavior in various processes, such as compression, expansion, and heating. For air, the typical value of γ is around 1.4, indicating the relationship between its pressure, volume, and temperature changes under adiabatic conditions;

- M represents the flight Mach number, which directly indicates the regime of flight experienced by the Sagittarius missile, and it is equal to 0.6;

- $T_{t_{air}}$, expresses or stagnation temperature, represents the temperature that air reaches when it is brought to a complete stop, or stagnation, due to the missile's movement. This temperature increase occurs because the kinetic energy of the moving air is converted into thermal energy as the air is slowed down. Essentially, $T_{t_{air}}$ reflects the temperature the air experiences at the point where it directly interacts with the missile, taking into account the effects of the missile's speed. This parameter is determined using the following formula [8]:

$$T_{t_{air}} = T_{air} + \left(\frac{(\gamma_{air} - 1)}{2} M^2 \right) \tag{8.54}$$

Where the variable T_{air}, representing the static air temperature, is simply the temperature of the air surrounding the missile as it would be measured by a stationary observer. It's the ambient temperature of the air through which the missile is traveling. Given the operational scenario of the Sagittarius missile, the static air temperature (T_{air}) is assumed to be $240K$, taking into account the altitude conditions. Therefore the stagnation temperature $T_{t_{air}} = 257K$; as expected, this value is very close to the static air temperature, given the flight regime under consideration.

8.5.3 Nozzle

In this section, the structure of the nozzle, a fundamental component of the chamber, is discussed. The nozzle is regarded as a highly critical structure, especially considering the elevated temperatures it faces in operational conditions. Its structure consists of three primary components:

- Heat Sink[8]: This component plays a crucial role in absorbing heat, it removes heat from the nozzle throat, dynamically cooling the affected area which experiences the highest levels of stress. The design and material selection for the Heat Sink are carefully chosen to possess high thermal conductivity properties for efficient heat transfer and absorption, thereby preventing excessive heating of the critical components of the nozzle. By effectively managing heat absorption, the Heat Sink component ensures the structural integrity of the nozzle, enhancing overall performance and reliability in extreme operating conditions;

- Liner [8],this layer, comprising a specific porous graphite material discussed in subsequent sections, acts as an insulator, contributing to maintaining temperatures within the limits allowed by the materials, reducing the temperature to $800 - 850K$, aligning with the characteristics of AISI 4340, the material of the third layer..

- Steel: A layer in AISI 4340 absorbs pressure loads on the structure. Additionally, this layer encounters a thermal gradient of 600 K, reflecting the temperature difference between the adiabatic wall temperature discussed in the case scenario and the temperature resulting from the layer in contact with the liner.

Following the application of each layer, a finite element analysis highlights temperature variations across the structure thicknesses. Furthermore, examination of pressure behavior within the chamber emphasizes stresses and maximum deformations resulting from the analysis.
The components will undergo analysis in an outermost-to-innermost layer sequence.

AISI 4340 Layer

In the dimensionalization of the third layer, a similar methodology as applied in the casing was utilized, with the primary considerations revolving around the pressure induced by combustion and the substantial thermal stresses stemming from the high temperatures within the chamber. It is crucial to acknowledge the variable cross-sectional nature of the nozzle structure, implying a dynamic thickness that corresponds to the evolving sections. The chosen design approach, albeit intricate due to structural implications, led to the decision to dimension the structure relative to its critical section, particularly the area with the smallest cross-section: the throat section. The material selected, as earlier indicated, is AISI 4340, with its specifications outlined in 8.4.

The dimensionalization of this layer was based on the design parameters derived from the propulsion phase, encompassing a throat diameter of $d_{th} = 16$ mm and a pressure of $p = 7$ bar, as established during the casing dimensioning process.

The identical code seen for the casing was reintroduced for this layer, as the reasoning behind it remains consistent, as previously asserted:

Script 8.13 MATLAB Code Steel Layer Thickness

```
t_u = (p*r_u)/(sigma_ya/1.5-sigma_Ta);
```

Where a thermal gradient of $600K$ was assumed, in line with the hypotheses outlined in the introductory section of the nozzle.
Therefore, a thickness of $t_{steel} = 1,5$ mm is obtained, thereby defining the specific design of the case as follows:

Figure 8.30 Steel Layer

FEM

The upcoming discussion will encompass the Finite Element Method (FEM) analyses of the combustion chamber, focusing solely on the Steel Layer responsible for supporting the structure against the pressure loads resulting from the combustion of the grain, as mentioned earlier. Subsequently, the CAD-designed structure in question will be presented in figure 8.31.

In analyzing this geometry within the FEMAP environment, the results obtained during the propulsion phase were taken into account. Additionally, other relevant lengths considered for analysis, apart from the dimensions used in the Case Grain design, include:

- Throat Diameter $d_{th} = d_{cr} = 16$ mm
- Convergent Length $l_{conv} = 24$ mm
- Divergent Length $l_{div} = 58,2$ mm
- Throat Length $l_{th} = 4$mm
- Exit Diameter $d_{exit} = 22$ mm

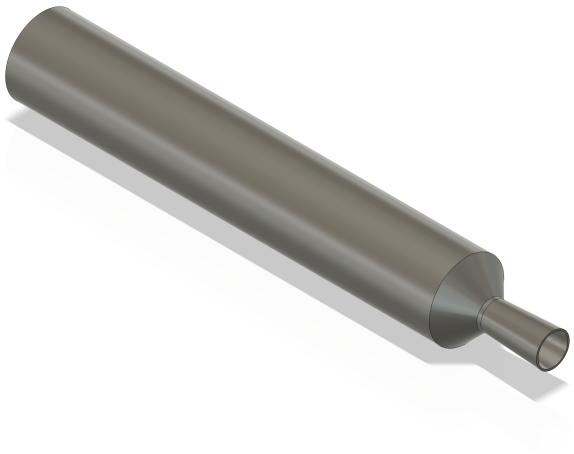


Figure 8.31 Chamber Steel Layer

It was decided not to analyze the nozzle in isolation but rather as an integral part of the chamber complex, which includes the Grain Casing. This decision was made based on the understanding that analyzing the nozzle independently is not particularly insightful as the pressures and temperatures it experiences are originated from processes within the casing. The entirety of the structure, as previously stated, is composed entirely of AISI 4340. Furthermore, the geometry was conceptualized in the FEMAP environment not through a rigid logic, but by considering the plate element, which accounts for node rotations and enables the evaluation of critical information such as Von Mises stresses. The discretized structure is presented in figure 8.27 prior to the application of loads and boundary conditions:

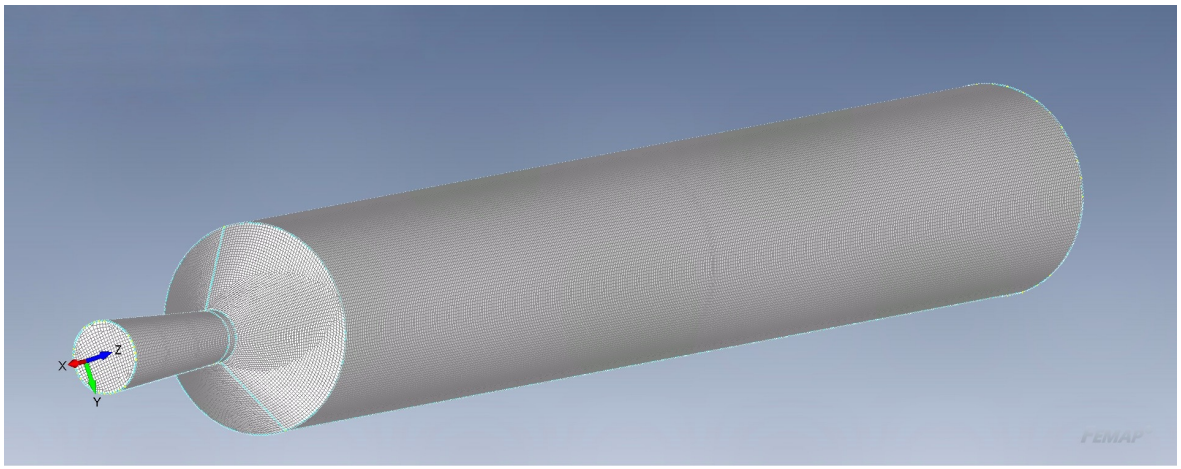


Figure 8.32 Chamber discretized

Regarding the chosen analyses for this structure, two types of FEM Analyses were idealized, focusing on the pressure loads acting on the chamber while differing in their boundary conditions:

- Double-Fixed: a logic similar to that of a "fixed-fixed beam" was applied, fixing the two end sections of the chamber. This analysis allows for the assessment of

stresses induced by the acting pressure.

- Single-Fixed: Following a logic akin to a "fixed-free beam," the free evolution of chamber deformation was envisioned. In this scenario, the end of the casing was fixed to demonstrate the translation of points along the chamber and the resultant deformation. This fixation was established due to the previously highlighted casing deformation and the Steel Layer being the most critically exposed component within the chamber complex.

Commencing with the illustration of the Double-Fixed analysis, as previously stated, the two ends of the chamber were fixed to assess the Von Mises equivalent stresses. The contour of the chamber is displayed in figure 8.33, through which the pertinent results will be demonstrated.

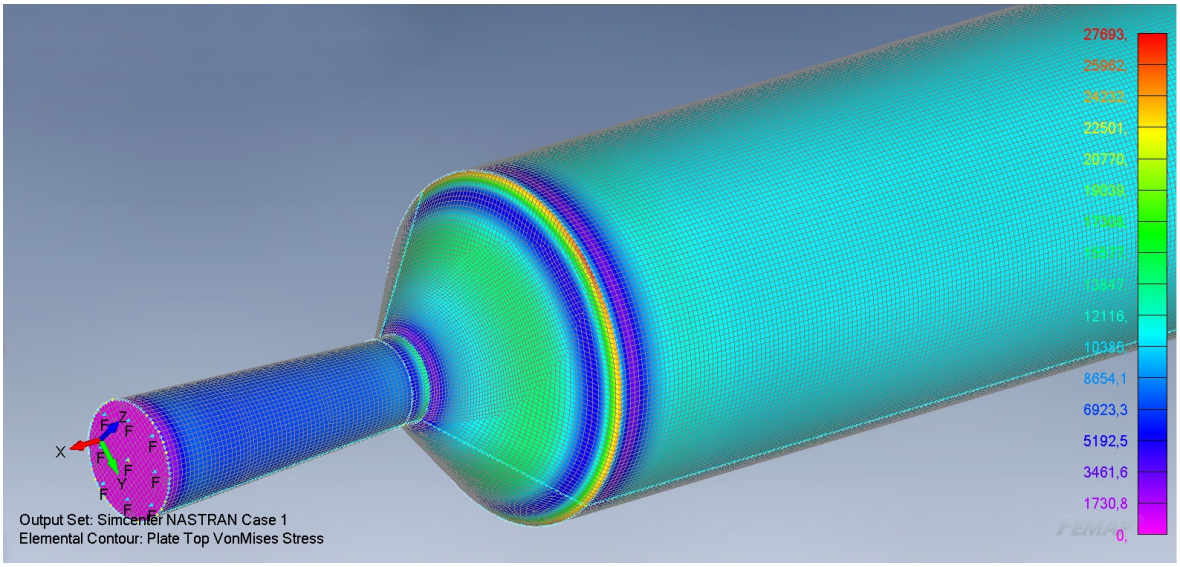


Figure 8.33 Double-Fixed Analysis

In discussing the obtained results, any imperfections near the end of the divergent section are deemed inconsequential. These minor irregularities stem from mesh errors resulting from surface overlap, a consequence of utilizing plate elements for the static analyses. This methodology introduces minor geometric discrepancies when surfaces are merged using the Solid-add command (however, deemed negligible for our discussions, as the study conducted primarily focuses on the vicinity of the throat section).
The Contour representation highlights the area most influenced by pressure loads, primarily coinciding with the nozzle throat and its neighboring sections (once again, the transitions between the explanatory convergent surfaces and the throat are considered negligible). Within this immediate surrounding, a discernible pattern emerges: the average stress value is approximately 10 MPa, while the most critical section contends with a pressure stress of 27 MPa, but this section represents the junction between the nozzle and the chamber; consequently, the primary stress is associated with the type of connection employed, specifically through FEMAP solid add comand. Had a weld been utilized between the two components via the connect command, this stress would have aligned with the others. Therefore, the throat section should be considered the most critical, where a stress level of approximately $13 - 14$ MPa is observed.

These values underscore the restricted impact of pressure on the design considerations, reaffirming the significance of subsequent Finite Element Method (FEM) analyses exploring the thermal evolution throughout the structure.
Furthermore, these stress values imply the structure's resistance to hypothetical fatigue behavior, aligning seamlessly with the design assumptions outlined in the Outer Shell dimensioning section. The inclusion of this FEM analysis aims for a comprehensive discussion, highlighting the forthcoming importance of discussing point translations facilitated through the Single-Fixed approach.

In presenting the results obtained using the previous approach, the focus now shifts to the Single-Fixed approach, as noted in the preliminary phase. This method enables the analysis of the structure's deformation in response to the applied loads. The results from this FEM analysis are depicted in 8.34.

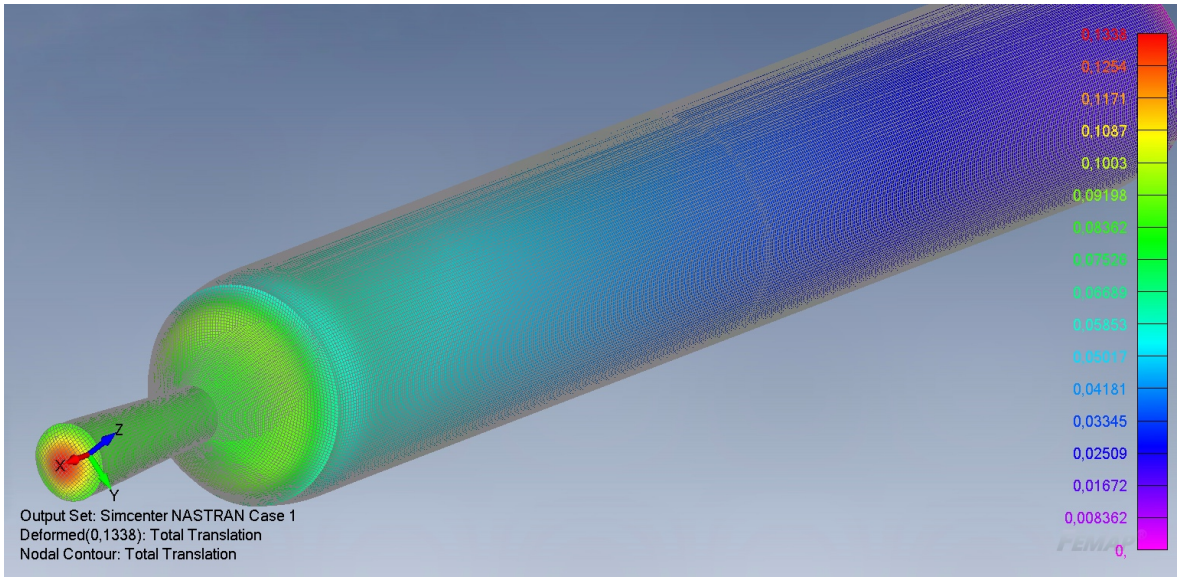


Figure 8.34 Single-Fixed Analysis

In considering the reference denoting the undistorted chamber condition marked by blue guidelines, the decision was made to apply pressure loads also nominally to the exit section of the chamber to simulate the expulsion of combusted gases. As expected, the results illustrate a more substantial deformation of the nozzle, primarily due to its heightened sensitivity within the chamber complex. A progressive translation is observed, reaching its maximum values near the nozzle's end section, as this deformation transmits through the nodes. The Contour visualization indicates that the highest deformation at the surface end is approximately 0.1 mm, a value deemed entirely acceptable within the operational context of the chamber. It is important to consider the deformation related to the closure of the convergent, as it is not structurally present. Its use, as previously stated, is solely intended to simulate the expulsion of the combustion gases. Regarding pressure loads, the structure demonstrates compliance with safety criteria. However, this assertion will be further reinforced in the subsequent three-dimensional Finite Element Method analysis of the chamber, incorporating considerations for the Heat Sink and Liner. As previously highlighted, this upcoming analysis will assess the structural safety against the thermal loads it encounters, facilitating an evaluation of temperature distribution across the various layers.

8.5.4 Thermal management System

Dealing with a solid propellant missile necessitates the implementation of a Steady-State cooling system to dissipate the heat transferred to the steel structural layer, which is responsible for absorbing dynamic loads, thus preventing the loss of its mechanical properties.
Regarding the cooling of the Grain Case, a thermal liner composed of porous graphite is employed, effectively preventing critical temperatures for the steel. In the nozzle, a combined cooling approach is utilized, featuring an initial layer of liner made from the same material used in the case, complemented by a copper mass with high thermal conductivity, named Heat Sink, designed to evenly distribute heat along its thickness. This strategy addresses the critical nature of the throat; consequently, the heat sink is thickened in proximity to this section due to its structural significance. For the remaining sections of the nozzle, a similar approach is applied, with the thickness of the heat sink gradually reduced as one moves away from the throat. Therefore, the nozzle insulating layers of the cooling system are sized starting from the nozzle throat section, which is subjected to the highest thermal stress.

Liner design

In light of the elevated temperatures generated by the combustion of grain, should steel be the sole element accountable for the heat absorption phase, it would be subject to structural problems and degradation of its properties. In alignment with the scenario scrutinized in this report, porous graphite has been selected as the material for the insulating layer.[11] The properties of this material are detailed in table 8.6.

Young Modulus E	150 GPa
Shear Modulus G	53GPa
Poisson Modulus ν	0,22
Compression limit stress σ_N	800 MPa
Shear limit stress τ_{max}	250 MPa
Expansion Coefficient a	$4 \cdot 10^{-6} C^{-1}$
Conducivity k	$0,400 W/mK$
Specific Heat c_p	$710 J/(Kg K)$
Density ρ	$1150\ Kg/m^3$

Table 8.6 Porous Graphite

The designated objective for this layer is to maintain the thermal gradient within the steel layer below the $600K$ threshold, a value discussed in the subsection on the steel layer, in order to limit the thermal stresses to which the outer layer is exposed. The second condition pertains to the temperature of the steel surface in contact with the liner, which must remain below $1400K$, beyond which AISI 4340 begins to deteriorate in its mechanical properties. It should be noted that this layer is responsible for withstanding the pressure stresses resulting from combustion. This process aims to provide preliminary results, which will later be verified and, if necessary, corrected through a finite

element analysis within the FEMAP environment, using the Thermal extension. As asserted in the subsection on the FEM case, this sizing took into account not only the steel but also the aluminum layer above it.

In the sizing of the Layer, an initial temperature of 855 K was assumed. The various layers composing the structure were considered as thermal resistances in series. This approach allows for the evaluation of heat flows and, consequently, the temperatures of the various layers. The formula for calculating the equivalent resistance[8] is introduced below:

$$R = \frac{1}{h_{air}} + \frac{1}{h_{gas}} + \frac{s_{liner}}{k_{liner}} + \frac{s_{steel}}{k_{steel}} + \frac{s_{shell}}{k_{alloy}} \tag{8.55}$$

- $h_{air} = 150 W/K \cdot m^2$, typical value of the convective heat transfer coefficient for air;
- $h_{gas} = 50000 W/K \cdot m^2$, convective heat transfer coefficient of the combusted gas, estimated using Bartz's formula:

  $$h_g = \frac{0.026}{D_t^{0.2}} \left(\frac{p_c}{c^*}\right)^{0.8} \left(\frac{D_{th}}{D}\right)^{1.8} c_p \mu_e^{0.2} \left(\frac{T_e}{< T >}\right)^{0.8-0.2w} \tag{8.56}$$

  For further clarifications regarding this formula, please refer to Sutton[8];
- $k_{liner} = 0,400 W/mK$, as reported in Table 8.6, representing the thermal conductivity of the specified graphite;
- $s_{steel} = 1,5$mm the result of the sizing of the steel layer, reported in subsection 8.5.3;
- $k_{steel} = 17 W/mK$, as reported in Table 8.4, representing the thermal conductivity of AISI 4340;
- $s_{shell} = 1,5$mm, the result of the sizing of the outer shell, reported in subsection subsection 8.5.1;
- $k_{alloy} = 200 W/mK$, as reported in table 8.5, representing the thermal conductivity of Alloy 7075.

Consequently, s_{liner} is the unknown variable in this reasoning, which will be determined in the Matlab environment through the following while loop.

Script 8.14 Matlab code Liner thickness

```
%% Liner
haria=150;
hgas=100000;
deltaw=t_c;
deltac=t;
kw=17;
ki=0.400;
kc = 200;
gamma_aria=1.4;
M_aria=0.6;
r=0.9;
```

```
T_aria=255.68;
T_t_aria=T_aria+(gamma_aria-1)/2*M_aria^2;
T_aw_aria=T_t_aria*(1+r*(gamma_aria-1)/2*M_aria^2);

Ti=3000;
N=10000;
deltaivec=linspace(0,0.030,N);
T2=10000;
T3=0;
i=0;
deltaT=T2-T3;
while deltaT >600  || T2>850
    i=i+1;
    deltai=deltaivec(i);
    R=1/(haria+1/hgas+deltaw/kw+deltai/ki+deltac/kc);
    q=(Ti-T_aw_aria)/R;
    T1=Ti-q/hgas;
    T2=T1-q*deltai/ki;
    T3=T2-q*deltaw/kw;
    T4 = T3-q*deltac/kc;
    deltaT=T2-T3;
end
t_i = deltai
```

The aforementioned assertion leads to a liner thickness of 4 mm. This dimensioning pertains to the cooling of the Grain Case, which occurs exclusively through the newly dimensioned layer.

As for the nozzle, a combined liner with a heat sink has been used, as stated in the introduction. It should be noted that this component is thickened near the throat due to the increased stress in that section because of its reduced size. Regarding the liner placed in the nozzle, a different sizing approach has been utilized, it is sized using a code that verifies that the Convective Heat Transfer Coefficient of the Porous Graphite thickness constituting the first layer is significantly smaller than the Boundary Layer Convective Heat Transfer Coefficient. After various tests, a graphite thickness (δ) of 4 mm was chosen, to ensure uniformity of thickness, thus guaranteeing simplicity in structural fabrication. Adhering to this rationale, the same type of graphite used for the liner in the Grain Case has been employed, with its characteristics detailed in Table 8.6.

The gas Convective Heat Transfer Coefficient (h_{gas}) is:

$$h_{gas} = 50000\, W/K \cdot m^2 \tag{8.57}$$

Porous Graphite thermal conductance is:

$$h_{\text{graphite}} = \frac{k_{graphite}}{\delta} = 250\, W/K \cdot m^2 \tag{8.58}$$

The code shows how the considered thickness meets the design requirements.

Script 8.15 Matlab code to calculate Convective Heat Transfet Coefficient of Porous Graphite layer

```
%% First Layer: porous graphite

```

```
delta1=0.004;  %thickness [m]

hg=k1/delta1; %Porous Graphite thermal conductance

if hg_eff/hg>150
    disp ('error')
end
```

In figure 8.35 is presented the CAD model of the nozzle, along with the corresponding thermal liner.

Figure 8.35 Nozzle CAD model

Heat Sink design

The Heat Sink constitutes the second insulating layer of the cooling system and is thickened, as already specified, only in the nozzle throat section for a length of 4 mm. The heat sink is idealized in copper[8], whose characteristics are reported in Table 8.7. The choice of material lies in its high thermal conductivity, as this component must evenly distribute the heat transmitted by the liner, minimizing its propagation to the structural steel layer and preventing the formation of hot spots that could undermine structural integrity.

Density ρ	8900 Kg/m^3
Thermal Conductivity k	360 W/mK
Specific Heat c_p	430 J/KgK

Table 8.7 Copper thermal characteristics

The thickness of the Heat Sink, like the liner, is the result of various tests that have led to the following value:

$$\delta_{copper} = 10\,\text{mm} \tag{8.59}$$

Copper thermal conductance is:

$$h_{\text{g Copper}} = \frac{k}{\delta} = 36000\,W/K \cdot m^2 \tag{8.60}$$

This value is acceptable as it is much higher than that of porous graphite ($h_{g\,eff}$), ensuring optimal heat diffusion.

Script 8.16 Matlab code to calculate Convective Heat Transfet Coefficient of Copper Layer

```
%% Second Layer: Copper (Heat Sink)

delta2=0.01; %thickness [m]

hg_2=k2/delta2; %Copper thermal conductance [W/K*m^2]

if hg_2/hg_eff<100
    disp ('error')
end
```

Knowing the temperatures involved, we can calculate the "Characteristic Time" related to heat diffusion. Such temperatures are:

- Adiabatic Wall Temperature (T_{aw}) = 3440 K
- Initial Heat Sink Temperature (T_0) = 300 K
- Copper Melting Temperature ($T_{melting}$) = 1360 K

The Characteristic Time (τ) is: 153.08 s. In conclusion, it is possible to determine the Heat Sink temperature at the end of the firing time (30 s) and verify that it is well below the melting temperature of copper. It is possible to illustrate the aforementioned in Figure 8.36, which depicts the temperature profile of the heat sink throughout the mission.

$$T_{end} = 858.82\,\text{K} \tag{8.61}$$

It can be observed that this temperature is comparable, if not identical, to that assumed by the steel layer.

Script 8.17 MATLAB code to calculate Final temperature of Copper Layer

```
tau=(rho2*c2*delta2*delta1)/k1; %Characteristic time [sec]

t_fin= 30; %Shooting time [sec]

tempo=0.01:0.01:t_fin;
T2=zeros(1,length(tempo));

for i=1:length(tempo)
    t=tempo(i);
T2(i)=T_aw-(T_aw-T0)*exp(-t/tau); %End temperature[K]

  if T2>T_fusione_Rame
      disp ('error')
  end
end
```

This result certifies the effectiveness of the cooling system. This will be elaborated in greater detail through a finite element model, which will illustrate the temperature distribution along the structure under the operational condition at the 15-second mark. This assumption aims to establish an average range and to emphasize the temperature

variations, particularly across the different layers of the nozzle.

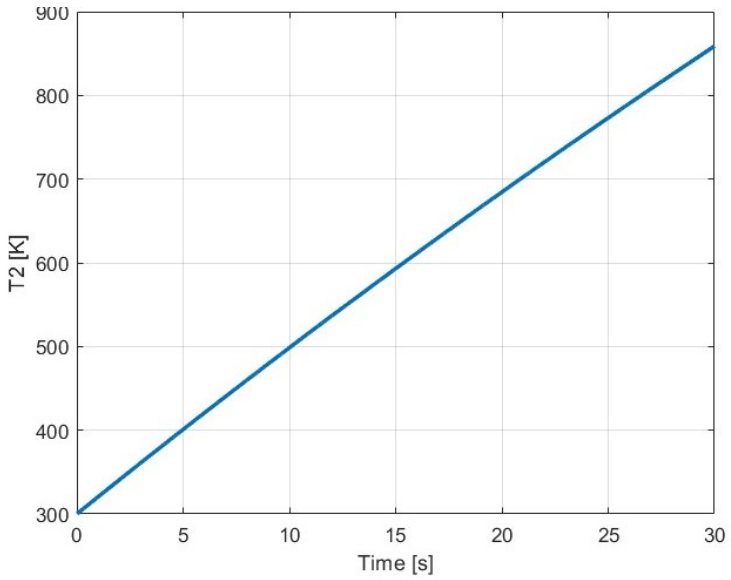


Figure 8.36 Heat-Sink Temperature during time

In figure 8.37 presents the CAD model of the final combustion chamber, complete with its associated cooling system:

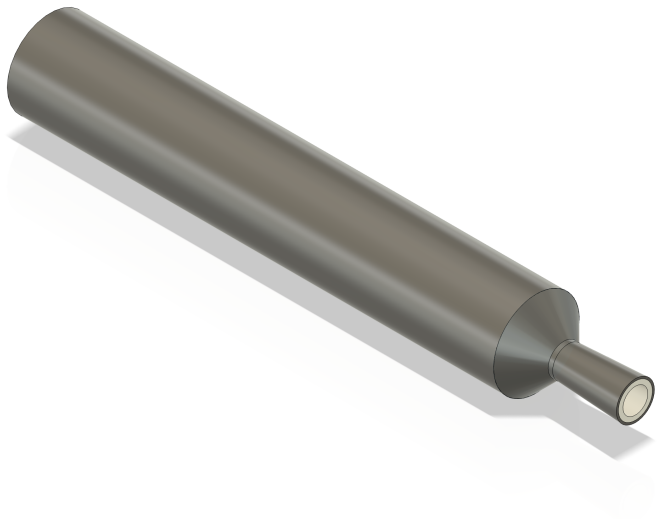


Figure 8.37 Full Chamber

Thermal FEM Analysis of Full Chamber

In this subsection, the most significant Finite Element Method (FEM) analysis regarding the missile's combustion chamber is detailed. As previously mentioned, this analysis considers the contributions of the Liner and Heat Sink, which have been extensively discussed in earlier sections. The materials outlined in preceding sections are to be incorporated into this analysis. This particular FEM analysis is the culmination of ongoing research aimed at enhancing the quality of observations, seeking to vividly illustrate how the chamber responds to combustion processes.
The analyses conducted are not static but rather Steady-State Heat Transfer, focusing not on structural deformations or translations but on verifying the proper absorption of heat across the various layers of the chamber. This analysis is critical as any incorrect temperature evolution could lead to a loss of mechanical properties in the materials constituting the layers. This phenomenon becomes more critical towards the exterior of the structure, near the steel layer, as high temperatures on the outer wall of this layer may lead to reduced electrical conduction capabilities of the wires, rendering the missile incapable of fulfilling its mission.
With the rationale behind our analysis now clarified, let us proceed by describing the process that led to the final analysis.

The first FEM analysis presented in this section concerns only the Grain Case, as the focus is to analyze the temperature evolution along this section. This is a preliminary analysis, intended to showcase in greater detail the trend observed in subsection 8.5.2, as referenced in figure 8.29. It should be noted that this analysis does not consider the tank cap; rather, it examines a surface constrained at the extremity. This choice is made to avoid a detailed depiction of the evolution of the entire structure, instead concentrating solely on the thickness of the steel layer and liner. Naturally, throughout the discussion of this analysis, the model's evolution, including the tank cap, will be considered to ensure a more accurate and comprehensive examination.
As previously stated, the analysis conducted is thermal in nature. To carry out this analysis, heat flows through the various layers were evaluated using Fourier's Law:

$$q = k \cdot A \left(\frac{\Delta T}{s} \right) \tag{8.62}$$

Of course, the underlying data will not be repeated here, as they have already been extensively covered in sections 8.5.4. Instead, the results obtained for these heat flows are presented below:

- Heat flow through the liner: $q_{liner} = 103.3519$ W
- Heat flow through the steel layer: $q_{steel} = 1021.12$ W

Additionally, it should be noted that, to verify the consistency of the structure with the established temperature standards, the preliminary phase temperatures were imposed along the surface extremities of the various layers:

- Internal extremity of the liner: 3000 K; this temperature should be approximately 3500 K; however, it is assumed that a gradient of $\Delta T = 500$ K is absorbed by the air before reaching the liner.

- Contact extremity between the liner and steel: 855 K
- Extremity between the steel and wiring: 255K

With these references in place, the temperature evolution can be delineated. It is worth noting that the use of the Thermal extension in FEMAP significantly enhanced the quality of the results obtained. Of course, this improvement is the culmination of numerous iterative cycles and efforts to determine the appropriate thermal gradient absorbed by the liner, thus avoiding excessive thermal stress on the steel layer.

The general overview of the initial analysis is illustrated in figure 8.38, serving as the foundation for our preliminary considerations.

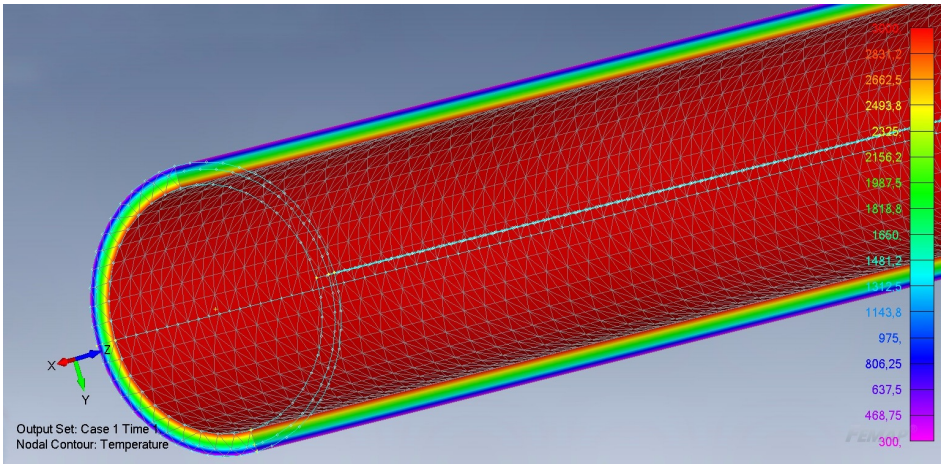


Figure 8.38 Case plus Liner FEM analysis

In this preliminary image, you can clearly see the curves marking the boundaries of the steel and liner layers. As expected, there's a consistent pattern throughout the structure, except for the cap, which we'll address in more advanced FEM analyses later in this section. Let's focus on a single section now and highlight the temperature distribution along it. Refer to figure 8.39 for those assumptions.

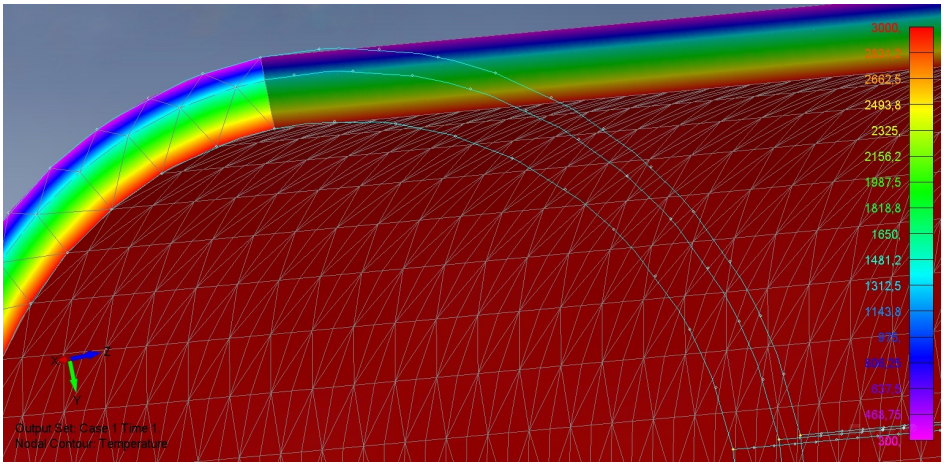


Figure 8.39 section

You'll notice that the first layer almost completely absorbs the temperature. The contour color shifts from red, indicating the highest temperature, to a bluish-violet hue, indicating the lowest. As discussed in earlier subsections, the steel layer's main job is to absorb the pressure stresses. To avoid overloading our structure with too much porous graphite, it also needs to handle a limited and consistent thermal gradient.
This analysis shows that the liner effectively absorbs the temperature, indicating that the liner's thermal dimensioning has been done correctly.

After outlining the initial analysis focusing on the Grain Case alone, let's now shift our attention to the entire combustion chamber. Here, it has been chosen to depict the sealed tank differently, not as a connecting surface, but by conceptualizing the corresponding cap within the FEM environment using the protrusion command (figure 8.40).

It's worth noting the heightened precision in modeling the geometry, especially near the junctions where the nozzle's presence adds to the intricacy. Given these nuances, now it is possible to introduce the upcoming FEM analysis's objective.
The aim of this study is to explore a preliminary endeavor in integrating the heat sink and nozzle. Regrettably, the intricate geometry and solid connections have posed challenges in clearly defining contours. However, it has been planned to address this issue down the line by leveraging the Surface command in FEMAP.

This approach will help streamline computational complexities by treating the two surfaces as distinct entities, with the liner affixed to the steel layer's surface. More insights on this process will follow in due course.

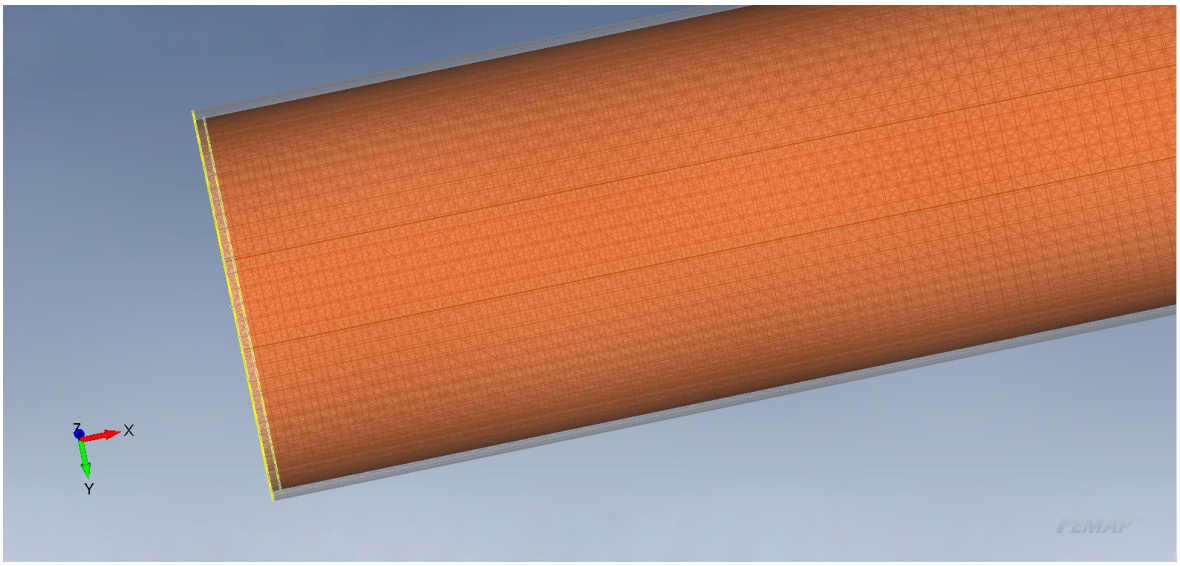


Figure 8.40 Cap tank

The structure in question is depicted in figure 8.41 for completeness, detailing the materials used:

- The Liner layer is represented in pink.
- The Steel layer is indicated in gray.

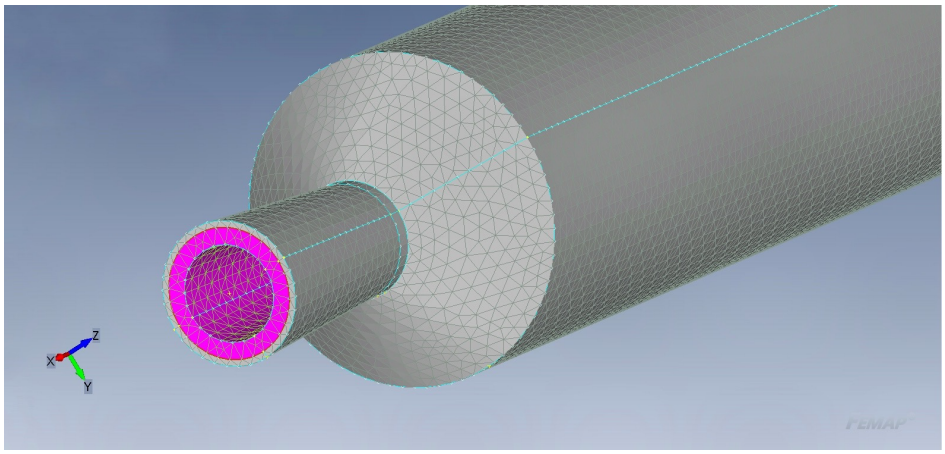


Figure 8.41 materials

It is observed that, unlike in subsection 8.5.3 where the tank was dimensioned using average surfaces, in this instance, the open final section of the diverter is considered. This decision is justified by the lack of interest in evaluating potential stresses or deformations.

After all appropriate modifications have been introduced to comprehend the results obtained, it is noted that the elevated geometric precision applied entails a rather reduced point mesh, resulting in a less visible contour. Referring to figure 8.42, displaying the outcome of the analysis at hand, it proves challenging to differentiate between the various temperature ranges. However, it does offer an initial demonstration of the function of the Heat Sink, as dimensioned in subsection 8.5.4.

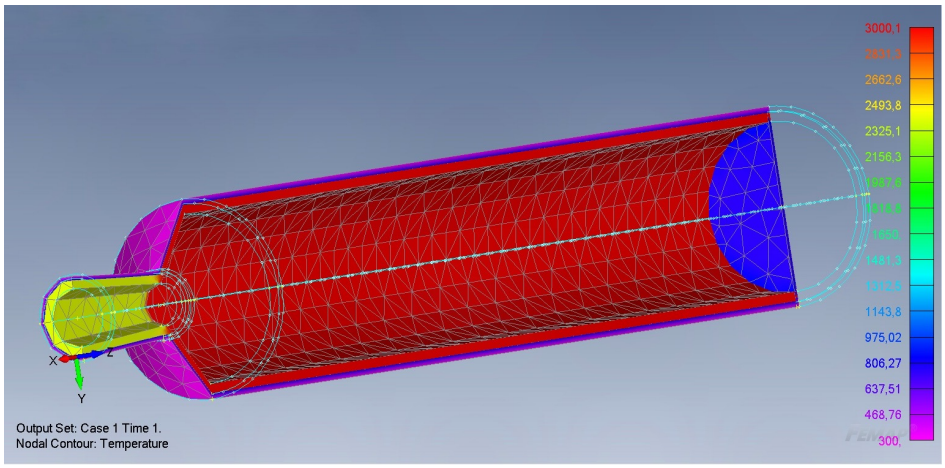


Figure 8.42 Pre chamber FEM

Concerning Fourier's Law from the preceding chapter, the following heat flows have been considered relative to the throat section alone, deemed the most critical structurally:

- Heat flow along the Liner layer: $q_{th_l} = 32$ W
- Heat flow along the steel layer: $q_{th_s} = 508$ W

Note that near the throat, the heat flux is greater than in the rest of the nozzle. However, due to the presence of the heat sink, which is responsible for heat absorption, the steel temperature is lower in the throat compared to other sections where it reaches 3000K. Unfortunately, due to the limited accuracy of the mesh, the behavior beyond the throat is not realistic; therefore, attention should be focused solely on the convergent sections. (A zoom is done in figure 8.43).
This behavior will be further analyzed in detail in the final FEM analysis concerning the combustion chamber. The objective of this current analysis was merely to showcase the improvements made in terms of geometry and the introduction of the Heat Sink and nozzle. Their rules will be further analyzed in detail in the final FEM analysis concerning the combustion chamber. The objective of this current analysis was merely to showcase the improvements made in terms of geometry and the introduction of the Heat Sink and nozzle.

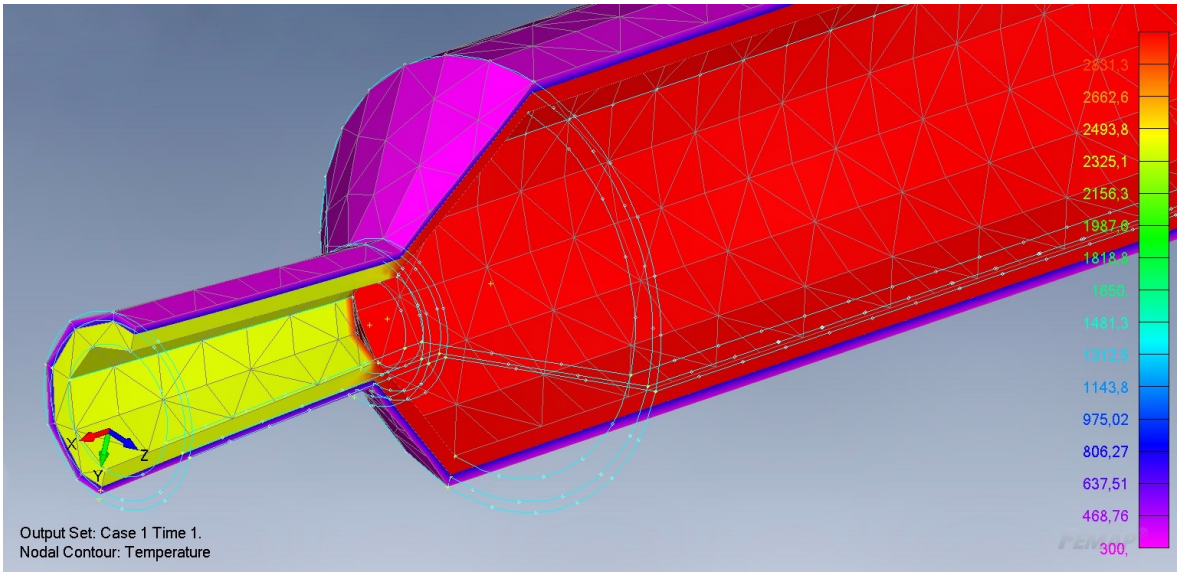


Figure 8.43 nozzle focus

The subsection culminates in the presentation of the most recent finite element analysis undertaken, which duly considers not only the heat flows assessed in accordance with Fourier's law but also the convective flows, thereby elucidating the behavior of the Heat Sink in relation to the cooling phenomena.

In the preceding analysis, a low degree of accuracy was noted attributable to the absence of a defined contact condition between the two surfaces. This concern was effectively addressed during the design phase, with the collaboration of the mechanical architect, by positing an adhesive bond between the liner and the steel layer. Such an approach facilitated the elimination of superfluous elements regarding the interaction between the surfaces, thereby significantly enhancing the overall accuracy of the model's mesh. The specifications pertaining to the contact parameters are delineated in figure 8.44.

As articulated in the introductory segment, to accurately simulate the behavior of the Heat Sink, the heat flows evaluated pursuant to Fourier's law were meticulously analyzed using Matlab and subsequently implemented in FEMAP.

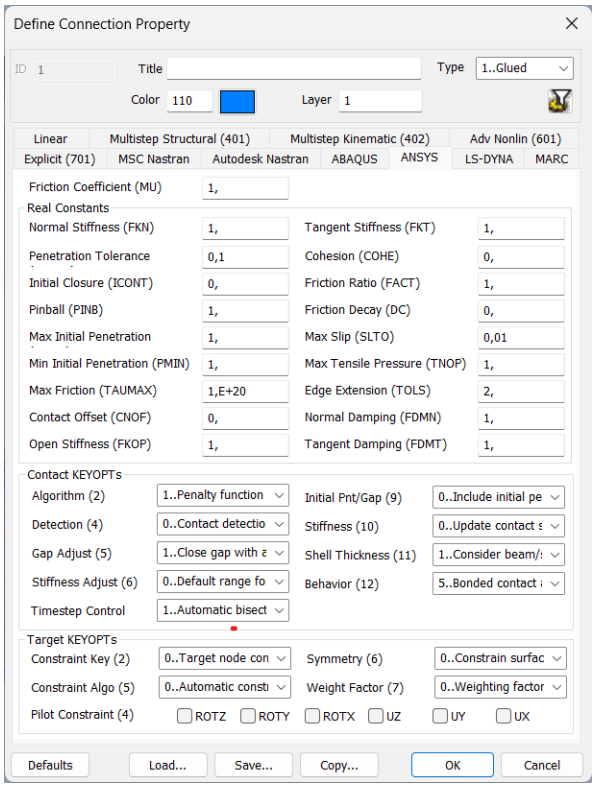


Figure 8.44 glued propriety

Furthermore, in order to better represent the thermal dynamics of the chamber, the convective heat transfer coefficients associated with the gas flux into the nozzle and the first liner layer has been considered; therefore, in order to consider the Copper mass behaviour, the conductive thermal coefficient has evaluated. However, it is important to note that this analysis did not encompass the two layers associated with cabling and aluminum, whose contributions had previously been acknowledged in the dimensioning of the liner.

Given the constraints that precluded the execution of a dynamic analysis necessary for an accurate depiction of the Heat Sink's behavior, a temporal snapshot at 15 seconds post-firing was selected for the finite element analysis. This strategic choice aimed to scrutinize the chamber's behavior to enhance the Heat sink function. In alignment with this rationale, the temperature of the internal surface of the steel layer was established to coincide with the temperature attained by the copper at that specific moment, $T = 594K$, a value extracted from the Matlab code detailed in subsection 8.5.4.
The remaining sections of the steel layer were prescribed a comparable temperature for the sake of homogeneity, approximately 700 K, while for the external case, a temperature of 855 K was designated, reflecting the temperature reached by the copper upon the conclusion of the firing cycle. Although this assumption may be deemed flawed, it was necessitated to facilitate the subsequent structural analysis under the most severe conditions for the steel. The interior of the chamber was set to a temperature of 3000 K, while the external surface of the steel, ideally situated in contact with the cabling, was established at a temperature of 255 K, as justified in prior analyses.

Concerning the new considered flows, these were judiciously introduced near the two surfaces of the steel layer to accurately simulate the Heat Sink's behavior.
The coefficients employed in this context are enumerated as follows:

- For the internal surface oriented toward the chamber, a convective heat transfer coefficient of $h_{gas} = 50000W/K \cdot m^2$ was applied at the inner liner subsection, thereby mirroring the behavior of the first liner layer;
- For the internal surface oriented toward the chamber, a conductive thermal coefficient of $h_{liner} = 250W/K \cdot m^2$ was utilized within the liner subsection, thereby mirroring the behavior of the first liner layer
- For the external surface, a conductive thermal coefficient of $h_{copper} = 36000W/K \cdot m^2$ was deployed, reflecting the behavior of the copper mass.

Refer to figure 8.45 for the image corresponding to the analysis under consideration.

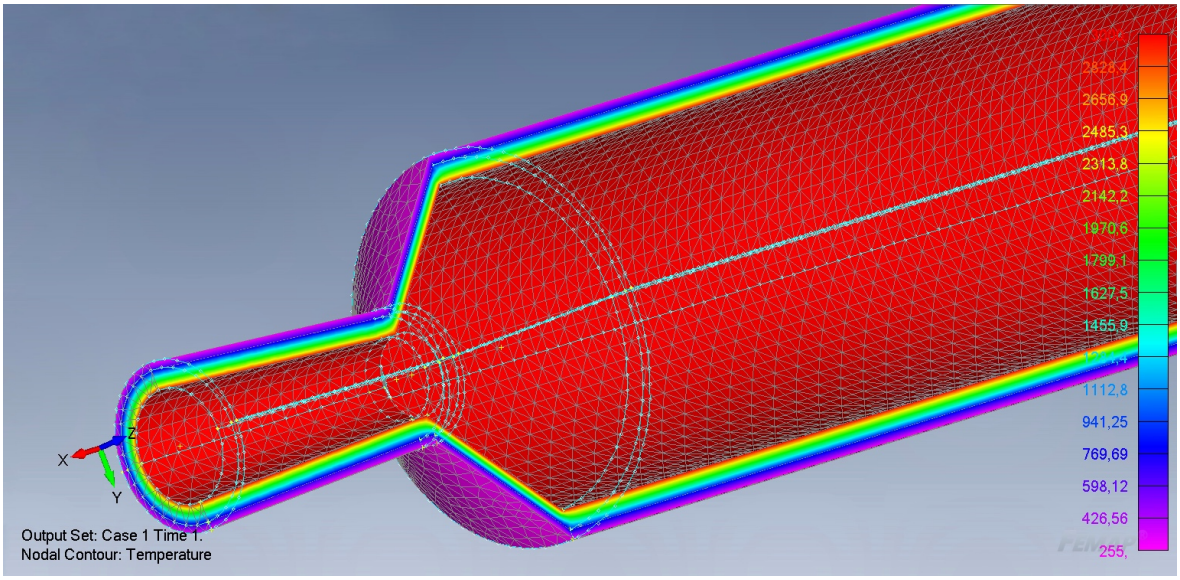


Figure 8.45 FEM camera

In general terms, it can be observed that the analysis is in perfect accordance with that conducted using Matlab computational codes, and that the temperatures attained do not result in any deterioration of the mechanical properties of the materials employed.

A more detailed analysis can be conducted using Figure 1.5, which allows for a closer examination of the nozzle throat section, where all the cooling agents of the chamber are located. First and foremost, note the uniformity of temperature across the thicknesses, which implies correct thermal sizing of the structure. This observation was already evident in the preliminary phase through the initial approximate FEM illustrated in figure 8.43. The necessity to improve the mesh quality is particularly pertinent to the throat section; in this regard, a focus on this section has been proposed in figure 8.46.

It is noteworthy that the steel layer at the throat absorbs less heat flux compared to other sections craft in AISI 4340, highlighting the proper functioning of the Heat Sink. Additionally, observe that the liner in this section absorbs a greater heat flux.

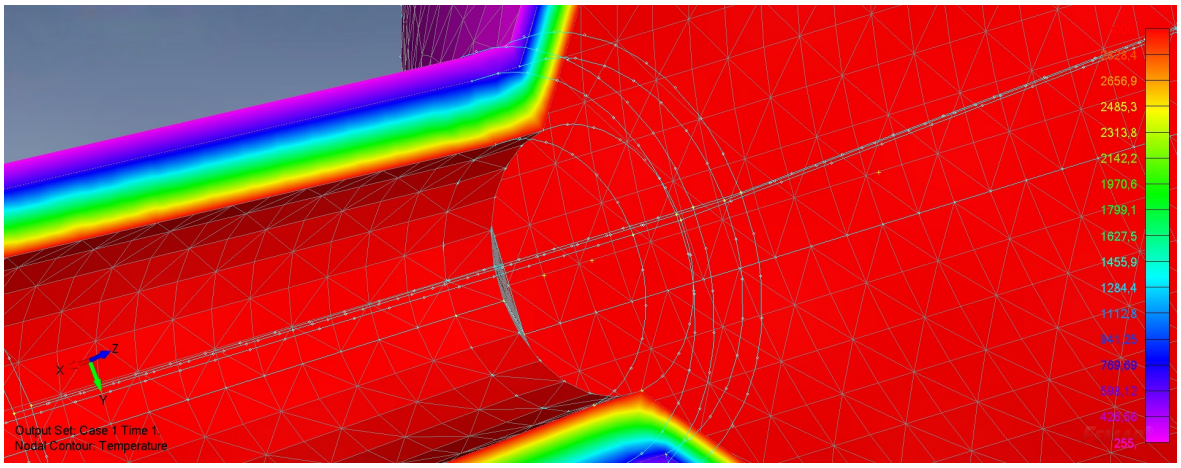


Figure 8.46 FEM camera, focus on throat section

Consequently, there is a correlation between the analytical data and the corresponding finite element analysis, verifying the proper operation of the combustion chamber and its resistance to both thermal stresses and pressure loads.

With these assertions, the structural phase of the combustion chamber is concluded, thus outlining the framework of the sustain and its correct functioning under the operational conditions established in the preliminary phase. Note that high precision was used in sizing the cooling system, considering that, due to the short mission duration, the presence of the Heat Sink might be deemed superfluous. Nevertheless, the decision was made to operate under the most critical conditions possible. This approach was consistently applied throughout the structural section.

8.5.5 Joints

Analyzing the main components from a structural perspective in subsections 8.5.1-8.5.2-8.5.3; it remains to examine the connection between the two macro-structures concerning the outer shell and the combustion chamber. To this end, joints known as thermal breakers[7] have been idealized, which have the task of:

- Withstanding the loads to which the entire missile assembly is subjected;
- Ensuring the adiabatic wall temperature around the outer shell, and consequently dissipating heat along the structure, thereby avoiding the previously mentioned issues of loss of electrical conductivity and loss of mechanical properties of the aluminum layer.

Due to their dual role, the porous graphite was chosen, cause it's a material with mixed characteristics, as illustrated in table 8.6, was considered for this purpose. It should be noted, in previous subsections, that a gap of 1.2 mm between the outer shell and alloy was assumed, consequently, our joint will have a height of the same value. Additionally, for simplicity in construction, these joints were idealized in a cylindrical shape, and their placement is illustrated below.

It should be noted that a dual connection was made, near the section of the cap and near the beginning section of the convergent. Focusing further on the connection, in order to ensure high stability, four joints were used for each section: two in the radial direction and two in the axial one. It is assumed that the surfaces are bonded with polyimide glue, which operates at a continuous service temperature of approximately 440 K, that is more than adequate for our needs. The sizing of these joints was idealized directly in the FEMAP environment; In fact, given the known height of the cylinder in question, a thickness of 1 mm was chosen, although excessive, for simplicity in construction; thus, an analytical approach was not used.

The CAD model representing the previously illustrated assertions is shown in figure 8.47:

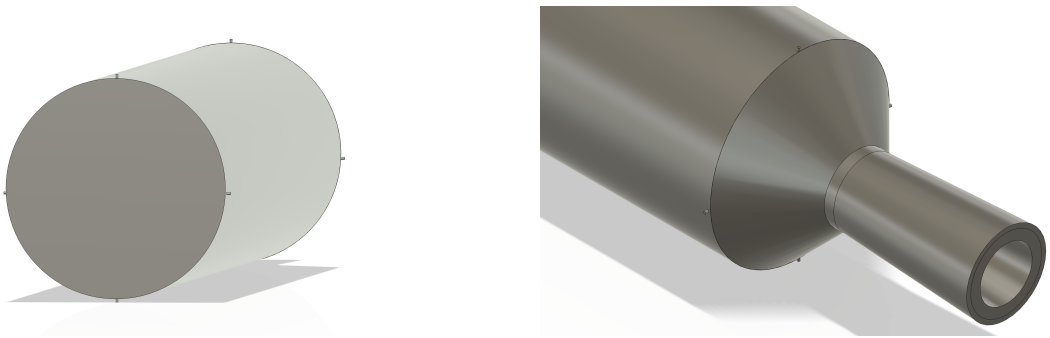


Figure 8.47 Thermal Breakers

Given the two arrangements of the thermal breakers, two different structural finite element analyses were conducted depending on the position of the joints.

A combined analysis was instead idealized concerning the temperature evolution along the structure under examination; this analysis is very similar to that conducted in the chamber.

Starting with the analysis of the vertical joints, these exhibit the characteristic of being subjected only to the stress due to the propulsive thrust, as the shear force seen in the section is discharged onto the constraints and absorbed by the structure. The analysis results are reported in figure 8.48:

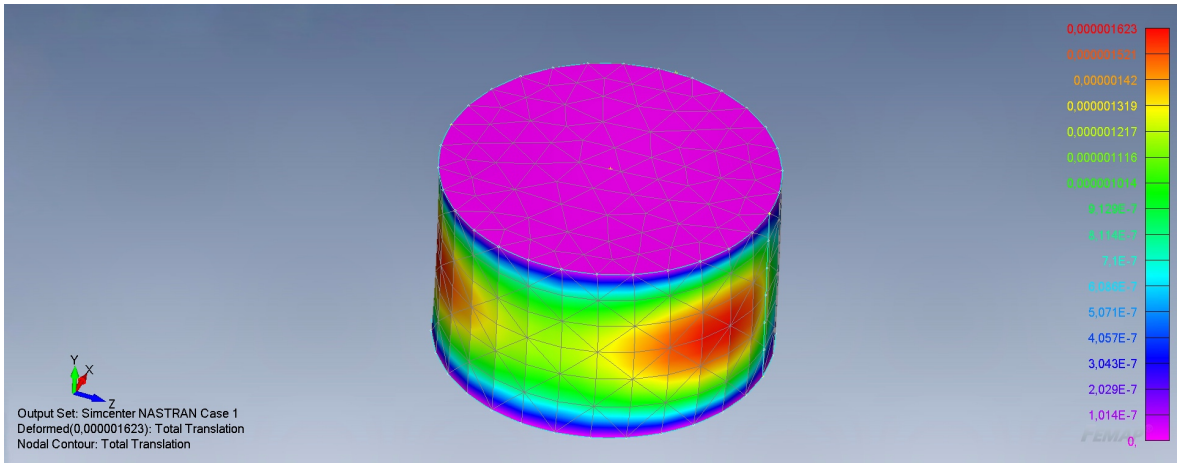


Figure 8.48 axial joints FEM

From the contour, it is evident that the deformations obtained are almost negligible, as was predictable given the thickening for ease of construction.
For the horizontal pylon, it is subjected to both the macro forces examined in subsection 8.5.1, acting as two shear forces. This results in a deformation slightly greater than the previous one, yet far below the elastic limits of the porous graphite, as shown in 8.51.

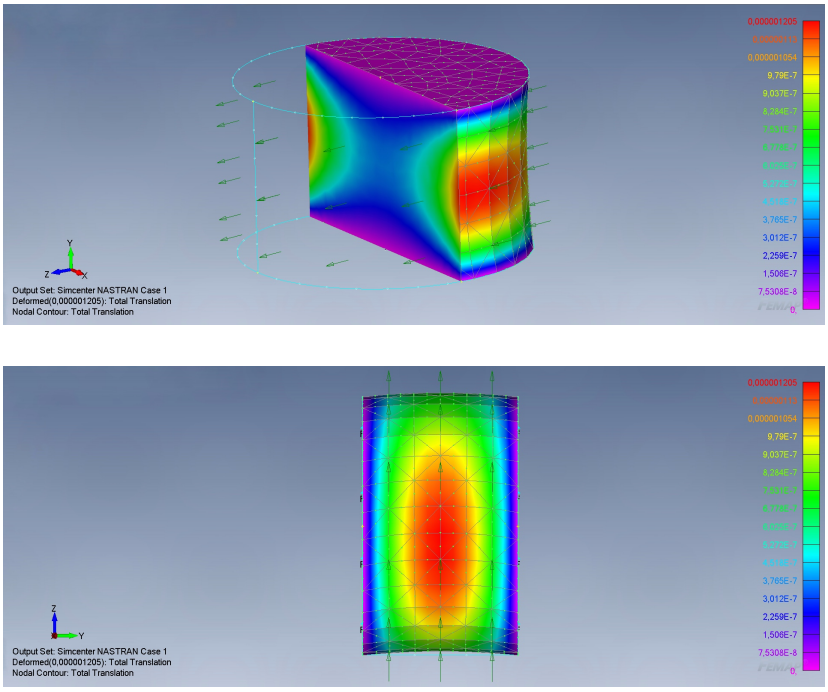


Figure 8.49 Radial Joints FEM

Regarding the thermal analysis, the most critical situation was exposed, considering that the steel is at a higher temperature than seen in the case study, around 400 K, and the aluminum is at a temperature of about 255 K. The analysis highlights the correct evolution of heat and, consequently, verifies the structure from a thermal perspective as well, as shown in figure 8.50.

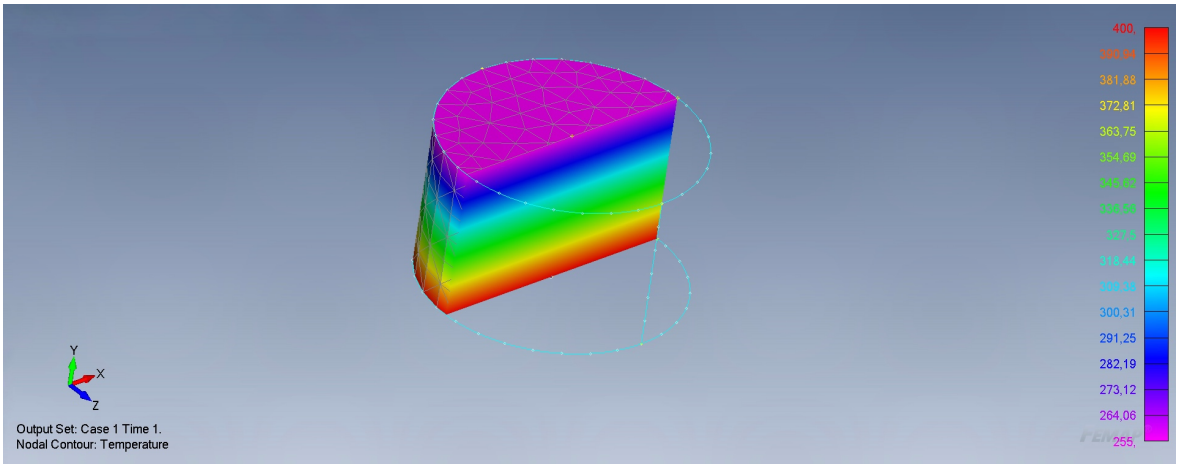


Figure 8.50 Joints Thermal FEM

8.5.6 Wings FEM

Regarding the wings, the sizing was carried out from a purely aerodynamic perspective, idealizing a Mach number of 30%, thereby increasing the applied force. The dimensions and corresponding thicknesses are detailed in the subsection 8.4.1 where the task is to verify the designed wing through finite element analysis. It is important to emphasize that the aerodynamic surface in question is hinged on the outer shell via a pin located 46 mm from the leading edge of the wing, allowing it to rotate freely around this hinge axis.

The material used is aluminum 7075, with its properties listed in 8.5. Additionally, to account for the full range of forces, the wing is subjected to a shear force applied at its center of pressure, and both the missile's translational acceleration and gravitational acceleration are considered to simulate the thrust force during the propulsion phase.

It should be noted that the following verification criterion was employed:

$$f < 5\% \cdot S \tag{8.63}$$

Where f denotes the maximum deflection of the structure, and S represents the span of the semi-wing.

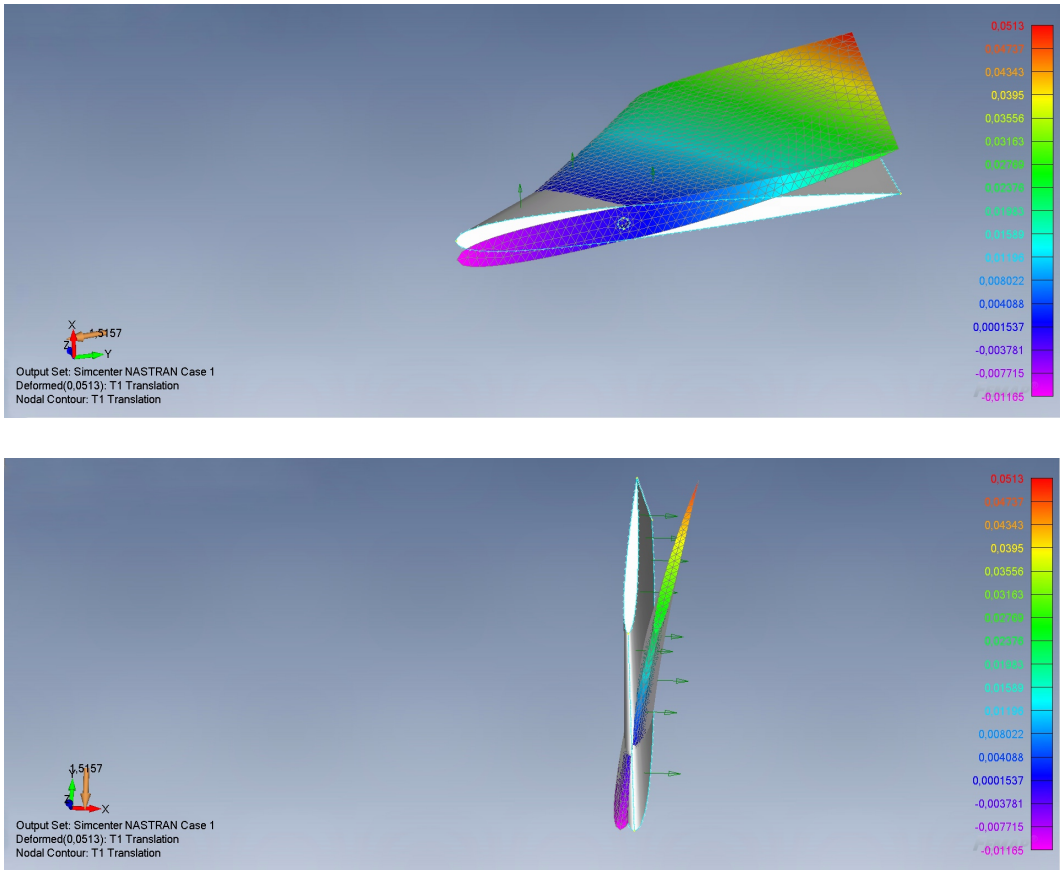


Figure 8.51 Wings FEM

The contour plot clearly shows that this criterion is more than adequately met.

Chapter 9

I stage

9.1 I stage + II stage Overview

The following is a general overview of the entire system, comprising both stages, where the dimensions and basic technical specifications are presented:

- **Total length**: 143.2 cm
- **Caliber**: 80.6 mm
- **Number of fins**: 8
- **Mass with loaded propellant**: 16.694 kg
- **Mass without propellant**: 13.969 kg
- **CG on X axis with loaded propellant**: 70.941 cm
- **CG on X axis without propellant**: 62.556 cm
- **Moment of Inertia I_{YY} with respect to CG with loaded propellant**: 1.711 $kg \cdot m^2$
- **Max load factor**: 6
- **Accelerometer position with respect to minum CG (L_acc)**: 22.556 cm

The caliber of the first stage is the same as that of the sustain stage to avoid the need for joints, which could lead to aerodynamic complications.
The total number of fins is 8 of which four for the first stage and four for the second stage. The fins on the first stage are solely for stabilizing purposes, given the reduced operating time so they do not present any actuating system.
The load factor of the complete missile is the same to that of the sustain stage.

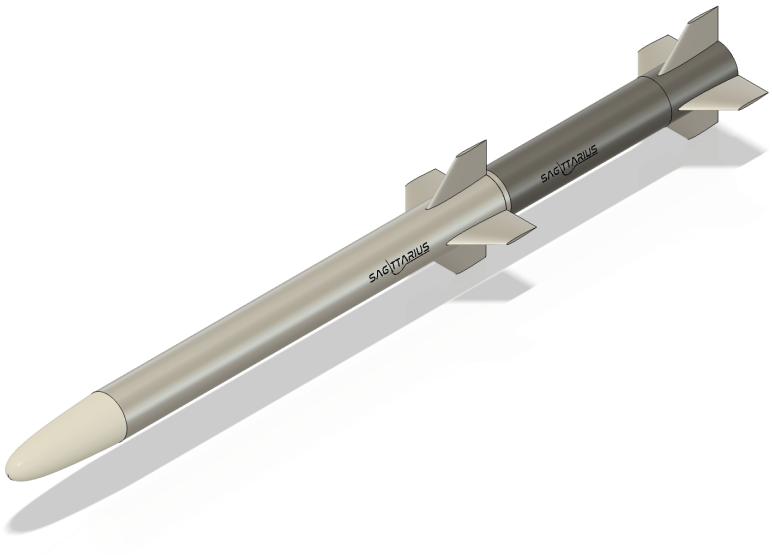


Figure 9.1 SAGITTARIUS missile 3/4 View

9.2 I stage: Mechanical Architecture

This paragraph provides a summary of the components of the first stage and explains the design choices related to them:

- Grain
- Nozzle
- Fins
- Inner shell

The propellant grain has been designed using custom Matlab codes, based on inputs provided by the CEA. Given the requirement for constant thrust and the need to keep combustion away from the walls due to the absence of a thermal liner, a star-shaped grain geometry was selected.
The nozzle has a truncated conical shape and designed to optimize axial outflow and avoid heating problems.
The fins are fixed and were designed using aerodynamic software to ensure the stability of the missile. The inner shell is a layer of steel that surrounds the entire first stage, providing structural coating.
An additional aluminum layer (outer shell) was not deemed necessary during the design phase due to the absence of cabling and the short firing time.

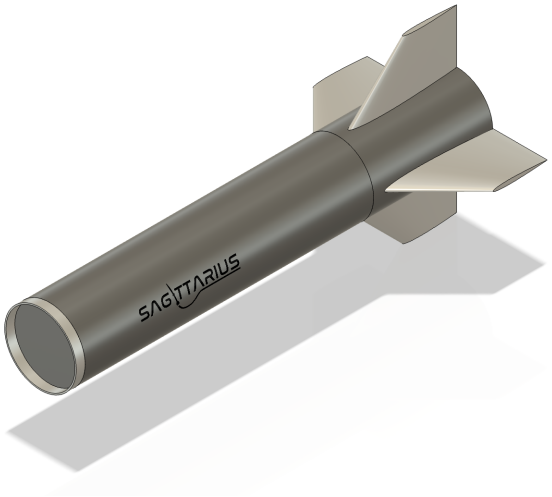


Figure 9.2 First stage 3/4 View

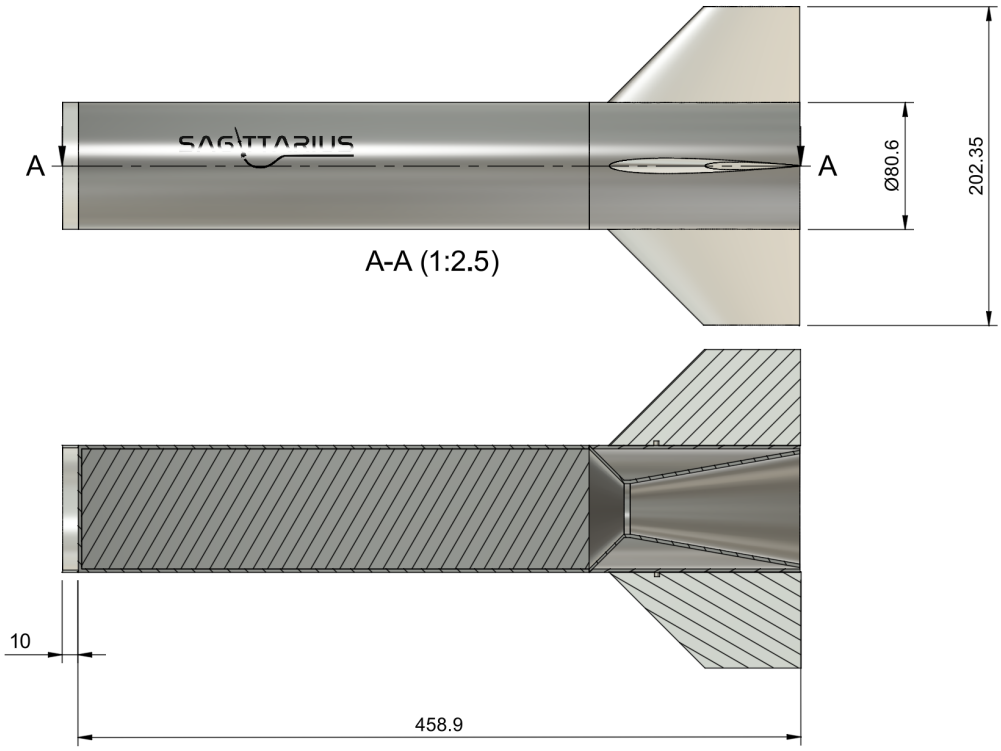


Figure 9.3 First Stage Lateral View and Section

9.3 I stage Propulsion

As said before, a two-stage missile configuration was chosen and the reasons that led to this choice have already been explained. The first stage, usually called the "Booster", is used for igniting the missile and reaching the predefined cruise altitude. The booster is essential to provide the initial thrust required to bring the missile to the desired speed and altitude in the shortest possible time.
The firing time, and consequently the operational life of this stage, will be on the order of seconds, highlighting the critical speed of action in the initial launch phases. In the design of the propulsion system of a booster, numerous factors must be carefully considered because they can lead to significantly different performances.
The choice of propellant and the geometric configuration of the grain are key elements that directly influence the booster's performance. Every detail of the design must be carefully considered to ensure maximum efficiency and reliability of the system.

9.3.1 Choice of Propellant

The process that led to the selection of HTPB-1912 propellant is similar to the one used for the sustain stage; therefore, it is not worth elaborating extensively on the fuel properties, which have already been thoroughly discussed. Although the booster and the sustain stage perform different tasks (namely, the former is responsible for liftoff from the launch base and reaching a certain altitude and cruising speed, while the latter must maintain this speed and adopt a trajectory capable of hitting the target) this propellant offers adequate performance for both purposes. [8]

9.3.2 Grain Geometry

On the other hand, the grain geometry, i.e., how the propellant is arranged inside the combustion chamber, can vary significantly between the booster and the sustain stage, that happens because it could strongly influences the thrust profile and, as it is have been specified, booster and sustain stage require different thrust profiles.
The grain geometry, additionally, can mitigate or exacerbate phenomena such as erosive burning, where unburned propellant pieces are entrained out of the engine nozzle, resulting in significant performance loss for the entire system. The chosen geometric configuration must therefore be optimized to minimize these effects and ensure uniform and controlled combustion.
As a consequence, a star-shaped configuration was selected for the booster, because it helps to achieve the desired Thrust Profile, which will be discussed below.
The star configuration helps maintain a controlled thrust, enhancing the booster's effectiveness in bringing the missile to the desired altitude.The choice of a star-shaped geometry was also driven by the ability to focus most of the combustion, as well as the presence of the flame, in the central part of the case, thus in an area away from the walls. The direct advantage of this choice is the possibility of having a simpler structural layer around the booster. In fact, such combustion, combined with a firing time of less than a second, allows for the avoidance of a thermal liner, which is instead found in the sustain stage, where the combustion is not radial, as the grain is cigarette-shaped and the firing

time is about 30 seconds.

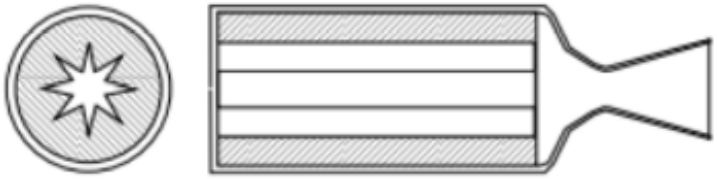


Figure 9.4 Grain with a star-shaped geometry [8]

9.3.3 Grain sizing

The next step in the design is to estimate the diameter and length of the propellant grain case, a calculation necessary for obvious structural and storage needs, but obviously achievable only through fluid dynamic and propulsion concepts. As will be clear, the chosen geometry for the grain will have a significant influence on the final results. The parameter to determine upstream of the analysis is the door area, i.e., the hollow section of the propellant grain case. To calculate it, an empirical formula is applied according to which it must be three times the critical area, in order to avoid erosive burning phenomena. Therefore, it will be equal to 0.0024 m^2. The wet area, i.e., the surface area of the propellant grain exposed to the flame, which appears in the calculation system, is determined by the following empirical formula, specific to a star-shaped section:

$$A = 0.9 \cdot d_c \cdot L_c \cdot \pi$$

In which:

- d_c is the diameter of the propellant grain case.
- L_c is the length of the propellant grain case.

The calculations are made thanks to a specific function compiled on Matlab:

Script 9.1 Matlab code used to calculate length and diameter of the grain

```
function F=sizegrain(x)

%Empirical Equations to find the Diameter and the Length for a Star Grain

global r rhop Aporta mdot mp

F1 = (pi/4)*((x(1))^2)*x(2) - Aporta*x(2) - (mp/rhop);

F2 = 0.9*x(1)*x(2)*pi - ((mdot)/(rhop*r));

F = [F1,F2];


end
```

We will have the following output values:

- $d_c = 76.2mm$
- $L_c = 32.21mm$

These final parameters confirm that the booster design is optimized to ensure maximum performance and reliability, in accordance with mission requirements. The detailed and accurate design of each booster component is essential for mission success, ensuring that the missile can reach its destination precisely and efficiently.

9.4 Thrust Profile

As already mentioned, the mission objectives for a sustain stage and a booster stage are different from each other. Specifically, the former requires a constant thrust profile over time to maintain cruise velocity, while the latter demands a maximum thrust value in the very early stages of operational life, as the missile must detach from the launch pad and quickly reach the required altitude and speed. Subsequently, the thrust profile must maintain nearly constant values over time until the end of its operational life. This happens because the booster is responsible for the first operational phase of the system, which goes from ignition and lift-off from the launch pad to reaching the desired speed and altitude, followed by the shutdown and separation of this stage. The consequence of all this is this specific thrust profile that allows for a very rapid gain in altitude and speed over time. In fact, the operational life of the booster is shorter than that of the second stage, in this specific case, it is 0.82 s. As a consequence, the star shaped geometry is the ideal solution in order to achieve the desired Thrust Profile. Now, it makes sense to identify the maximum value of the Thrust generated by the system, derived as:

$$T = c \cdot \dot{m}$$

And this thrust is 3626.5 N. Thanks to the graphic, it is possible to observe a slight de-

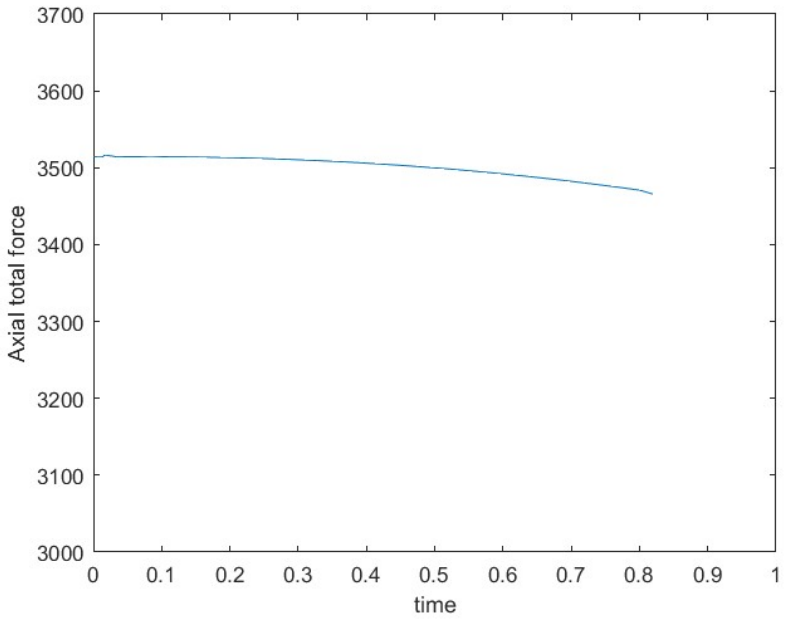


Figure 9.5 Net Axial Force (I stage)

crease in net axial force in the final stages of the first stage operational life, which is attributable to the increasing contribution of the Drag.

9.5 Rocket motor design

9.5.1 CEA Application: Equivalent exhaust velocity and Thrust

The input parameters have been obtained by conducting an analysis using the CEA (Chemical Equilibrium Applications) software provided by NASA. By inputting the combustion chamber pressure and the propellant mixture, the equivalent exhaust velocity (c) has been obtained. The CEA software provides an output c of 2565.5 m/s, but this is multiplied by a correction factor of 0.9. The physical reason behind this process lies in the fact that this software is specific to liquid propellant mixtures, whereas the current case involves a solid propellant. Therefore, the equivalent exhaust velocity is 2309 m/s. It should be noted, for completeness, that the equivalent exhaust velocity is also the result of the product of two parameters, namely the thrust coefficient (c_f) and the characteristic velocity (c^*), both of which are excellent indicators of the performance of this system. Therefore, the following were calculated:

$$c^* = \frac{p_0 \cdot A_{cr}}{\dot{m}}$$

The thrust coefficient is calculated as follows:

$$c_f = \frac{c}{c^*}$$

The characteristic velocity is 1532.1 m/s, while the thrust coefficient is 1.507. These values are crucial for evaluating the efficiency of the engine and ensuring that the booster can effectively perform its task.

9.5.2 Propellant flow rate

The booster engine in question has a chamber pressure of $30*10^5$ Pascal. Since this engine is responsible for lifting the entire system off the ground and bringing it up to 500 meters, depending on launch angle. For practical reasons, the thermofluidodynamic parameters are evaluated at sea level. This approach simplifies calculations. The first step in the development of the booster design was to determine the critical area, using specific fluid dynamic formulas where the adiabatic expansion coefficient (γ) is assumed to be 1.2, knowing the ratio between the stagnation pressure and the exit pressure, which as mentioned above, is the same as the external pressure. The critical area is a fundamental parameter that determines the geometry of the nozzle and directly affects the thrust and efficiency of the engine. Another fundamental parameter for calculating the critical area is the exit area of the nozzle, which, after various iterations, was set to 0.0038 m^2 in order to ensure the correct functioning of the nozzle. With these parameters known, it is possible to estimate the critical area, which turns out to be $8{,}02*10^{-4}$ m^2. The critical area has been calculated thanks to a specific thermofluidodynamic equation which involves the "gamma function", that is a specific function which involves the adiabatic expansion coefficient. Accurate determination of these areas is crucial to ensure efficient combustion and optimize booster thrust. Moving forward in the booster propulsion design, it became clearly necessary to estimate the propellant flow rate, i.e., the mass of propellant

consumed per unit time. This was derived using the following formula:

$$\dot{m} = \frac{p_0 \cdot A_{cr} \cdot f(\gamma)}{\sqrt{R \cdot T_0}}$$

where:

- p_0 is the stagnation pressure.
- A_{cr} is the critical area.
- $f(\gamma)$ is a specific function of the adiabatic expansion coefficient.
- R is the specific gas constant, assumed to be 287,05 J/(kg·K).
- T_0 is the stagnation temperature.

The propellant flow rate is estimated to be 1.5706 kg/sec.

9.5.3 Fuel quantity, Firing time and Axial Velocity

After calculating the propellant flow rate, the design moves on to determining fundamental parameters such as the required fuel quantity, firing time, equivalent exhaust velocity, thrust, as well as the length and diameter of the case containing the propellant grain. These parameters are essential to ensure that the booster can effectively perform its task. Before calculating the necessary propellant mass, an estimate of the total mass of the entire system (m), assumed to be 16 kg, should be made. This estimate includes all components of the system, from the propellant to the engine, to the missile structure itself. The objective is to minimize the mass of propellant required for the system's operation. To achieve this, a dedicated calculation code has been written that solves the equations of motion through an iterative process. In particular, this code solves the equations of motion for five variables and does so iteratively until the minimum propellant mass required to achieve the desired final velocity is found. The equations of motion are the following:

$$\dot{V}_{\text{axial}} = \frac{\dot{m} \cdot c}{m - \dot{m} \cdot t} - \frac{\frac{1}{2}\rho S C_D V_{\text{axial}}^2}{m - \dot{m} \cdot t} - g \sin(\vartheta) \tag{9.1}$$

$$\dot{V}_{\text{normal}} = g \cos(\vartheta) \tag{9.2}$$

$$\dot{x} = V_{\text{axial}} \cos(\vartheta) + V_{\text{normal}} \sin(\vartheta) \tag{9.3}$$

$$\dot{z} = V_{\text{axial}} \sin(\vartheta) - V_{\text{normal}} \cos(\vartheta) \tag{9.4}$$

$$\dot{\vartheta} = 0 \tag{9.5}$$

where:

- V_{axial} is the velocity along the wind axis, directed towards the nose of the missile.
- V_{normal} is the velocity orthogonal to V_{axial}.
- c is the equivalent exhaust velocity, it would be calculated below.
- t is the firing time, that is calculated iteratively.

- S is the frontal surface area of the missile.
- C_D is the drag coefficient of the missile, derived from the OpenRocket software.
- ϑ is the inclination angle of the trajectory, obtained from the specific 3DoF simulation.

The calculation code also includes an iterative estimation of the firing time, assumed as the ratio between the fuel mass and the fuel flow rate. The simulation inputs a cruising speed to avoid complications due to compressibility effects, and this axial speed is 200 m/s. As already said, near this value, the iterative procedure stops. This approach allows for the optimization of the booster design based on mission requirements. For a better overview, the listing of this code and its implementation in the main, using an ode45 function, is reported :

Script 9.2 Matlab code for calculation of the fuel mass

```
%% Searching mdot by dynamic equations integration

% iteration parameters
m0 = 16.035;                              % Total mass of the rocket at
    instant zero [kg]
for mp = 0.1:0.001:10                   % Propellant of the booster [kg]

    % Propellant Mass [kg]
    t_fin = mp/mdot;                       % time of booster separation

    state0 = [V0_axial, V0_normal, x0, z0, theta0_rad];

    [vTime,mState] = ode45(@SpintaFun,[0 t_fin],state0);
    if mState(end,1)>=200
        mState(end,1)
        mp
        break
    end
end
```

Script 9.3 Matlab code for the equations of motion

```
function [dStatedt] = SpintaFun(t,State)

global g mdot S c m0 Cd

% Initial statements
V_axial= State(1);
V_normal = State(2);
x = State(3);
z = State(4);
theta = State(5);


% position or time functions
V = sqrt(V_axial^2 + V_normal^2);
[~,a,p,rho] = atmosisa(-z);          % gas thermodynamic state as a
    function of altitude

% derivatives
V_axial_dot = (mdot*c)/(m0-mdot*t) - 0.5*rho*S*Cd*V_axial^2/(m0-mdot*t) -
     g*sin(theta);
V_normal_dot = g*cos(theta);
x_dot = V_axial*cos(theta) + V_normal*sin(theta);
z_dot = V_axial*sin(theta) - V_normal*cos(theta);
theta_dot = 0;

dStatedt = [V_axial_dot; V_normal_dot; x_dot; z_dot; theta_dot];
end
```

The final iteration returns the following outputs:

- Fuel mass $(m_p) = 1.288$ kg
- Firing time $(t) = 0.8201$ s
- Axial velocity = 200.15 m/s

9.5.4 Nozzle Sizing

The selected nozzle for the booster engine in question is a conical nozzle. More complex geometries, such as a Rao nozzle or a spike nozzle, have been judged excessively burdensome from a structural and technological point of view, while a conical nozzle, as will be seen shortly, guarantees sufficiently adequate performance for the mission profile with lower complexity and cost. As already specified, an exit pressure equal to the ambient pressure has been assumed; consequently, the nozzle will be adapted to sea level. This design choice makes sense, given the low operating altitudes of the booster and the significantly reduced firing time. Moving on to sizing, it depends on a series of factors:

- Inlet diameter
- Throat area
- Exit area
- Converging section angle

- Diverging section angle

The inlet diameter is known, being equivalent to the diameter of the propellant grain case

$$d_{\text{inlet}} = d_{\text{case}} = 0.0762\,\text{m}$$

The critical diameter is easily derived from the already known critical area:

$$d_{\text{crit}} = \sqrt{\frac{4 \cdot A_{\text{crit}}}{\pi}} = 0.032\,\text{m}$$

The converging section angle is assumed to be $45°$, to avoid issues related to excessive thermal heating. Given these data, it is possible to derive the length of the converging section:

$$L_{\text{conv}} = \left(\frac{d_{\text{inlet}}}{2} - \frac{d_{\text{crit}}}{2}\right) \tan\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = 0.0221\,\text{m}$$

The length of the throat section is assumed to be 4 millimeters to allocate the necessary space for an adequate cooling system (heat sink) and for the stabilization fins, as well as to avoid flow recirculation. Regarding the diverging section, it is necessary to know the exit area, easily derived from the exit diameter (equal to 0.07 m):

$$A_{\text{exit}} = \left(\frac{d_{\text{exit}}}{2}\right)^2 \pi = 0.0038\,\text{m}^2$$

The diverging section angle is assumed to be $10°$. The divergent angle is small to avoid performance losses due to non-axial outflow; however, it cannot be too small because the flow in a very long duct would experience losses due to the boundary layer. Given these data, it is possible to estimate the length of the diverging section:

$$L_{\text{div}} = \frac{d_{\text{crit}}\left(\sqrt{\frac{A_{\text{exit}}}{A_{\text{crit}}}} - 1\right)}{2\tan\left(\frac{\pi}{18}\right)} = 0.1078\,\text{m}$$

Now it is finally possible to calculate the overall length of the nozzle:

$$L_{\text{nozzle}} = L_{\text{conv}} + L_{\text{throat}} + L_{\text{div}} = 0.1339\,\text{m}$$

In conclusion, using an empirical formula specific to conical nozzles, it is possible to calculate the axial thrust coefficient (λ) and consequently the percentage thrust loss (T_{loss}).

$$\lambda = 0.5\left(1 + \cos\left(\frac{\pi}{18}\right)\right) = 0.9924$$

$$T_{\text{loss}} = 1 - \lambda = 0.0076$$

With a thrust loss of 0.76%, it can be confidently asserted that the nozzle is highly efficient, with thrust losses due to non-axial flow being negligible.

Figure 9.6 Nozzle of the booster

9.6 Aerodynamics

In this chapter, an aerodynamic analysis of the booster phase will be conducted, starting from the design of the fins, followed by the results from RASAERO, and concluding with a series of final observations.

9.6.1 Wings Second stage

The fins selected for the booster phase are identical to those used in 8.4.1. This decision was feasible due to the short duration of the analyzed phase, allowing the use of non-steerable fins. This choice provides beneficial results in terms of aerodynamics and stability.

9.6.2 Rasareo First + Second Stage

Using the data obtained from 8.4.3, the RASAERO program was updated to include considerations for the booster phase as well. The results are as following:

General Characteristics (I + II Stage):

$$C_p = 98.3cm$$
$$C_{g,initial} = 70.94cm$$

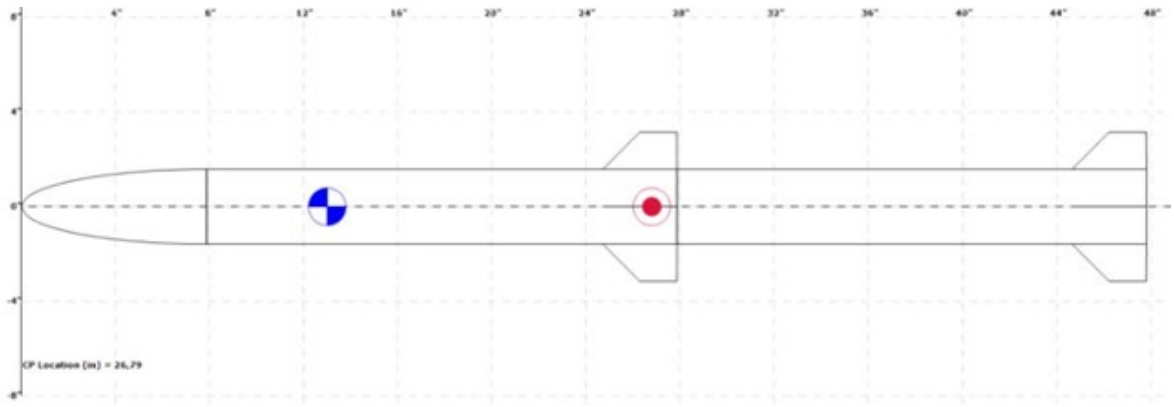


Figure 9.7 Rasaero analysis

RASAERO Data:

$$C_D = 0.35 \quad \text{(Drag coefficient)}$$
$$C_{Z,\alpha} = 7.2\,\text{rad}^{-1}\,(412.53\,\text{deg}^{-1})$$
$$C_{M,\alpha} = C_{Z,\alpha} \times \frac{C_{g,\text{position}} - C_{p,\text{position}}}{d} = -24.44\,\text{rad}^{-1}\,(\text{-}0.425\,\text{deg}^{-1})$$

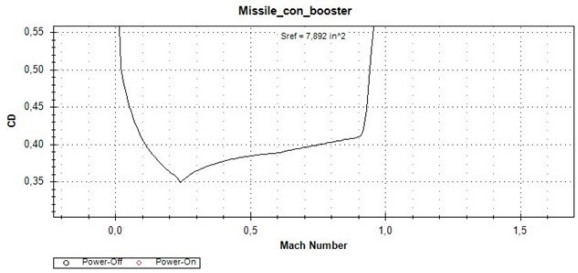


Figure 9.8 CD vs Mach

9.6.3 Observations

- The configuration of the fins for both the sustain and booster phases has been selected in a cruciform arrangement. This design choice was made to minimize structural issues and enhance maneuverability in subsonic conditions;
- This configuration is particularly advantageous given that the missile utilizes a skid-to-turn maneuvering system;
- Initially, the booster phase fins were staggered relative to those on the sustain phase. However, subsequent theoretical analysis indicated that aligning the fins for both phases does not introduce significant problems in terms of induced drag or load distribution;
- Utilizing the same fins for both phases also potentially reduces costs; however, cost analysis is not covered in this context.

9.7 structure

Regarding the structural sizing of the first stage, the procedures used and the Matlab codes implemented are entirely analogous to those discussed in subsections 1.2 and 1.3. The only difference to note is that, unlike the first stage, the booster does not have an outer aluminum layer. This is because the booster accompanies the first stage for a very short time interval, making it unnecessary to idealize a structure responsible for absorbing aerodynamic loads.

In the booster, the chamber pressure is $p = 30$ bar, which necessitates thickening the nozzle and casing. Specifically, considering a chamber diameter of 76.2 mm, as and a throat section diameter of approximately 32 mm, as results from the subsection 9.3.3 and 9.5.4, it's obtained the following dimensions:

- For the grain case, a thickness of 2.2 mm is determined. This value is also practically justified since, in the absence of an outer shell, the steel layer must have the same dimension as the aluminum layer to connect to the second stage;
- For the nozzle, a similar thickness of 2.2 mm is achieved, providing a homogeneous structural layer within the overall chamber.

No cooling system is provided for this chamber because, as previously mentioned, this stage has a short duration. Therefore, it is unnecessary to burden the structure with potential thermal liners, which would complicate the design and incur significant economic costs.

9.7.1 Wings FEM

As reported in subsection 9.6.1, the first stage wings, unlike those examined in the second stage, are hinged to the structure and are unable to rotate. The material used is the same as in subsection 8.5.6, aluminum 7075, with its characteristics listed in Table 8.5. For completeness, the FEM analyses are also reported, although it should be noted that the criterion discussed in the subsection 8.5.6 is fully met.
It should be emphasized that the load conditions applied and analyzed are the same as those considered during the sustain phase. the results are shown in figure 9.9

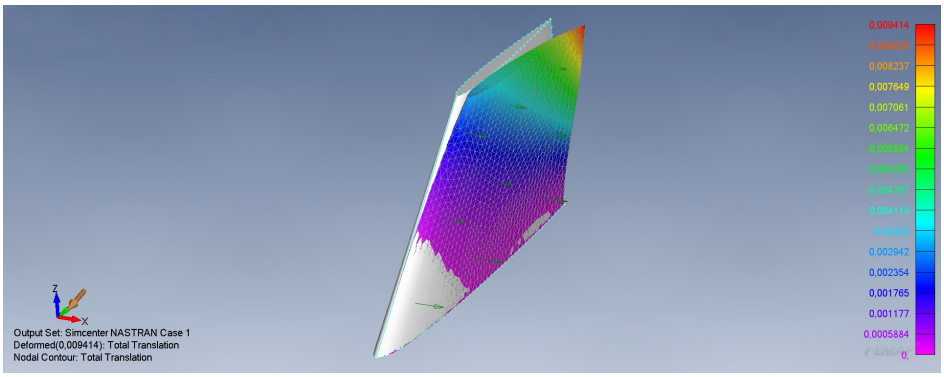


Figure 9.9 I stage Wings

Chapter 10

Off design Coonditions

As discussed in previous chapters, the requirements derived from the Simulink simulation were based on the reference scenario described in Chapter 3. In this chapter, off-design scenarios will be investigated to understand how the interceptor behaves in different warfare scenarios. As mentioned in Chapter 3, this chapter will display the graphics of the second stage cruise only. The initial conditions of this phase are the final conditions of the boosting phase, as explained in Chapter 5.

10.1 Off-design Scenario 1

The first off-design scenario involves a high-altitude launch where the target is detected near the zone to be protected. In this case, the target has an initial ground distance of 3000 meters and an initial altitude of 5500 meters, with a constant horizontal velocity of 60 m/s, flying towards the zone to be protected.

Initial ground distance	3000 m
Initial altitude	5500 m
Velocity	60 m/s

Table 10.1 Target parameters

In this scenario, it is evident that the missile intercepts the threat as the distance to go reaches zero. From the mass versus time plot, it can be observed that the propulsive burn ends nearly when the target is about to be intercepted. However, the airspeed drops rapidly once the thrust is nullified. This occurs due to the high pitch angle, θ, as shown in the corresponding graph. The parallel but opposite direction of the gravity acceleration and the velocity vector causes the missile airspeed to drop quickly.

As the missile approaches the target, the P_k value increases. In this case, unlike the reference scenario, the increase in P_k is slower. This may be due to the lower velocity of the missile and, more specifically, the lower relative speed, since in the reference scenario, the missile and the target fly towards each other, whereas here they fly perpendic-

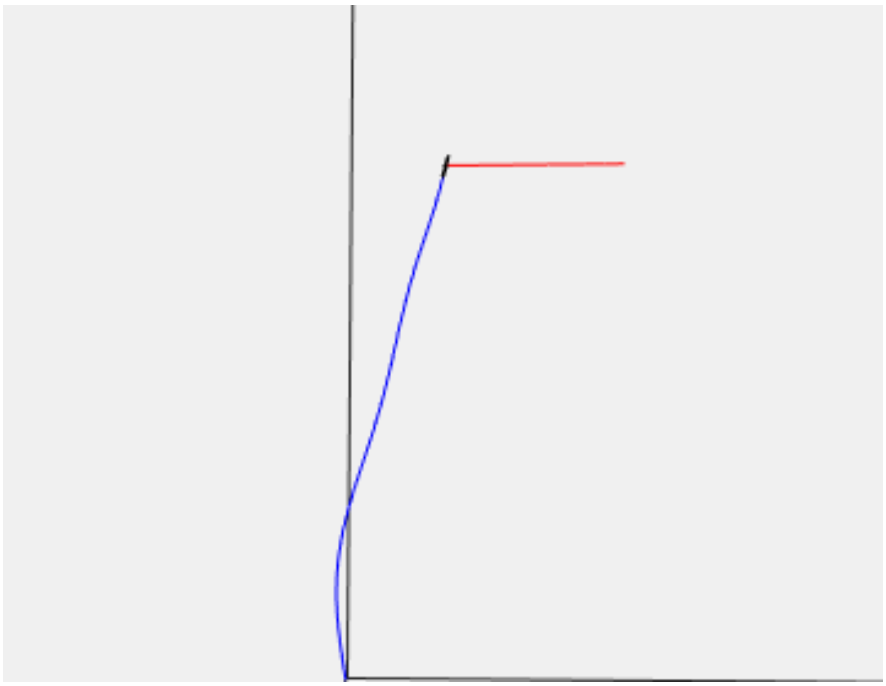


Figure 10.1 Second stage trajectory

ularly. Furthermore, the P_k shows a more precise determination, with denser and higher fidelity values, because the pitch angle is greater than 45 degrees, allowing for more accurate hypotheses, as explained in Chapter 7.

In contrast to the reference scenario, this scenario shows higher oscillations in the angle of attack and pitch angle. This is due to the faster corrections needed as the target is nearer and the error from the auto-aiming function, which is calibrated for the reference scenario. As in Chapter 3, the divergence at the end of the simulation can be noted in the plots of the demanded acceleration and the look angle. Here, the activated acceleration follows the demanded acceleration.

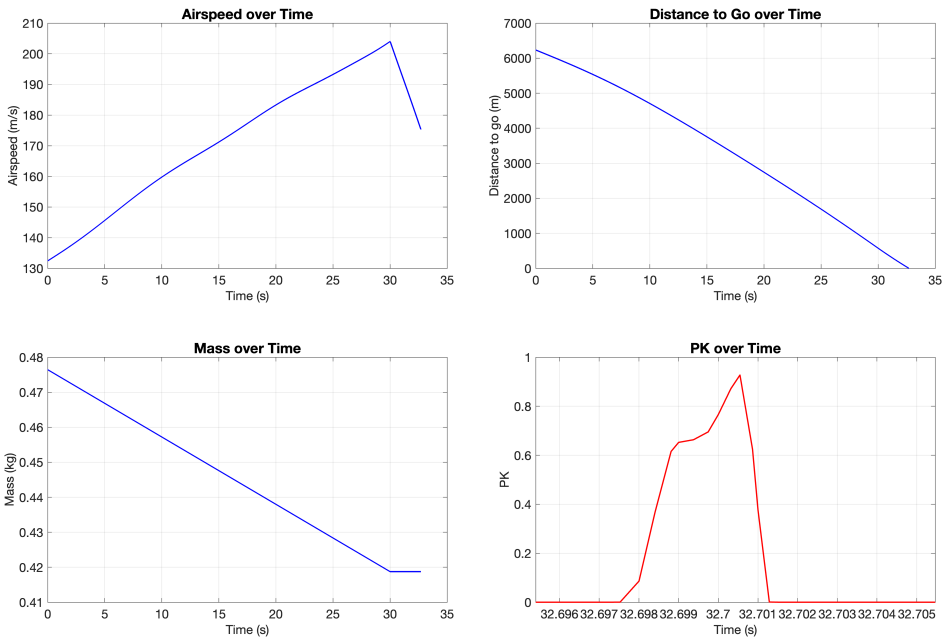


Figure 10.2

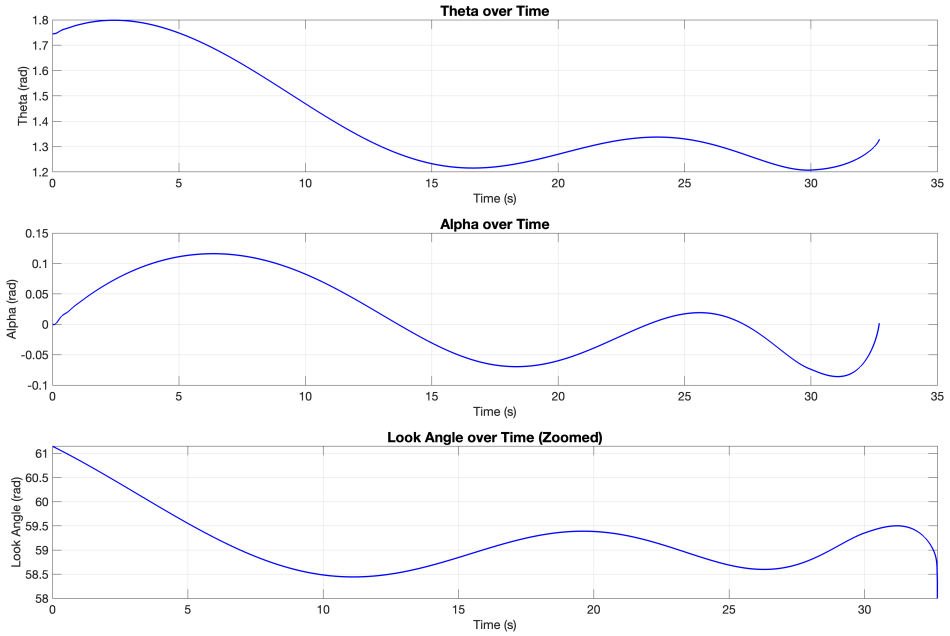


Figure 10.3

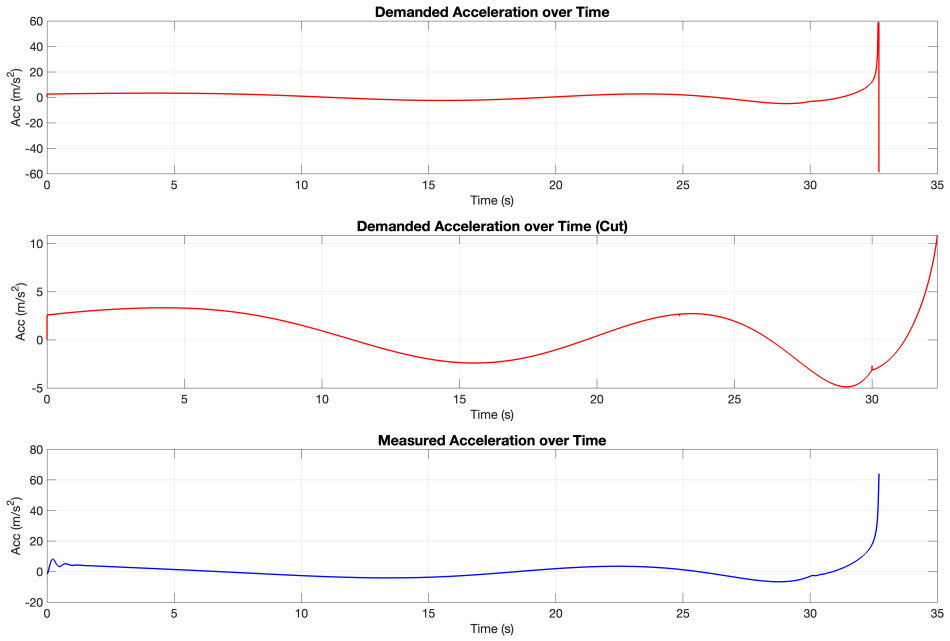


Figure 10.4

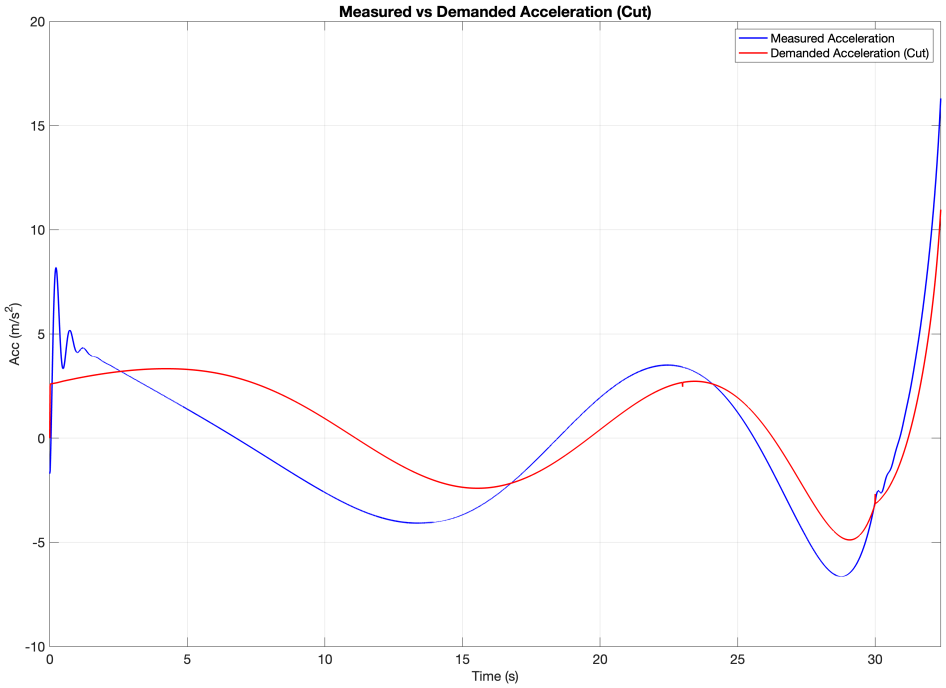


Figure 10.5

10.2 Off-design Scenario 2

This scenario investigates the case where the target is at a lower altitude but further away. Initially, the target is at a ground distance of 10000 meters and an altitude of 2000 meters, and as in all cases, it is flying horizontally at a constant speed of 60 m/s towards the zone to be protected.

Initial ground distance	10000 m
Initial altitude	2000 m
Velocity	60 m/s

Table 10.2 Target parameters

In this scenario, the missile is capable of intercepting the threat as the distance to go reaches zero, and non-zero P_k values are obtained. As in the other test cases, the burn finishes before the missile reaches the target, resulting in a flight portion where there is no thrust, and the missile is subjected only to drag and gravity effects. Here, the effects of these forces are weaker, as indicated by the slower decrease in airspeed compared to the other cases. This suggests that the major contributor to the missile's deceleration in the reference scenario and the previous off-design scenario is gravity, due to the significant pitch angle.

In this scenario, the pitch angle is smaller, and the primary contributor to deceleration is drag, as higher speeds are reached. The higher speed is due to the missile not expending fuel to gain altitude but propelling itself horizontally to a higher velocity, nearing transonic speeds. The trajectory is nearly a straight line, similar to the reference scenario, indicating that the auto-aiming function is more accurate for low angles. Consequently, there are minimal demanded and actuated accelerations. The divergence

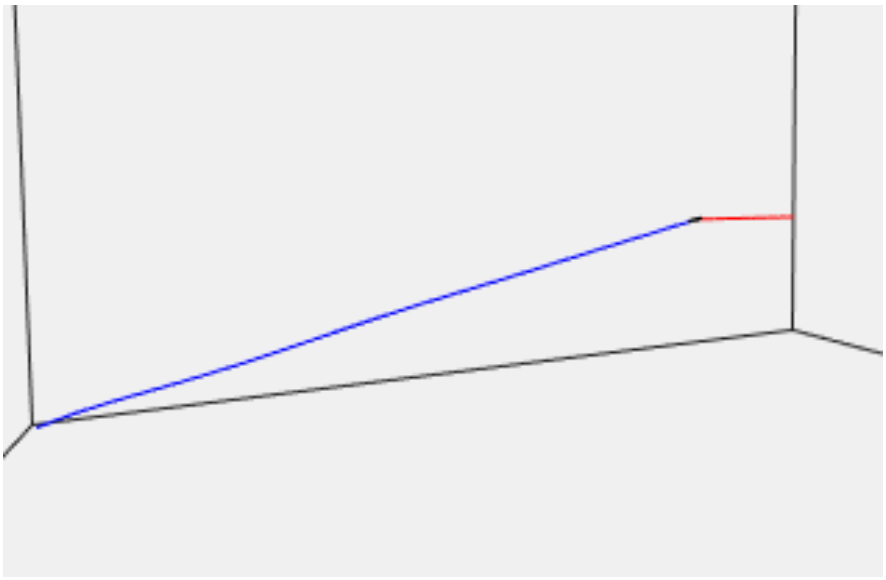


Figure 10.6 Second stage trajectory

trend shown in the other test cases for the demanded acceleration and look angle persists here, reinforcing the explanation of the phenomenon.

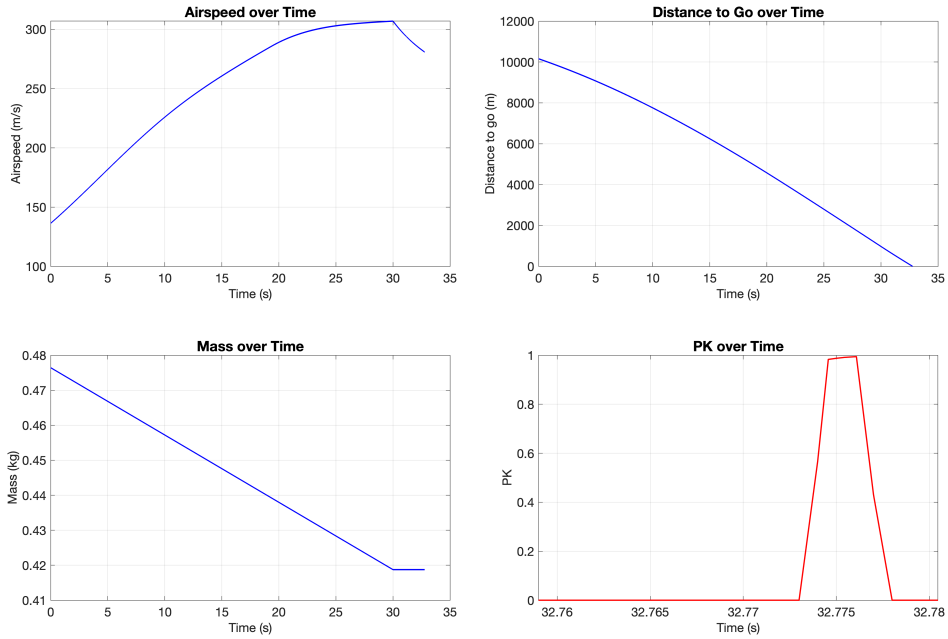


Figure 10.7

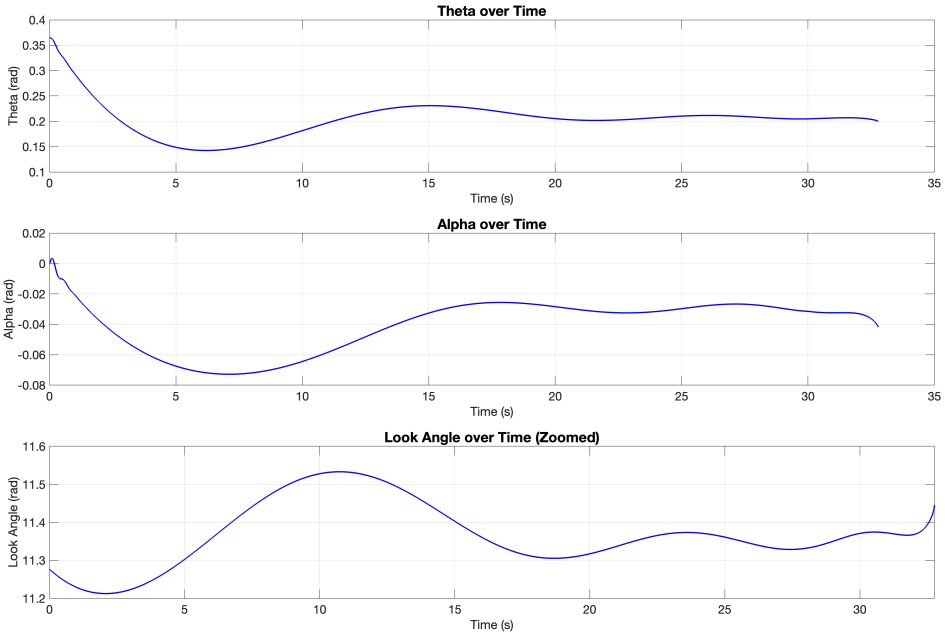


Figure 10.8

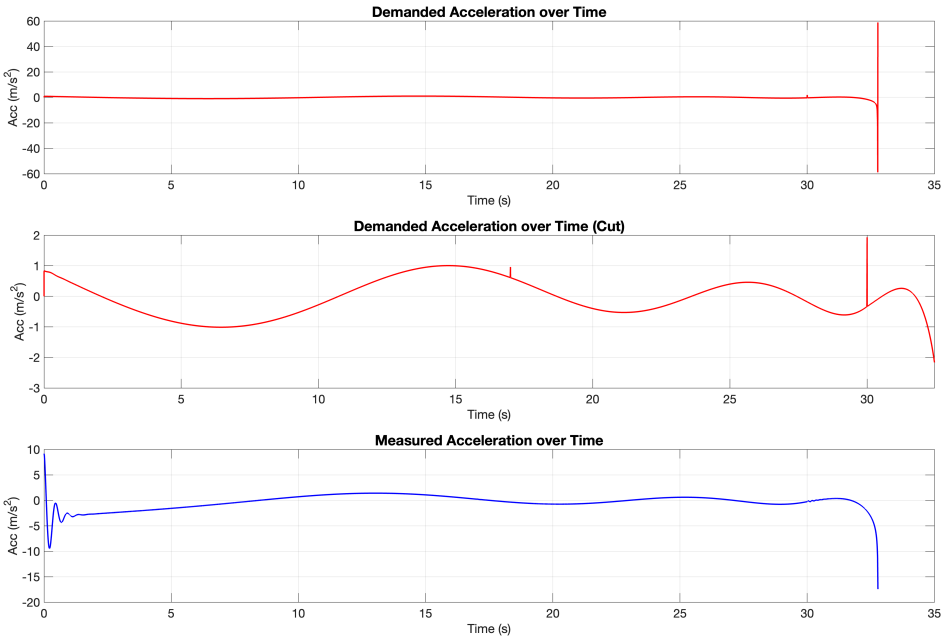


Figure 10.9

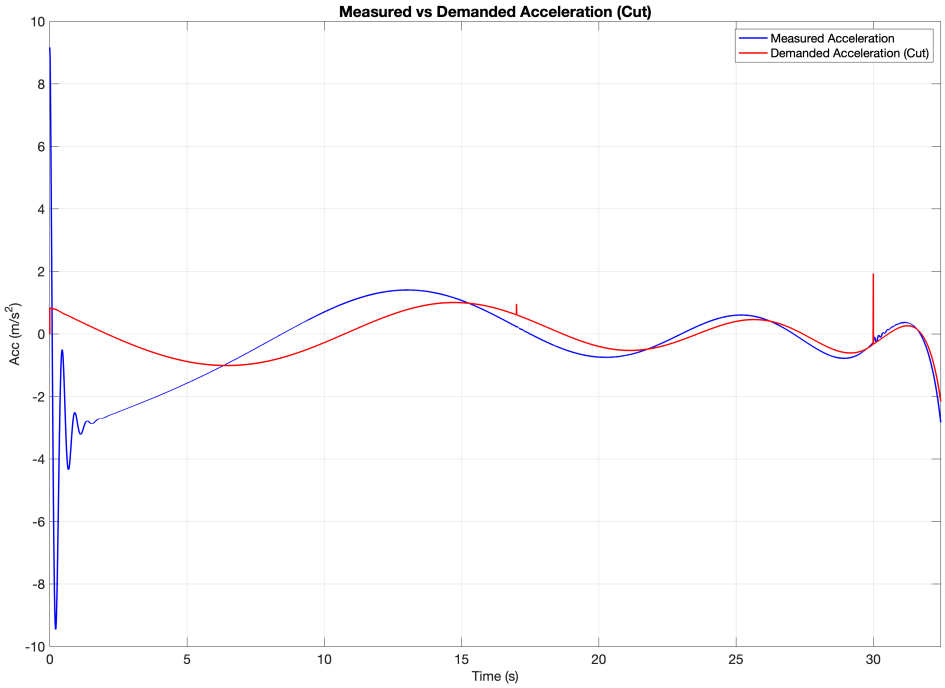


Figure 10.10

10.3 Off-design Scenario 3

This scenario investigates the case where the target is at a lower altitude but further away. Initially, the target is at a ground distance of 10000 meters and an altitude of 2000 meters, it is flying at a constant speed of 60 m/s towards the zone to be protected but after 20 seconds it performs a dive with a 30 deg angle.

Initial ground distance	10000 m
Initial altitude	2000 m
Velocity	60 m/s
Dive angle	30 deg

Table 10.3 Target parameters

In this scenario it is possible to see, that also here the missile ends the burn before intercepting the threat, nevertheless it succeed as the distance to go goes to zero. The P_k reaches the value of nearly one demonstrating a high efficiency of the system. In the angles plots it is possible to note how an oscillation begins as soon as the dive of the target begins. Also the demanded acceleration varies immediately.

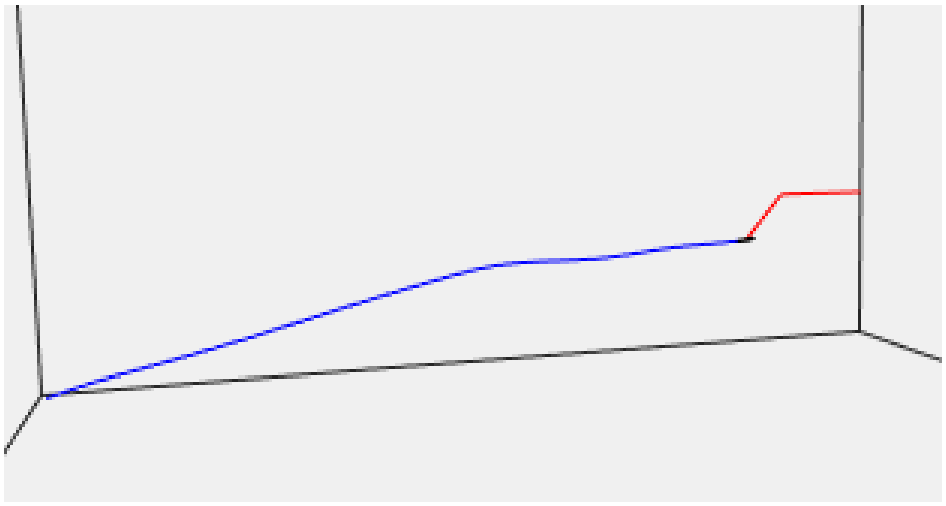


Figure 10.11

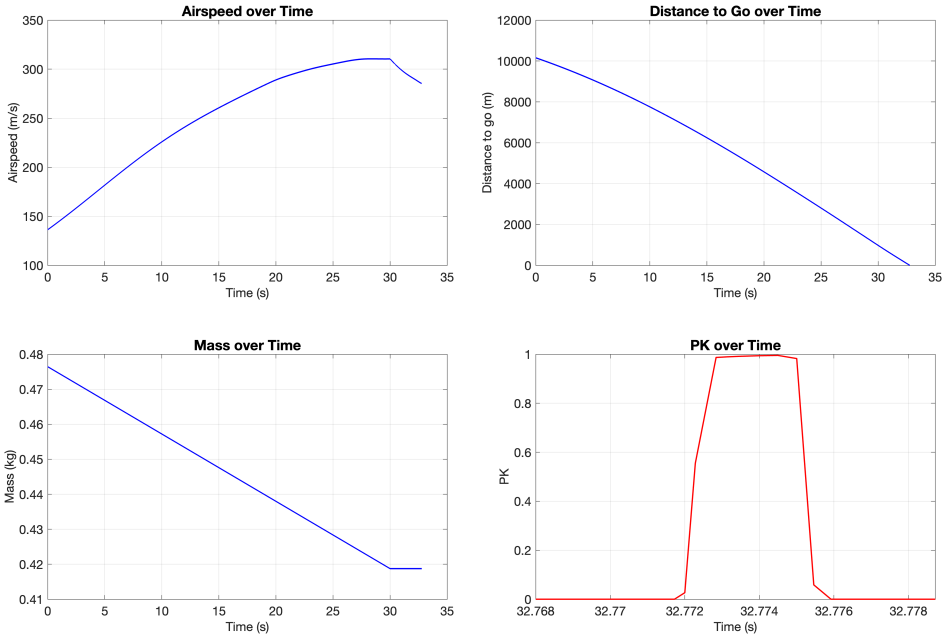


Figure 10.12

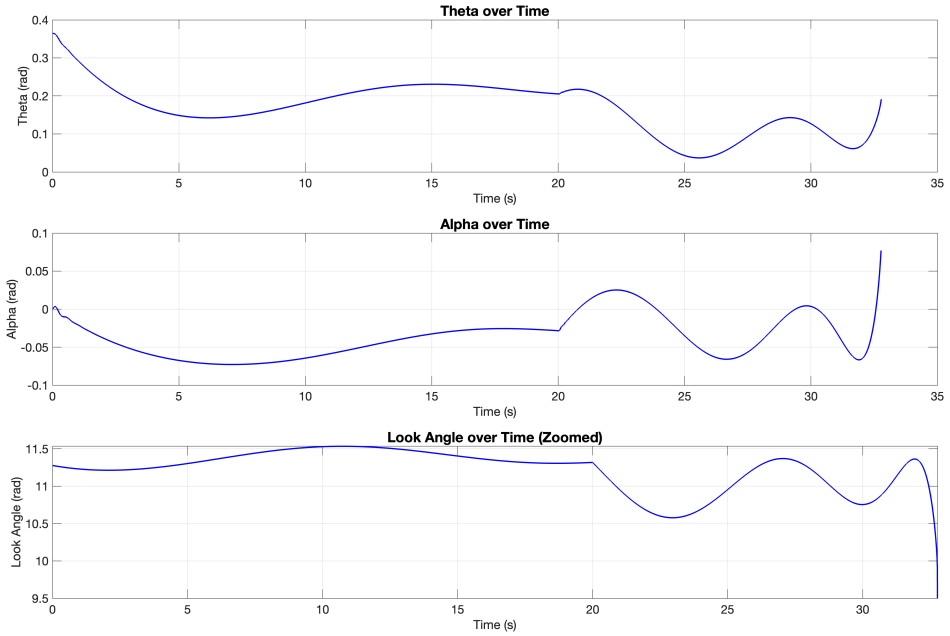


Figure 10.13

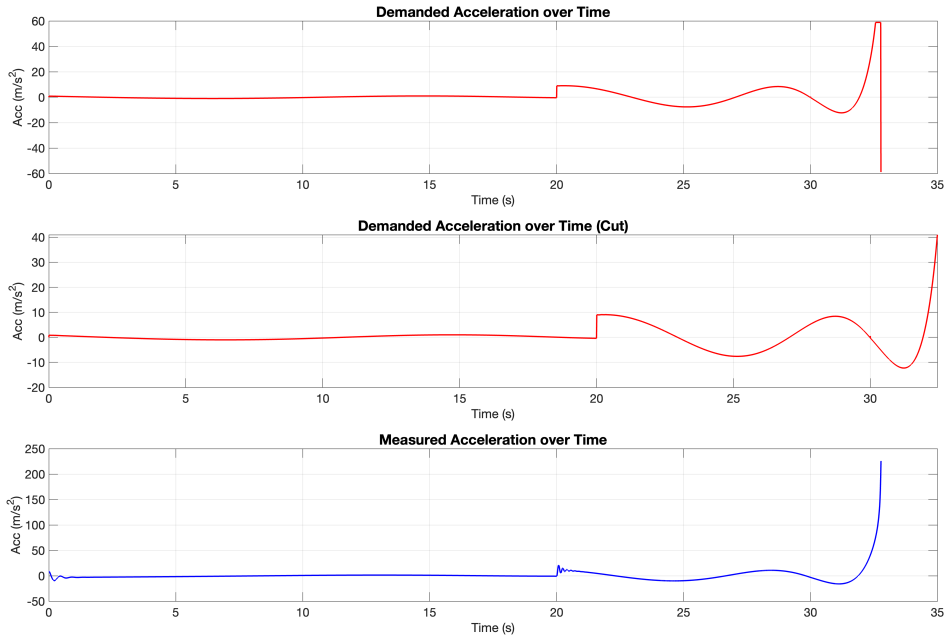


Figure 10.14

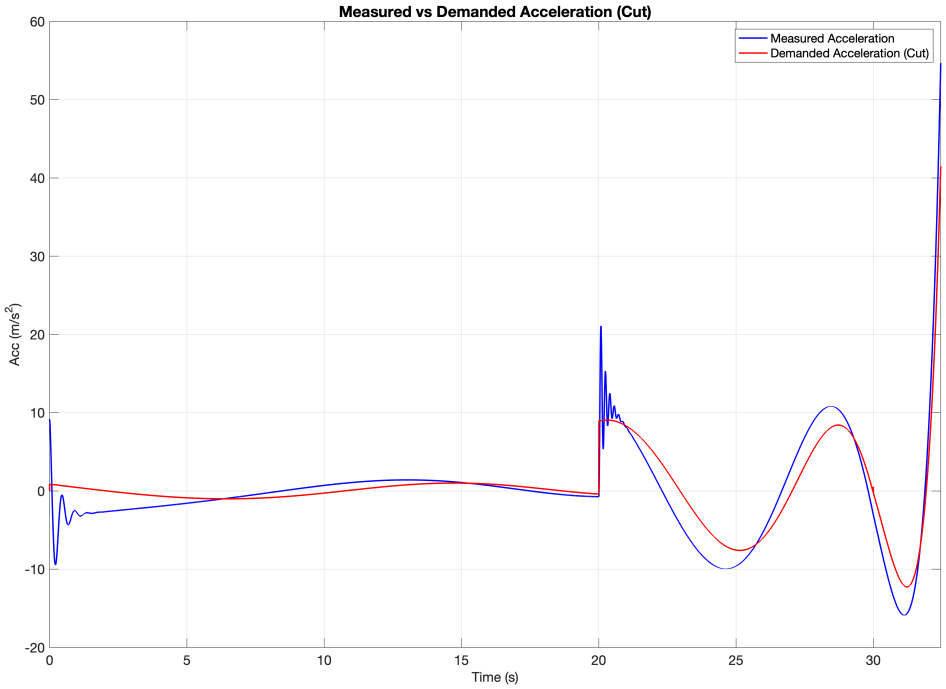


Figure 10.15

10.4 Off-design Scenario 4

This scenario investigates the case where the target is at a high altitude and near to the zone to protect. Initially, the target is at a ground distance of 3000 meters and an altitude of 7000 meters, it is flying at a constant speed of 60 m/s towards the zone to be protected but after 20 seconds it performs a surge with a 30 deg angle.

Initial ground distance	3000 m
Initial altitude	7000 m
Velocity	60 m/s
Surge angle	30 deg

Table 10.4 Target parameters

In this scenario it is possible to see that the missile doesn't have enough energy to reach the threat as it performs a surge to evitate it.

Figure 10.16

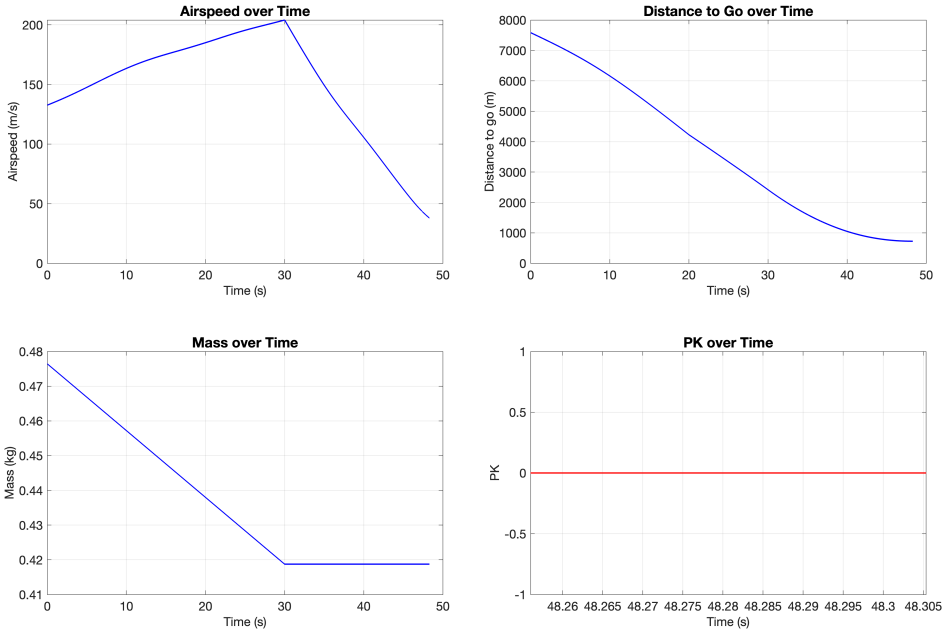


Figure 10.17

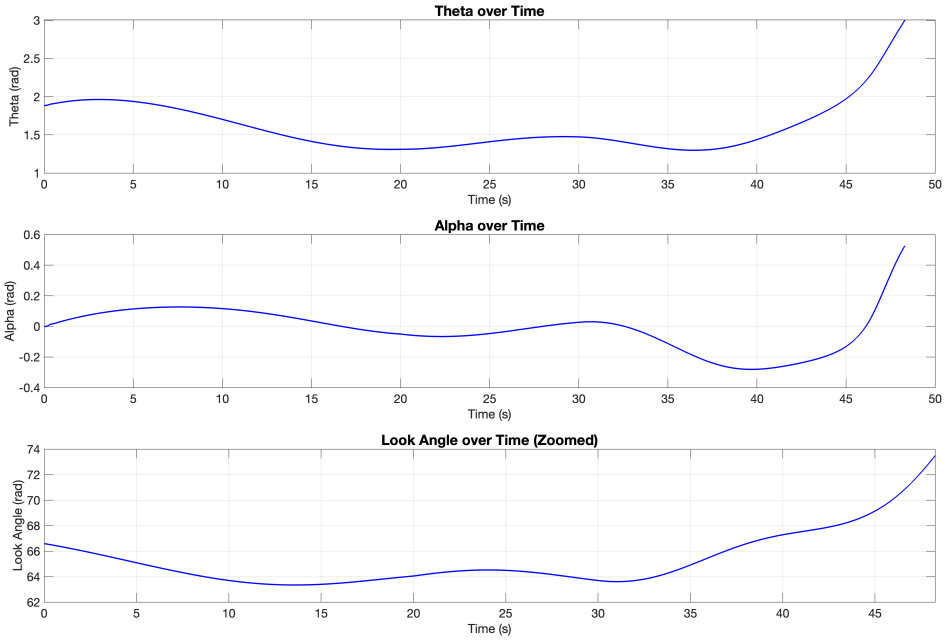


Figure 10.18

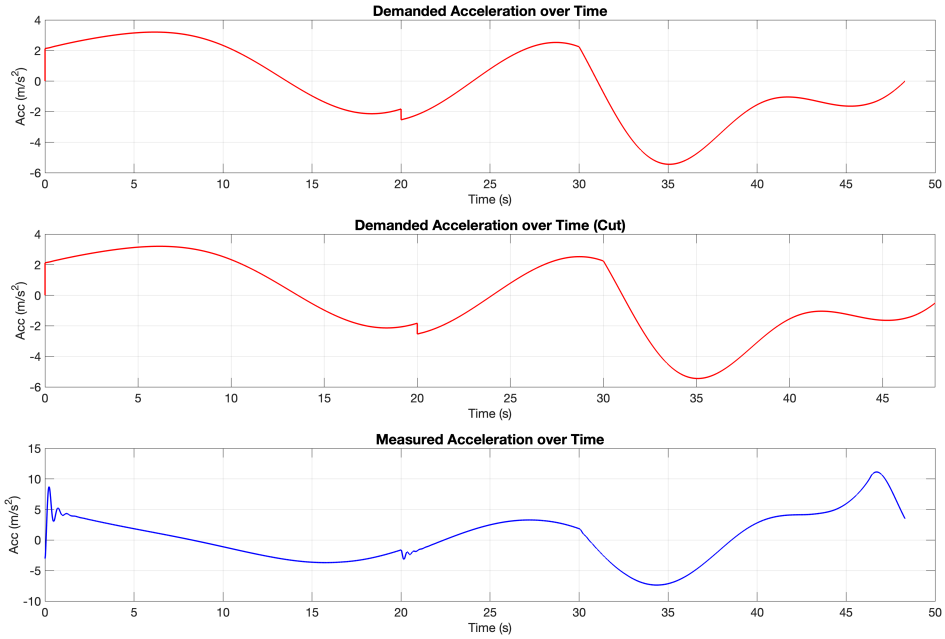


Figure 10.19

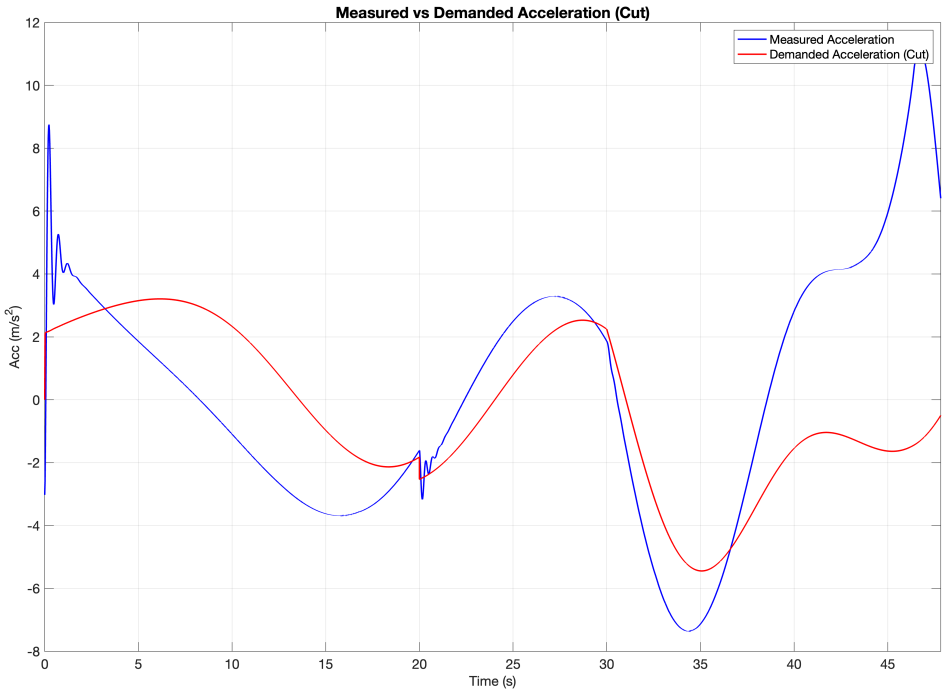


Figure 10.20

10.5 Off-design Scenario 5

This scenario investigates the case where the target is at the design scenario initial position and performs a dive after 20 seconds form the beginning of the simulation. Initially, the target is at a ground distance of 8000 meters and an altitude of 5000 meters, it is flying at a constant speed of 60 m/s towards the zone to be protected but after 20 seconds it performs a dive with a 30 deg angle.

Initial ground distance	8000 m
Initial altitude	5000 m
Velocity	60 m/s
Dive angle	30 deg

Table 10.5 Target parameters

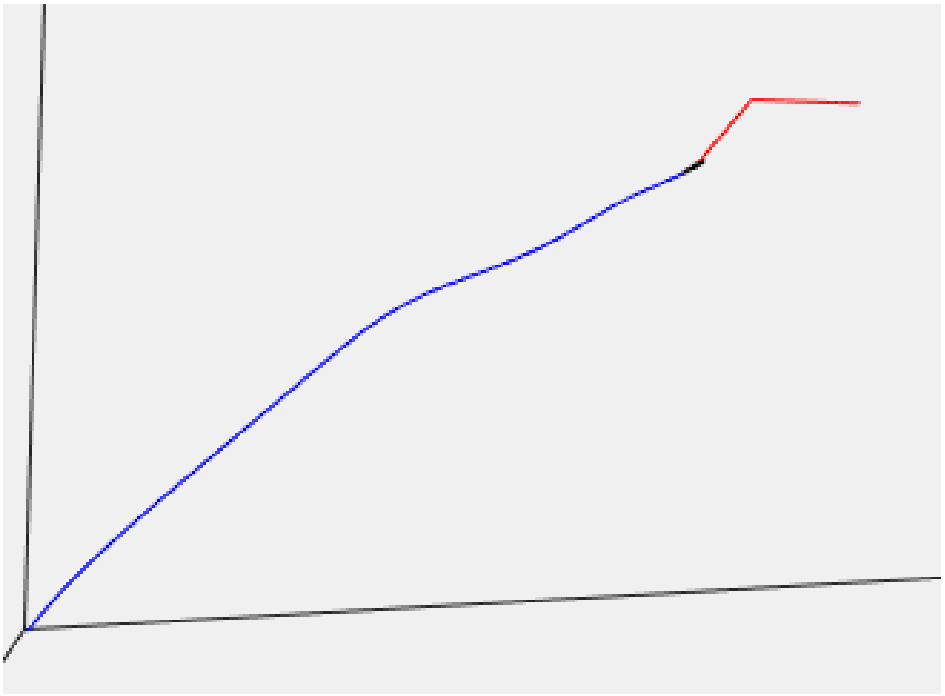


Figure 10.21

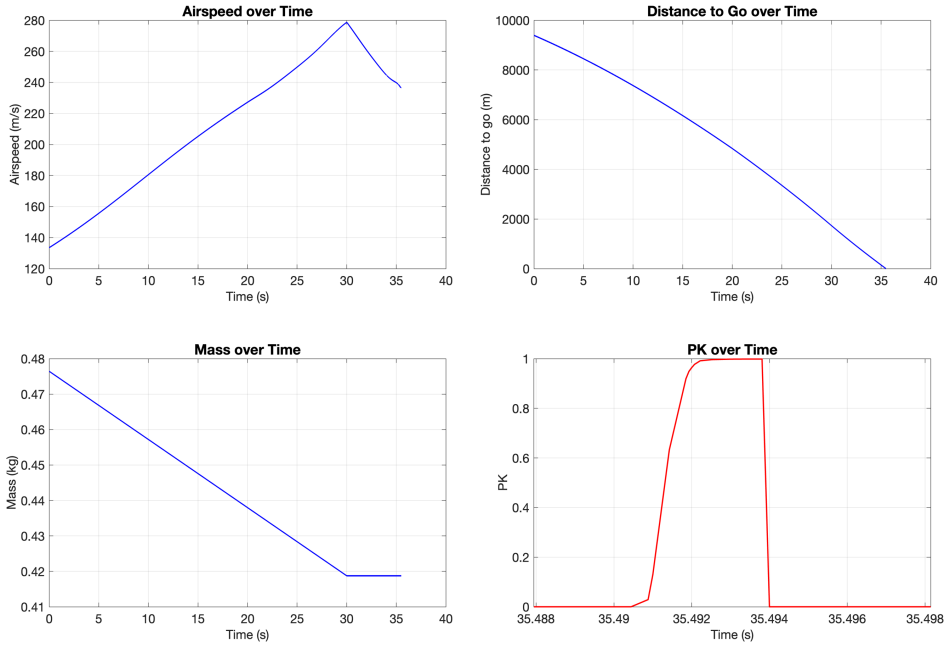


Figure 10.22

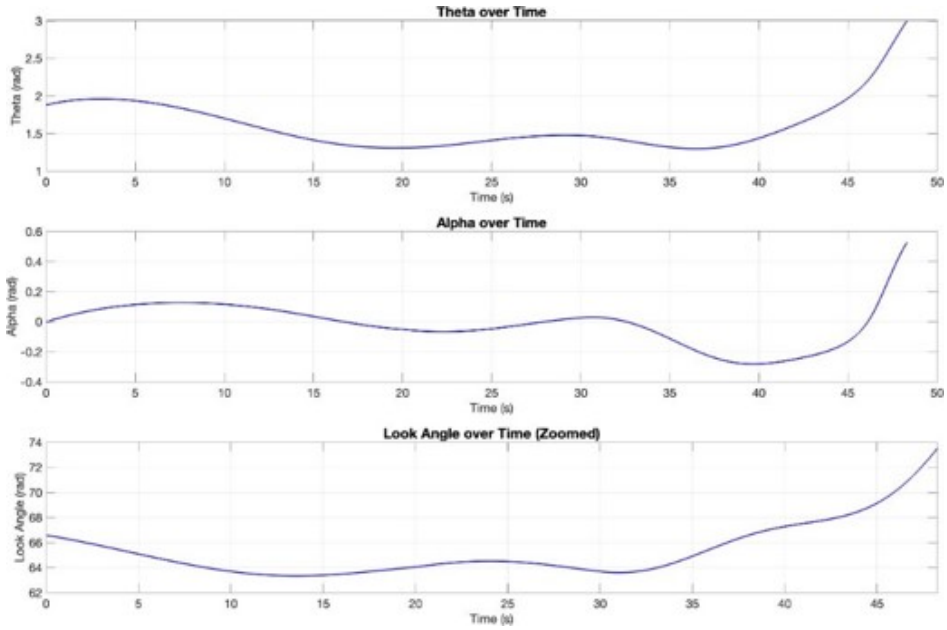


Figure 10.23

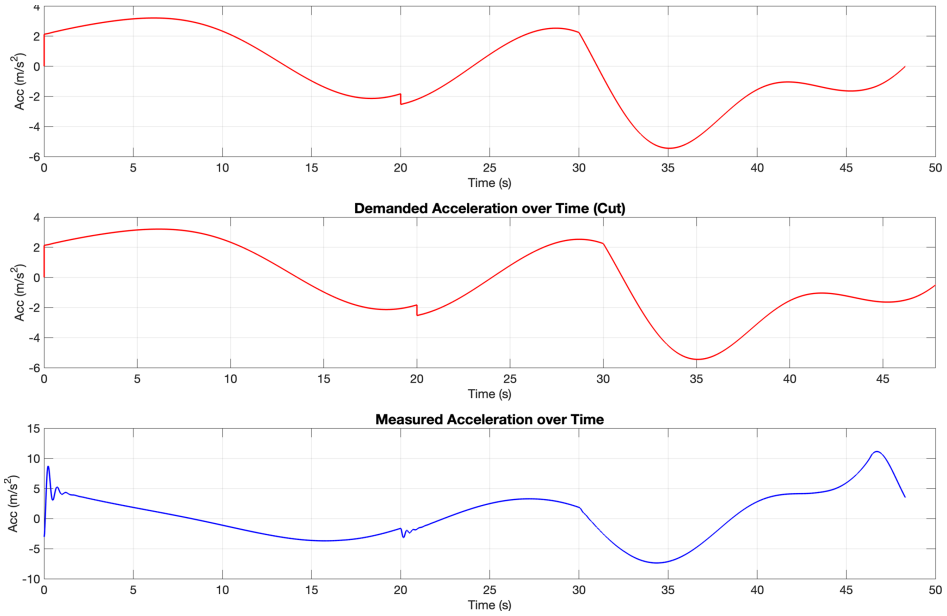


Figure 10.24

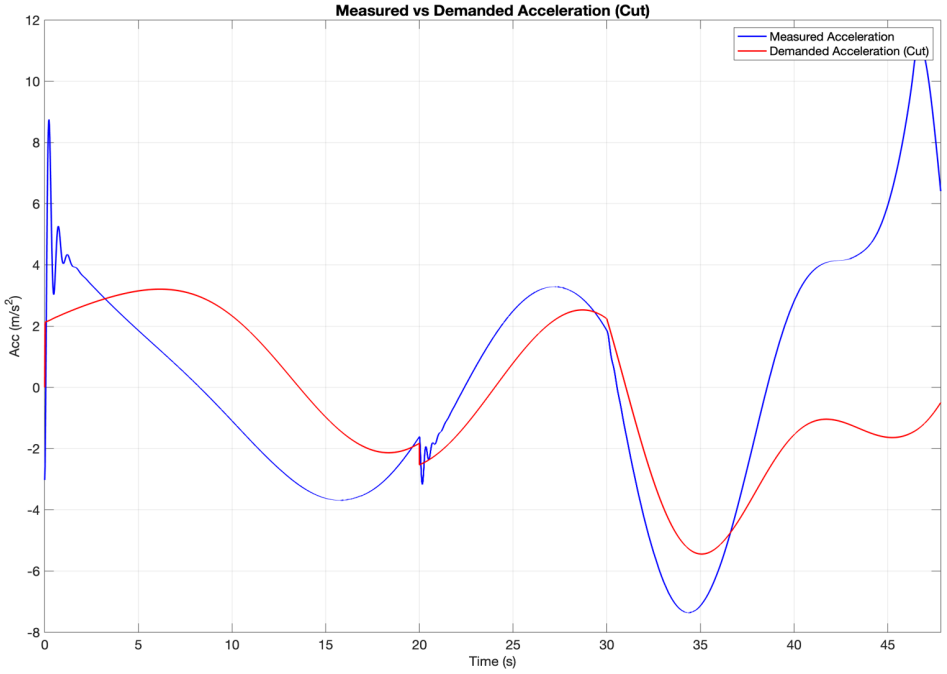


Figure 10.25

10.6 Off-design Scenario 6

This scenario investigates the case where the target is at the design scenario initial position and performs a dive after 20 seconds form the beginning of the simulation. Initially, the target is at a ground distance of 8000 meters and an altitude of 5000 meters, it is flying at a constant speed of 60 m/s towards the zone to be protected but after 20 seconds it performs a dive with a 45 deg angle.

Differently form the previous off-design scenario, it is possible to note that here we only reach a P_k value of approximately 0.5. This reveals a lower efficiency of this system when the dive angle is big, this result is something we expect to have since we have proportional navigation, and since it is harder to reach the perfect position for the missile,

Initial ground distance	8000 m
Initial altitude	5000 m
Velocity	60 m/s
Dive angle	45 deg

Table 10.6 Target parameters

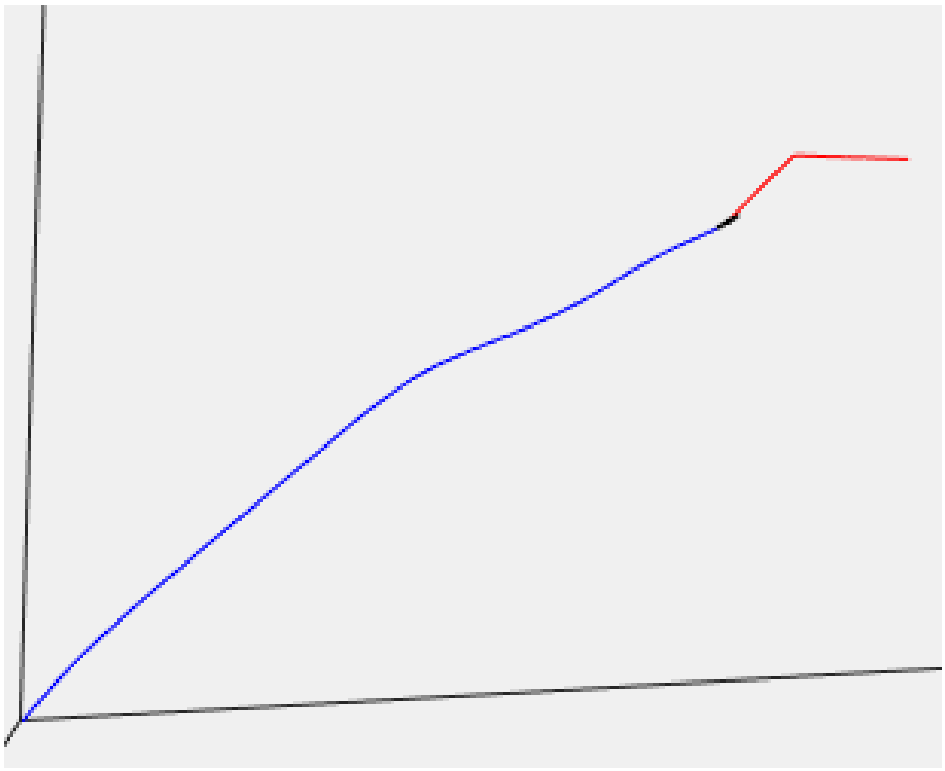


Figure 10.26

that is on the below-right of the target, as explained in the aiming correction paragraph.

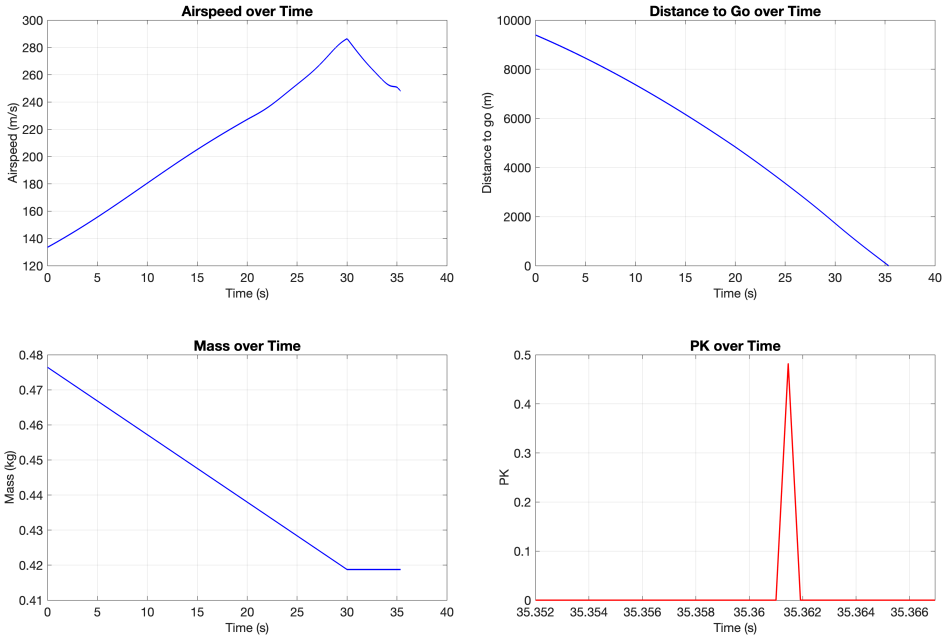


Figure 10.27

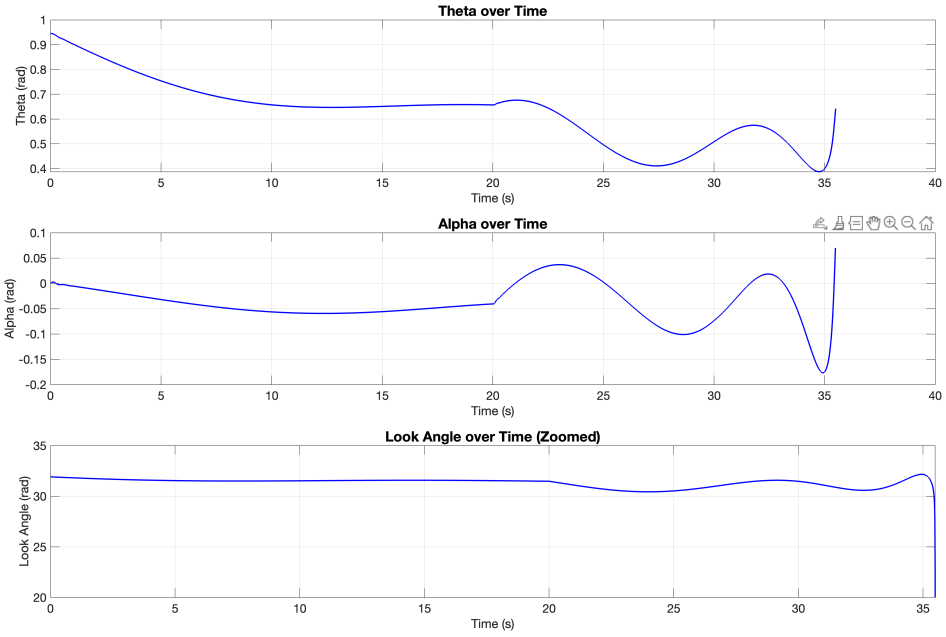


Figure 10.28

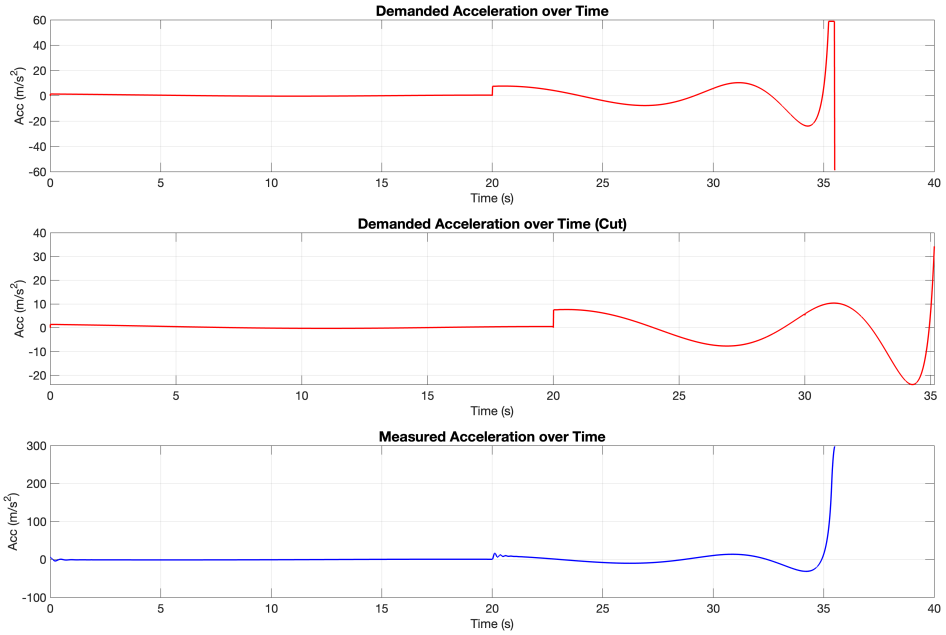


Figure 10.29

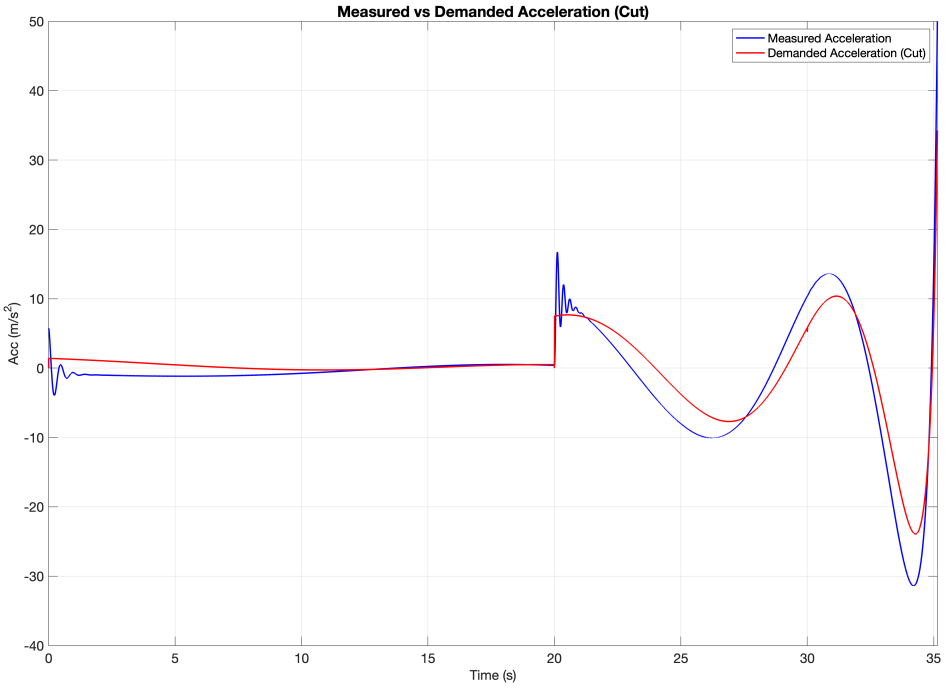


Figure 10.30

10.7 Off-design Scenario 7

This scenario investigates the case where the target is at the design scenario initial positions and performs a sinusoidal trajectory. Initially, the target is at a ground distance of 8000 meters and an altitude of 5000 meters, it is flying at a constant speed of 60 m/s towards the zone to be protected the flight proceeds as sinusoid of Amplitude of 0.8 rad with a frequency of 0.1 $s^{-}1$.

Initial ground distance	8000 m
Initial altitude	5000 m
Velocity	60 m/s
Amplitude	0.8 rad
Frequency	0.1 s^{-1}

Table 10.7 Target parameters

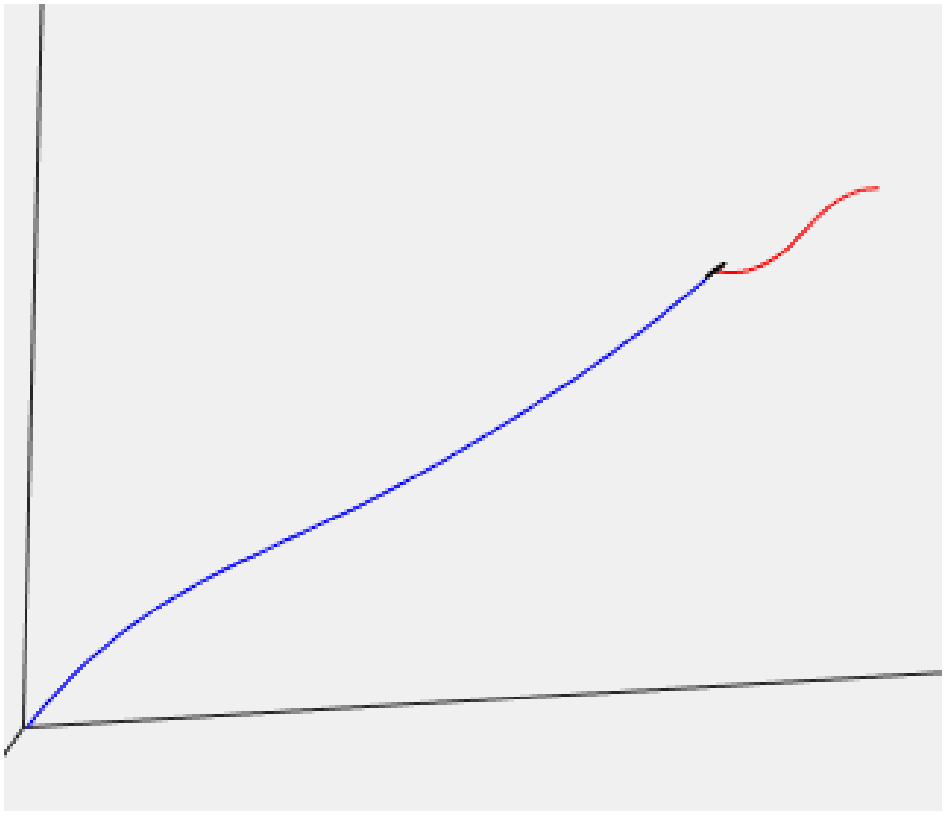


Figure 10.31

It is possible to underline the robustness of the system as also in this scenario we have high values of the P_K Also the guidance system, even if it presents a bit of oscillations seems to preview quite well the future target position.

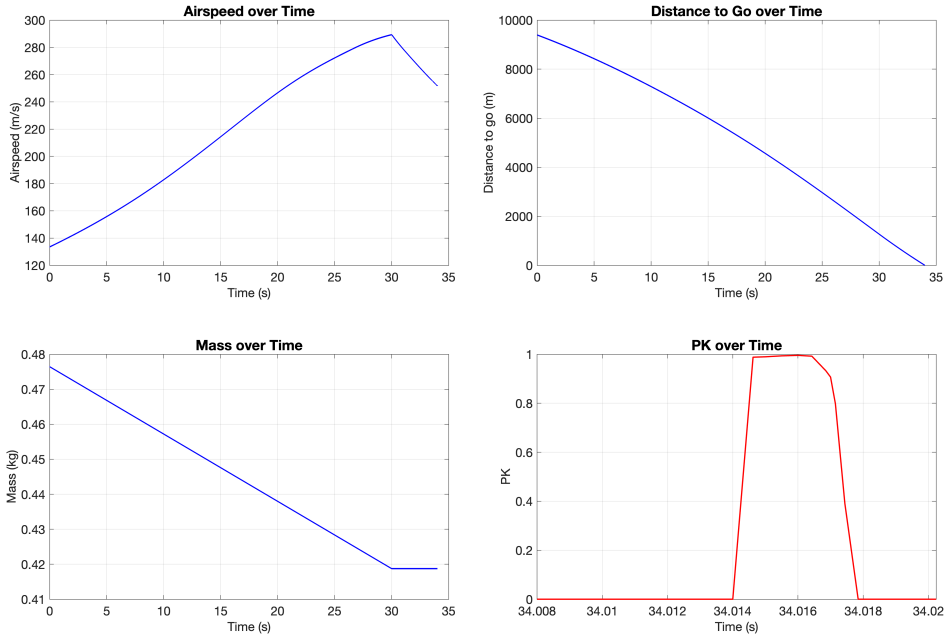


Figure 10.32

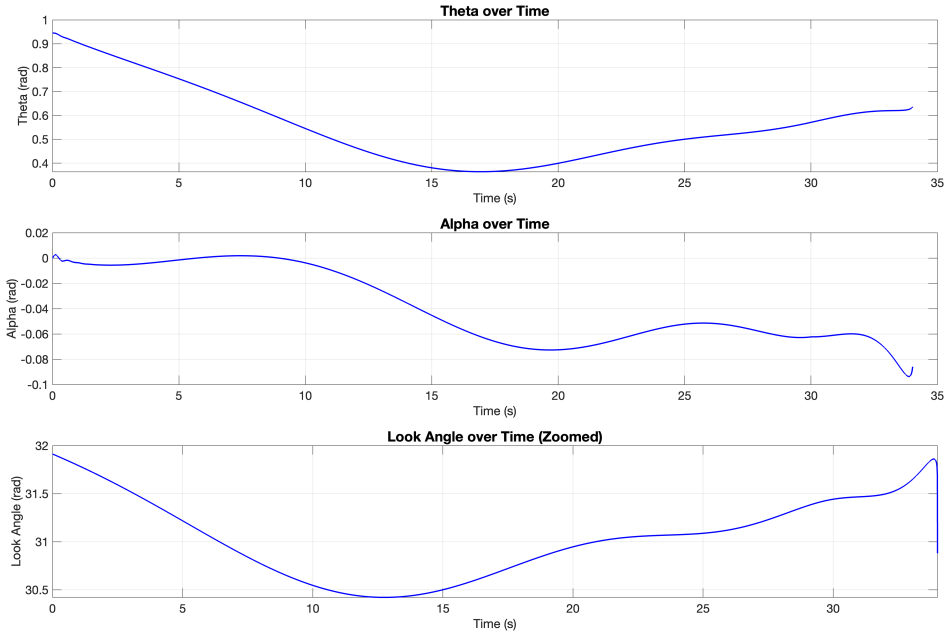


Figure 10.33

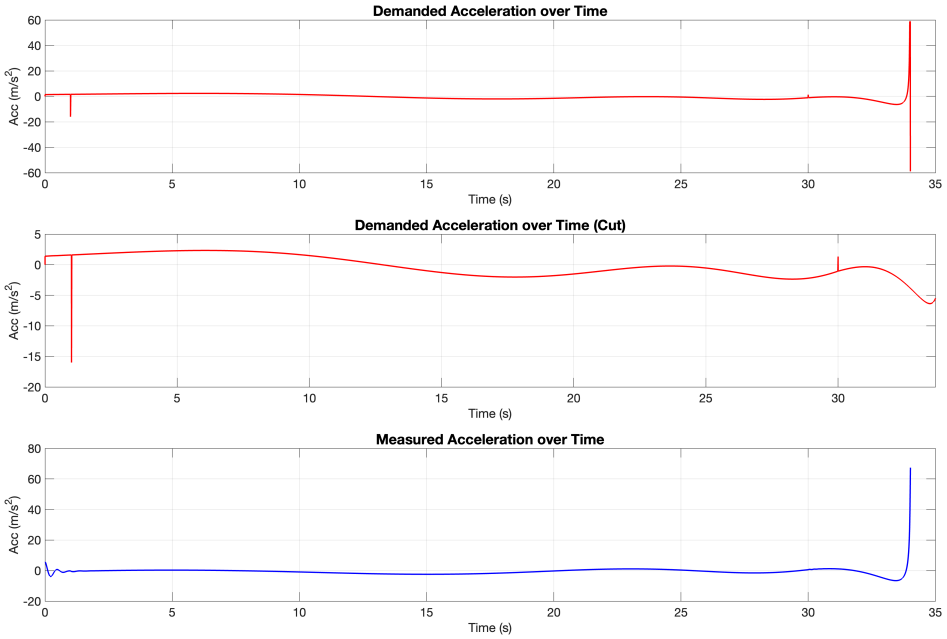


Figure 10.34

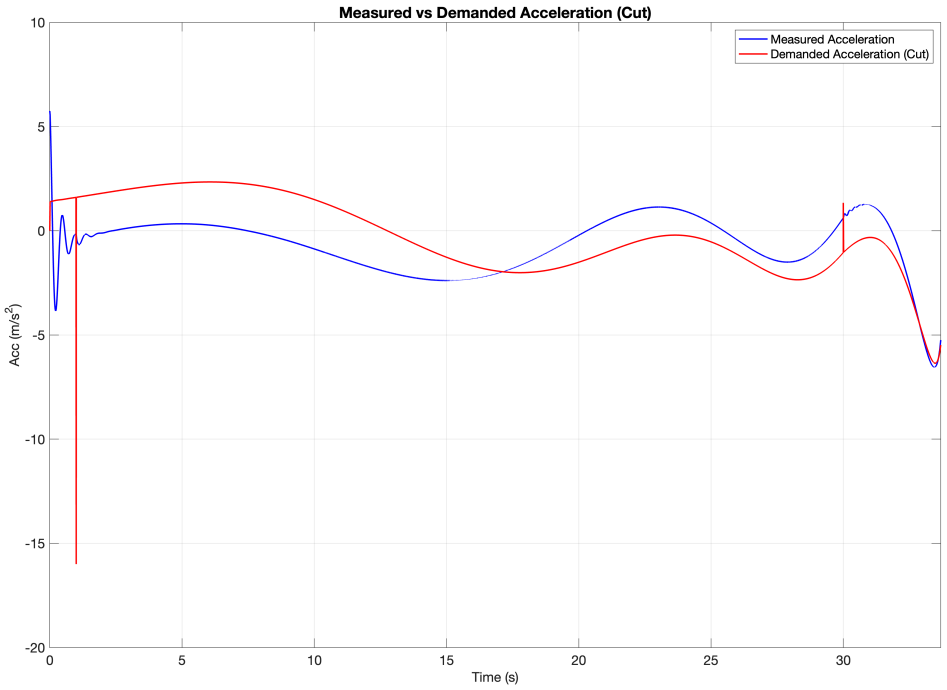


Figure 10.35

10.8 Off-design Scenario 8

This scenario investigates the case where the target is at the design scenario initial positions and performs a sinusoidal trajectory. Initially, the target is at a ground distance of 8000 meters and an altitude of 5000 meters, it is flying at a constant speed of 60 m/s towards the zone to be protected the flight proceeds as sinusoid of Amplitude of 0.8 rad with a frequency of 0.5 $s^{-}1$.

Initial ground distance	8000 m
Initial altitude	5000 m
Velocity	60 m/s
Amplitude	0.8 rad
Frequency	0.5 s^{-1}

Table 10.8 Target parameters

Here, differently from the previous scenario the frequency of the target oscillations seems to be too high, causing a failure of the system. It is possible that this is due to the fact that the missiles goes over the target and this is not included in our PK description model.

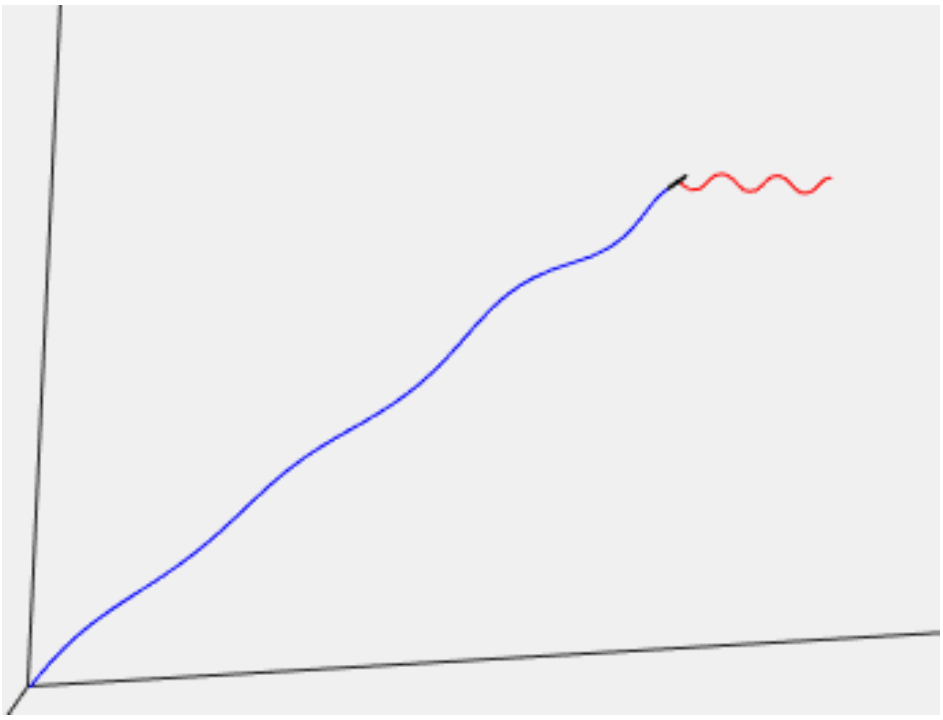


Figure 10.36

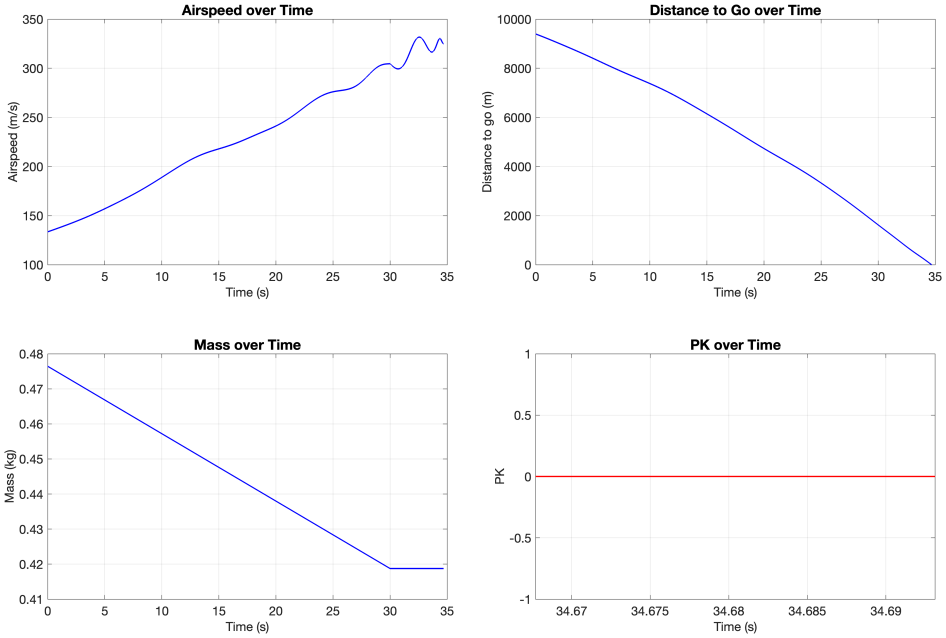


Figure 10.37

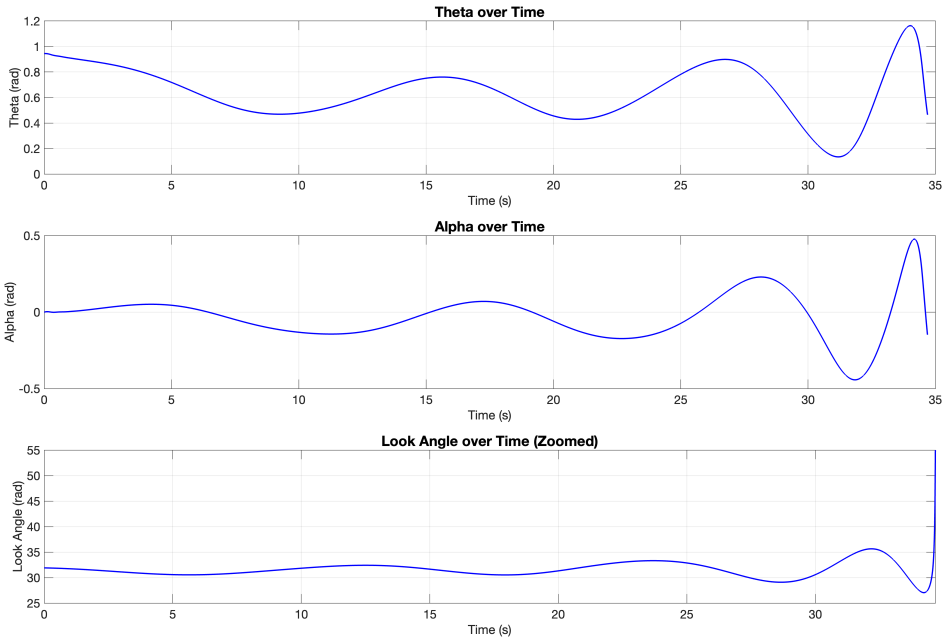


Figure 10.38

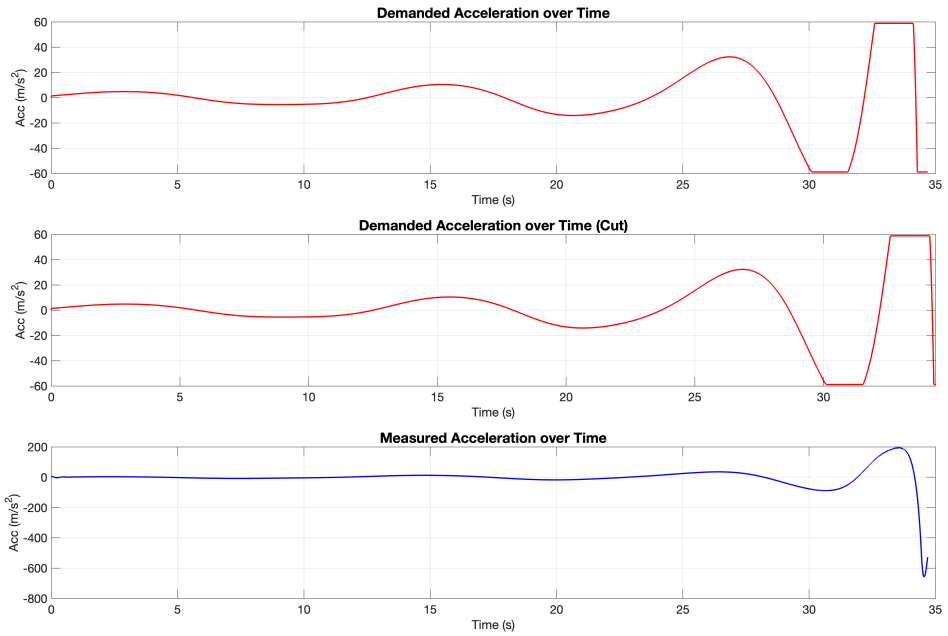


Figure 10.39

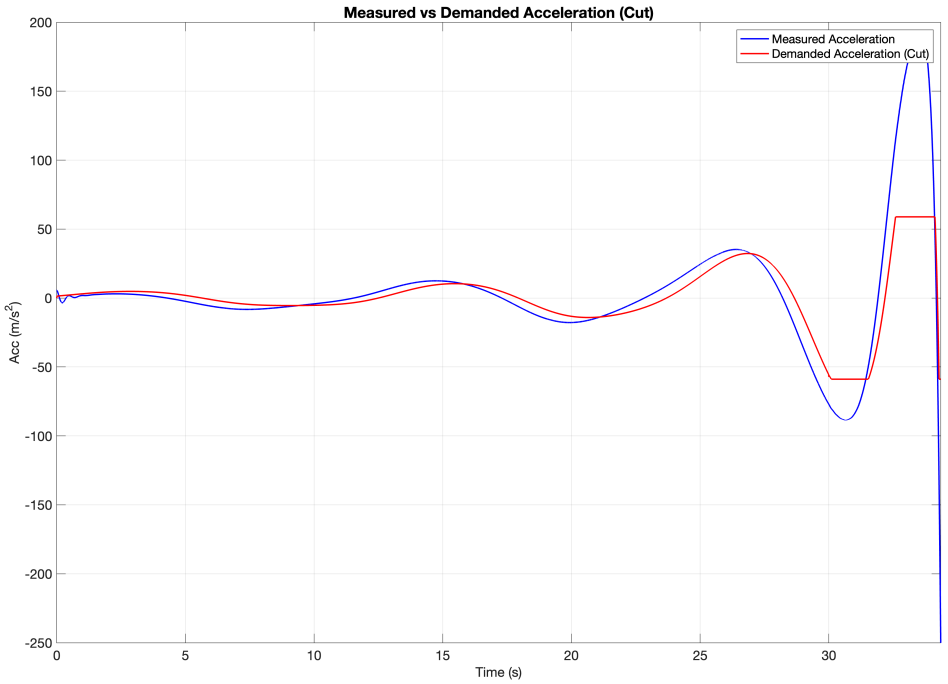


Figure 10.40

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